

Single-Parameter Gauge-Covariant Loop-Flow Control for Phase-Shifting Transformers and Meshed Converter Networks

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Abstract

An *active-phase link element* is a branch whose series phase offset θ_ℓ is set independently of its endpoint bus angles: a phase-shifting transformer (PST), a UPFC or other FACTS device, a back-to-back voltage-source converter pair, an MMC configured as a B2B, a dual-active-bridge DC interface, or a grid-forming inverter in a meshed low-inertia fleet. Meshed networks containing such elements — the European transmission system and its PST-coordinated border flows being the largest deployed instance — carry a loop- level coordination degree of freedom currently managed by four layered mechanisms (per-converter virtual impedance, per-loop dedicated circulating-current control, centralized droop-offset dispatch, day-ahead PST optimal power flow), none closing the loop-local-real-time gap between them.

We identify this loop-flow coordination problem as an instance of the *actuated-edge regime* of cellular-sheaf coordination [1], [2], [3] — a problem class on the same cellular-sheaf substrate as the recently developed node-actuated coordination framework, but with the restriction-map parameters as control inputs and the bus-angle node states evolving under autonomous physics. The gauge-invariant loop observable Θ_C that drives the controller is the small- deviation form of the topological winding number established for passive meshed power grids by [4] and collaborators; the contribution here is to place it under active control on actuated meshes. For this regime, the cycle-space dimension $b_1(G) = m - n + c$ gives a tight two-sided bound on the number of scalar feedback parameters required to asymptotically stabilize the trivial-connection equilibrium, attained by the canonical $\kappa_j(1 - \cos \Theta_{C_j})$ feedback per basis cycle — where Θ_{C_j} is the loop holonomy, the gauge-invariant observable identified physically with circulating reactive power. The framework composes cleanly with graph-Laplacian Kron reduction [5], and a Migdal–Kadanoff renormalization-group argument on the compact- $U(1)$ Higgs action collapses the $b_1(G)$ per-cycle parameters to a single network- wide κ at long wavelengths.

On a three-bus back-to-back triangle, the controller delivers $\sim 3\times$ suppression of circulating power against an uncoordinated droop baseline across load-step, setpoint-error, and stochastic-noise scenarios, with identical bus-frequency regulation. A six-test acceptance suite on PST-grounded deployed-scale topologies — a 3-area Germany–Poland–Czech style mesh, the IEEE 14-bus system under Kron reduction, and a head-to-head comparison against Verboomen / Van Hertem OPF coordination on IEEE 118-bus with five PSTs — is executed in full; the decisive head-to-head yields 45.9% fast-event loop-flow reduction at 94% of the physical bound set by the placement-reachability theorem, with exact static non-displacement against the OPF equilibrium. The same gauge invariance that underwrites the control law delivers a calibration-robust monitoring observable: Θ_C computed from raw PMU angle data is exactly invariant to per-bus calibration errors, verified to machine precision under $\pm 2^\circ$ stress, enabling cross-TSO coordination without calibration harmonization. Topological phase slips and finite-resolution ($N \geq 64$) quantization are analyzed explicitly and shown to sit inside engineering margins. Non- Abelian (SU(2)) extension for unbalanced operation and stacked-3D Fradkin–Shenker phase structure are documented in appendices as avenues for future work.

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1 The loop-flow coordination problem

An *active-phase link element* is a branch on a power network whose series phase offset θ_ℓ is set independently of its endpoint bus angles. Phase-shifting transformers (PSTs) are the largest deployed instance — several hundred installed across the European transmission system, the PJM footprint, and interconnections worldwide, specifically to manage *loop flow* between control areas. Unified power flow controllers (UPFCs) and other FACTS devices provide a more agile realization of the same physical primitive. Back-to-back voltage-source converter pairs, modular multilevel converters configured as B2B units, and dual-active-bridge converters realize the same primitive on DC-decoupled AC ports, where the phase offset between the two AC sides is a fully independent control variable. Grid-forming inverters in meshed low-inertia fleets add a third physical realization. These are different devices with different dynamics, but they share a single structural feature: each carries an independent link variable on a mesh branch that passive transmission does not.

A *meshed* network with active-phase link elements — a PST-equipped transmission mesh, a multi-terminal HVDC system with redundant DC ties, a microgrid with B2B interconnections between AC islands, a DC distribution system with dual-active-bridge converters on parallel paths, an inverter-rich distribution feeder — carries a loop-level coordination degree of freedom. When the link phase offsets around an independent loop are not mutually consistent, real and reactive power circulate around the mesh, loading devices without delivering useful work, dissipating in losses, and saturating thermal and current headroom. In European transmission the symptom is *border flow*: scheduled power between two control areas transits through a third, driven by dispatch asymmetries (notably German renewable output flowing through Poland and the Czech Republic). In multi-terminal HVDC and meshed microgrids the symptom is reactive circulation on DC-decoupled loops; in data-center DC distribution it is shared-bus imbalance across parallel DAB paths. The consistency condition is a single constraint per independent loop — a loop-level observable that is not a function of any individual bus or any individual device.

Current engineering practice manages this loop-level degree of freedom through four mechanisms, layered by coordination scope and timescale, with no mechanism currently occupying the *loop-local-real-time* layer:

Virtual impedance. Each converter synthesizes an additional impedance term in its control law. The impedance magnitude is tuned per converter to suppress its contribution to the circulating component. Inter-loop cross-coupling makes this a per-commissioning tuning problem that does not generalize across mesh topologies.

Dedicated circulating-current controllers. A separate control loop decomposes the converter output into useful and circulating components (double-frequency negative-sequence for paralleled inverters, DC-component extraction for B2B loops) and actively suppresses the circulating component. Each loop requires its own tuned controller; the approach does not scale naturally to meshes of more than a few loops.

Droop-offset coordination. A centralized dispatch layer (EMS/SCADA) periodically adjusts the droop setpoints on each converter to minimize steady-state loop flow. The most developed methodology for sizing these droop constants is the *pilot-voltage droop* concept of [6], which distributes a communicated common DC-voltage signal $V_{dc,pilot}$ to all participating terminals and sizes droop constants D_k via load-sharing constants T_k . The approach achieves precise, pre-planned power redistribution after contingencies and is the current engineering state of the art for coordinated multi-terminal droop. Berggren’s own framing (§I, [6]) is instructive:

“Comparing with power drooped against frequency control in ac systems, the advantage of having a common signal (the frequency at steady state) across the system is evident.”

Pilot voltage is the engineered substitute for what AC frequency provides naturally — a gauge-invariant scalar shared across the system. The scheme addresses static conditions well but requires communication for the common signal, responds slowly to transients, and does not extend to reactive circulation on non-trivial loops.

Network-wide optimal power flow over active-phase elements. At the widest coordination layer, a transmission system operator solves an optimal-power-flow problem over all PST tap positions (and, where present, FACTS setpoints) on a day-ahead or intra-day schedule horizon of 15 minutes to one hour. The

canonical formulation is the border-flow-control OPF of [7], with coordinated-PST sizing methodology developed in [8]. This is the current engineering state of the art for loop-flow management across the European transmission system and is the direct analog at the transmission layer of Berggren’s pilot-voltage droop at the HVDC layer: both take a network-wide view, both are centralized, both execute open-loop with respect to the coordinated set points between scheduling runs. Between OPF solves, a load change, generator trip, or renewable ramp produces loop flow that is not corrected until the next schedule cycle.

All four mechanisms are effective in the regimes they are designed for, and their joint tuning is a commissioning-intensive problem for any network larger than a handful of nodes. Laid out by coordination scope and timescale:

Layer	Mechanism	Scope	Timescale
Device-local	Virtual impedance	Single converter	ms
Loop-local	Dedicated CC controllers	Single loop	10–100 ms
Network-global (droop)	Pilot-voltage droop (Berggren)	MTDC grid	seconds
Network-global (optimization)	PST / FACTS OPF (Verboomen)	Transmission system	15 min – 1 h
Loop-local real-time	currently unoccupied	Independent cycle	Local-control

The shared structural pattern across the existing four is visible once stated plainly: *one coordination scalar per subsystem, acting at a slower timescale than local power-flow regulation*. What pilot-voltage droop constructs by communication, what PST OPF constructs by centralized optimization, and what frequency droop inherits from a global physical observable, this paper recognizes as a consequence of the network’s cycle topology. The loop holonomy $\Theta_C = \sum_\ell \theta_\ell$ of § 2 is the natural gauge-invariant scalar on non-trivial mesh loops: topologically determined rather than engineered, locally measurable rather than communicated, and invariant under per-bus reference-frame drift rather than sensitive to it (§ 2.4). Frequency is the trivial-loop gauge-invariant scalar; Θ_C is the non-trivial-loop gauge-invariant scalar. Pilot voltage and PST OPF both sit in between — engineered approximations of the former on networks that have no structural access to the latter.

The contribution is three-fold. *First*, the loop-flow coordination problem on meshed networks with active-phase link elements is identified (§ 2.2) as an instance of the *actuated-edge regime* of cellular-sheaf coordination [1], [2], [3] — a problem class distinct from the node-actuated regime those works develop, in which the data inside the restriction maps are the control inputs and the bus-angle node states evolve under autonomous physics. *Second*, the control-theoretic specialization gives a *tight two-sided bound* at $b_1(G) = m - n + c$ on the number of scalar feedback parameters required to asymptotically stabilize the trivial-connection equilibrium (§ 3.5), attained by the canonical $\kappa_j(1 - \cos \Theta_{C_j})$ feedback per basis cycle; a renormalization-group argument on the Migdal–Kadanoff recursion (§ 4) collapses the $b_1(G)$ per-cycle parameters to a single network-wide κ at long wavelengths, licensing the single-parameter controller deployed in the engineering instantiation. *Third*, the engineering instantiation — single-scalar gauge feedback layered on top of existing PST OPF coordination — suppresses circulating power on the 3-bus B2B triangle by $\sim 3\times$ against an uncoordinated droop baseline (§ 6.3); the six-test acceptance suite ($\alpha' - \zeta$) on PST-grounded deployed-scale topologies concludes with a head-to-head comparison against the Verboomen–Van Hertem OPF coordination baseline on IEEE 118-bus with five phase-shifting transformers (§ 6.4) yielding 45.9% fast-event loop-flow reduction at 94% of the physical bound set by the placement-reachability theorem of § 7.5. The same invariance also delivers, as a direct consequence, a *calibration-robust monitoring observable*: Θ_C computed from raw PMU angle readings is exactly invariant to per-bus calibration errors, making cross-TSO coordination feasible without harmonized calibration and enabling reference-independent model-validation residuals (§ 7.2).

The engineering contribution is *compositional*: outer-layer coordination (PST OPF, pilot-voltage droop, day-ahead dispatch, N-1 security) remains intact and unchanged, and gauge feedback closes the real-time loop those layers structurally cannot close. An evidence-status table mapping each load-bearing claim to

its validation tier is in § 6.5; all framework, intrinsic-controller, and deployment-scale claims (Tests α' – ζ) are demonstrated with computed numerical results at submission. Positioning relative to the closest prior work — cellular-sheaf coordination, consensus-on-Lie-groups control, group synchronization, coordinated-PST control, and discrete gauge theory on graphs — is in § 8.

2 The loop-level degree of freedom

2.1 Loop-level invariants in meshed power grids

Before laying out the mathematical framework this paper uses, we fix the most direct intellectual antecedent for the observable Θ_C . A line of work beginning with [4] — and continued by [9], [10], [11], and most recently [12] — establishes that steady-state flows on meshed transmission networks carry integer topological invariants: the loop flows that distinguish one stable phase-locked state from another are labelled by integer winding numbers $q_C \in \mathbb{Z}$, one per independent cycle C of the underlying graph. The winding-number partition classifies the multistable equilibria of the Kuramoto-type oscillator models of AC power grids; loop flows that carry nonzero winding are *topologically protected* in the sense that they cannot be continuously deformed to zero loop flow without passing through a desynchronization. The algebraic object — an integer-valued invariant defined as a signed sum around a cycle of the network graph — is the same object that will appear in this paper as the loop holonomy Θ_C .

Two differences separate the present paper from that thread. First, the Coletta–Delabays–Jacquod program studies *passive* meshed networks: the branches are fixed couplings set by line impedances, with no independent control variable on any edge. The winding number q_C is a property of the equilibrium *state* of the oscillator model. This paper studies *actuated* meshed networks: branches carrying phase-shifting transformers, UPFCs, or back-to-back converter pairs have independently commandable phase offsets θ_ℓ , and the loop holonomy $\Theta_C = \sum_{\ell \in C} \theta_\ell$ is a property of the *control inputs* rather than the equilibrium state. This is a structural distinction — actuation turns the cycle-space invariant from a classification label into an observable and control target. Second, the Coletta–Delabays–Jacquod work characterizes multistability, phase-slip energetics, and transfer-capacity consequences of the winding-number partition; the present paper takes the existence of the loop-level invariant as a starting point and develops a feedback controller that drives it back to zero, with engineering validation against a Verboomen-class OPF baseline. The contribution of the present paper is the actuated-edge control framework on top of the loop-invariant structure established by the prior line — not the observation that loop-level invariants exist in meshed power grids, which is due to Jacquod and collaborators.

The continuous analogue $\Theta_C \in (-\pi, \pi]$ used in this paper and the integer winding number $q_C \in \mathbb{Z}$ used in the Coletta–Delabays–Jacquod line are the same object at two scales: Θ_C is the small-deviation (continuous) holonomy within a single winding sector, and q_C labels the sector itself. Phase slips (§ 5.2) correspond to transitions between sectors; small-disturbance loop flow (§ 6.4) corresponds to Θ_C deviations within a sector. The unified object — cycle-space-valued and gauge-invariant by construction — is what the next three subsections formalize within the broader cellular-sheaf / discrete- gauge framework.

2.2 Background: cellular sheaves, connection graphs, discrete gauge theory

The mathematical object underlying this paper — a graph with group-valued edge parameters that transform covariantly under node-wise relabeling — has been studied under three names in three communities. This subsection fixes terminology and cites the background; the rest of § 2 identifies the regime this paper operates in and what the power-systems application contributes that these literatures do not.

Cellular sheaves on graphs [1], [13]. A cellular sheaf $\mathcal{F}$ on a graph $G = (V, E)$ assigns vector spaces (or, more generally, objects in a chosen category) to each vertex and edge, together with *restriction maps* $\mathcal{F}_{v \triangleleft \ell} : \mathcal{F}(v) \rightarrow \mathcal{F}(\ell)$ from each vertex stalk to each incident edge stalk. The coboundary $\delta : \bigoplus_v \mathcal{F}(v) \rightarrow \bigoplus_\ell \mathcal{F}(\ell)$ compares the two restrictions at each edge, and the sheaf Laplacian $L_{\mathcal{F}} = \delta^\top \delta$ generalizes the graph Laplacian. Hansen and Ghrist develop the spectral theory; their framework subsumes matrix-weighted graphs and connection graphs as special cases.

Connection graphs and graph connection Laplacians [14], [15]. The special case $\mathcal{F}(v) = \mathcal{F}(\ell) = \mathbb{R}^d$ (or $\mathbb{C}$) with restriction maps valued in a matrix group $\mathcal{G}$ gives the *connection Laplacian*, studied as vector diffusion maps in signal processing and as a foundation for group synchronization in computer vision [16].

Discrete gauge theory on graphs [17], [18]. Treating $\mathcal{G}$ -valued edge parameters as *connection 1-forms* on a discrete principal bundle, the curvature 2-form encodes the loop holonomy and the obstruction to

trivializing the connection by node-wise gauge transformations. Jiang’s “gauge theory on graphs” establishes the Weitzenböck formula and Yang–Mills functional in this discrete setting; Leok–Marsden–Weinstein give the foundational discrete-geometry treatment.

Coordination sheaves and distributed control [2], [3]. Cellular-sheaf coordination treats the sheaf structure as a representation of the coordination geometry and drives agents at the *vertex stalks* into harmonic sections through nonlinear sheaf diffusion or ADMM-style distributed optimization. Consensus, formation control, and distributed convex optimization all instantiate this framework.

Consensus on Lie groups [19], [20]. When agent states live on a compact Lie group $\mathcal{G}$ and the coordination task is to drive them to consensus through gradient-flow dynamics of a gauge-invariant cost, the closed-loop system descends to an autonomous ODE on the quotient manifold. This is the node-actuated antecedent of the controller developed in § 3.

These are not competing frameworks — they describe the same algebraic object at different levels of abstraction, with cellular sheaves the most general, discrete gauge theory the most geometric, and consensus-on-Lie-groups the control-theoretic case closest to the present paper. The next two subsections identify which case of this framework the paper operates in and fix the power-systems-specific structure that distinguishes it.

2.3 The actuated-edge regime

Existing distributed-control work on cellular sheaves [2], [3] and on Lie-group consensus [20] addresses the *node-actuated* case: the sheaf structure (or equivalently, the connection) is fixed by the coordination geometry, and the controllers drive the node states $x_v \in \mathcal{F}(v)$ into harmonic sections by diffusion of the sheaf Laplacian. The coordination target is $\delta x = 0$: zero covariant mismatch at every edge.

The power-systems application treated in this paper is dual. Active- phase link elements — phase-shifting transformers, UPFC and other FACTS devices, back-to-back voltage-source converters, MMC-as-B2B, dual-active bridges, grid-forming inverters with independent output- phase control — place the *edge parameters* (the data inside the restriction maps) under direct actuator control. Bus voltage angles evolve under generator, load, and line physics that the actuator does not command. The controller’s degrees of freedom are the link parameters θ_ℓ , not the node states δ_n ; the coordination target is $\Theta_C = 0$: zero holonomy around every independent cycle.

This is the *actuated-edge regime* of the cellular-sheaf framework. It is not a special case of the standard node-actuated regime; it is the dual problem on the same cellular-sheaf substrate. It is also not the passive-mesh regime studied by [4] and collaborators: where that work characterizes the multistable equilibria of a mesh with fixed couplings, the actuated-edge regime places the cycle-space invariant under direct control and derives closed-loop dynamics for driving it to zero. Table 2.1 summarizes the distinction against the node-actuated (cellular-sheaf) regime; the passive-mesh (winding-number) antecedent is treated in § 2.1.

	Standard coordination sheaf (Hanks et al.; Zhao et al.)	Actuated-edge regime (this paper)
<i>Sheaf structure</i>	Fixed by coordination geometry	Parameterized by actuation variables
<i>Control inputs</i>	Vertex stalk values x_v	Restriction-map parameters (link variables θ_ℓ)
<i>Node states</i>	Actuated by local agents	Evolve under autonomous physics
<i>Target</i>	Harmonic section: $\phi_\ell = 0$	Flat connection: $\Theta_C = 0$
<i>Characterization of equilibrium</i>	$x \in H^0(G, \mathcal{F})$	$\theta \in \ker(Z^\top)$
<i>Cycle-space role</i>	Obstruction to harmonic section	Parameter space of gauge equilibria

The distinction matters operationally: a PST/FACTS/B2B controller cannot drive bus angles to consensus — generator inertia, load dynamics, and line impedance set the angles — but it can drive the link holonomy

to zero, eliminating circulating power around mesh loops. The actuated-edge regime captures this structure precisely. The rest of § 2 develops the regime’s key structural properties before § 3 derives the controller.

2.4 Why passive networks carry no loop observable

In a purely passive three-phase network — transmission lines, transformers, passive distribution — the phase drop across a branch between buses m and n is simply $\delta_m - \delta_n$, a difference of node quantities. Summing around any closed loop,

$$\sum_{\text{loop}} (\delta_m - \delta_n) = 0,$$

identically. No loop-level degree of freedom exists. The coupling between buses through passive impedances is linear and sequence- diagonal (Fortescue), with no mechanism for generating an independent loop variable. Prior investigations into candidate mechanisms that might circumvent this — two-machine dipole-dipole exchange as a “one- photon” coupling, the DFT of cubic iron-core saturation nonlinearities as a candidate non-Abelian vertex — each arrive at the same structural conclusion and are archived in [research-notes/archive/](#).

2.5 What active-phase link elements add

An *active-phase link element* (§ 1) breaks this constraint. On a branch ℓ carrying such an element — a phase-shifting transformer injecting a series quadrature voltage, a UPFC or other FACTS device with series compensation, a back-to-back voltage-source converter pair with no through-DC phase coupling, an MMC configured as a B2B, a dual-active-bridge with an isolated DC stage, a grid-forming inverter with independent output-phase control — the branch phase offset θ_ℓ between the two endpoint buses is *not* a difference of bus angles but an independent degree of freedom, set directly by the device’s control law. For a PST this is the series phase injection; for a B2B VSC pair it is the mismatch between the two converters’ independently commanded output phases; the physical mechanism differs but the mathematical object is the same.

Define the *covariant branch mismatch* and the *loop holonomy*:

$$\phi_\ell = \delta_{s(\ell)} - \delta_{t(\ell)} - \theta_\ell, \quad \Theta_C = \sum_{\ell \in C} \theta_\ell,$$

where $s(\ell), t(\ell)$ are the branch endpoints and C is an independent loop. Under a node-wise angle offset $\delta_n \rightarrow \delta_n + \alpha_n$, the branch mismatch is invariant *if and only if* the link variable transforms as

$$\theta_\ell \rightarrow \theta_\ell + \alpha_{s(\ell)} - \alpha_{t(\ell)}.$$

This is the standard $U(1)$ link transformation rule of lattice gauge theory / the connection-graph update rule of Singer & Wu / the restriction-map reparameterization of a cellular sheaf — three names, one structure. The loop holonomy Θ_C is gauge-invariant by construction (the α ’s telescope around the loop).

The dimension of the cycle space — the free abelian group of independent loops — equals the first Betti number $b_1(G) = m - n + c$ for a graph with m edges, n nodes, and c connected components. This is the first cohomology $\dim H^1(G, \mathcal{F})$ of the constant $U(1)$ connection sheaf [1], Thm. 6.2.1, [17] (see § 4). For an actuated-edge network, $b_1(G)$ therefore counts simultaneously: the number of independent gauge-invariant loop observables, and — as § 3 will show — the minimal number of scalar feedback tunings required to coordinate them.

Proposition 2.1 (Invariance under time-varying per-bus reference drift). *Let $\alpha_n(t)$ be an arbitrary, independent, time-varying phase-reference drift at each bus n . Under the substitution $\delta_n(t) \rightarrow \delta_n(t) + \alpha_n(t)$ with the corresponding link transformation $\theta_\ell \rightarrow \theta_\ell + \alpha_{s(\ell)} - \alpha_{t(\ell)}$, the loop holonomy $\Theta_C = \sum_{\ell \in C} \theta_\ell$ is unchanged.*

Proof. Along a closed cycle C , each bus n appears exactly once as $s(\ell)$ and once as $t(\ell)$ across the two incident links. Summing the link transformation around C : $\Delta\Theta_C = \sum_{\ell \in C} (\alpha_{s(\ell)} - \alpha_{t(\ell)}) = 0$, since each α_n contributes once with each sign and cancels identically, for any $\alpha_n(t)$ — including uncorrelated drifts across terminals. $\square$

The content of [the proposition above](#) is not that telescoping holds — that is immediate — but that the invariance extends to the full operationally relevant class of reference-frame perturbations: per-terminal, time-varying, uncorrelated. No analogous invariance holds for pilot-voltage droop [6], local DC-voltage droop [21], or any coordination scheme whose scalar is not a loop sum: a reference-frame drift of magnitude δV produces steady-state sharing error of order $\delta V/D_k$ at each terminal. The loop holonomy is structurally immune to this failure mode. [Test \$\alpha\$](#) of [Appendix F](#) verifies the proposition on the full nonlinear 3-bus dynamics against injected per-bus drift (attenuation $\Delta\Theta_C/\alpha_{\max} = 0.123$, with the telescoping identity itself verified to machine precision); [Test \$\alpha'\$](#) of § 6.4 extends the verification to a PST mesh with discrete-tap actuation and multi-TSO reference drift.

Physical identification. $\Theta_C \neq 0$ if and only if the mesh carries circulating power around loop C : the link phase settings around the loop cannot be made simultaneously consistent with any bus-angle assignment. The holonomy is the standard “circulating-current” (or, on transmission meshes, “loop-flow”) quantity in a slack-bus-independent, gauge-invariant form. In particular, Θ_C is directly computable from telemetry already streaming out of each active-phase link element — tap positions from every PST in a substation telemetry feed, commanded phases from each converter in a B2B or MMC installation, series-injection values from each FACTS device — and can be monitored on deployed systems today *without* any control change on any device; a point we return to in § 7.

The fact that a few structural terms — “link variable,” “gauge transformation,” “holonomy” — now apply to this control problem is not a loose analogy; it is an algebraic identity. A full Rosetta dictionary spanning cellular sheaves, lattice gauge theory, and the converter-network model — together with compact entries for several further instantiations of the same structure (Kuramoto networks, photonic MZI meshes, RF phased arrays, clock synchronization, formation control, tensor-network circuits) — is given in [Appendix A](#).

2.6 Composition with Kron reduction

Large transmission and distribution networks contain many *interior* buses that carry no converter and no dynamic load and serve only to connect passive branches. The standard graph-theoretic operation for eliminating such buses and computing an equivalent reduced admittance matrix on the remaining *boundary* nodes is Kron reduction — the Schur complement of the nodal admittance matrix with respect to the interior index set — whose formal properties (Laplacian preservation, effective-resistance preservation, order-independence) were given a rigorous graph-theoretic treatment by [5].

Kron reduction composes cleanly with the gauge framework. A cycle of the original graph that does not pass through any interior node is a cycle of the reduced graph, and a cycle that does pass through an interior node contracts to a cycle of the reduced graph whose link set retains all active-phase link elements on the original cycle. The cycle space — the abelian group of independent loops — is preserved by Kron reduction, and therefore so is the loop holonomy Θ_C on each independent cycle. In the passive setting addressed by Dörfler and Bullo the link variables vanish identically and the holonomy is zero; when active-phase link elements are present on the boundary mesh, Θ_C is nontrivial and survives reduction unchanged. The stiffness V^2/X_ℓ that will appear in § 3 is, for compound branches produced by reduction, the effective-resistance stiffness on the reduced graph in the sense of Dörfler and Bullo. The gauge framework is therefore the active-link extension of graph-Laplacian Kron reduction, not a replacement for it: passive interior structure is eliminated by their machinery, and the gauge observable on what remains is the object of this paper.

3 The gauge-covariant controller

3.1 Canonical Wilson form

The total stored-plus-coordination energy of a meshed network with active-phase link elements decomposes into two manifestly gauge-invariant terms, parameterized by per-branch couplings $\beta_\ell > 0$ and per-loop stiffnesses $\kappa_j > 0$ over a chosen cycle basis $\{C_j\}_{j=1}^{b_1(G)}$:

$$\mathcal{H} = \underbrace{\sum_{\ell} \beta_{\ell} (1 - \cos \phi_{\ell})}_{\text{branch: matter-link coupling}} + \underbrace{\sum_{j=1}^{b_1(G)} \kappa_j (1 - \cos \Theta_{C_j})}_{\text{plaquette: loop curvature}}, \quad \beta_{\ell} = V^2/X_{\ell}.$$

The first term is the standard reactive-power energy written in covariant form — it reduces to the familiar $(V^2/X_{\ell})(1 - \cos \phi_{\ell})$ per branch with ϕ_{ℓ} the covariant mismatch defined in § 2.4. The second term is a loop-curvature penalty on mutually inconsistent converter settings: $\kappa_j > 0$ is the per-loop stiffness on basis cycle C_j , Θ_{C_j} is the loop holonomy of § 2.4, and the sum runs over a chosen basis of independent mesh loops.

The per-loop κ_j parameterization is the general form consistent with the cycle-feedback budget derived in the “sufficiency” and “necessity” subsections below; § 4 (“From per-loop tunings to single-parameter control”) then shows that at long wavelengths the per-loop distinctions are renormalization-group irrelevant and a single network-wide $\kappa = \kappa_j$ governs the effective theory. The engineering controller deployed in § 6 uses the single- κ form; the per-loop form here is the analytically correct starting point from which the single- κ reduction is derived rather than assumed.

3.2 Control law

The gauge-covariant control law is the gradient-flow equation of motion for the link variable under $\mathcal{H}$:

$$\tau \dot{\theta}_{\ell} = \underbrace{\beta_{\ell} \sin(\phi_{\ell})}_{\text{conventional droop}} - \underbrace{\sum_{j=1}^{b_1(G)} Z_{\ell j} \kappa_j \sin(\Theta_{C_j})}_{\text{new: plaquette feedback}}, \quad \beta_{\ell} = V^2/X_{\ell},$$

where $Z \in \{-1, 0, +1\}^{m \times b_1(G)}$ is the cycle-incidence matrix of the chosen basis: $Z_{\ell j} = +1$ if link ℓ traverses cycle C_j in the positive orientation, -1 in the negative, and 0 if $\ell \notin C_j$. The first term is the conventional power-flow law already in place on each active link: frequency droop on a converter, OPF-scheduled tap position on a PST, series-injection setpoint on a FACTS device. The second term adds scalar feedbacks proportional to the sines of the loop holonomies — exactly one scalar tuning parameter κ_j per element of the cycle basis, distributed back to each participating link through the cycle-incidence sum. The controller occupies the loop-local-real-time coordination layer left unoccupied by the four existing mechanisms enumerated in § 1, with $b_1(G)$ scalar tunings in the general case and a single network-wide κ under the long-wavelength reduction of § 4.

3.3 Sufficiency of the canonical Wilson form

The canonical Wilson form with any strictly positive choice of $\{\kappa_j\}$ suffices to locally asymptotically stabilize the trivial connection $\Theta = 0$:

Proposition (Sufficiency). For any $\beta_{\ell} > 0$ ($\ell \in E$) and $\kappa_j > 0$ ($j = 1, \dots, b_1(G)$), the trivial connection $\Theta = 0$ is a locally asymptotically stable equilibrium of the gradient-flow dynamics projected to the holonomy coordinates.

Proof. The holonomy-sector linearization of the gradient flow at the equilibrium $\phi^* = 0$, $\Theta = 0$ is governed by the Hessian of the loop potential $W(\Theta) = \sum_j \kappa_j (1 - \cos \Theta_{C_j})$. Direct calculation:

$$\nabla^2 W(0) = \sum_{j=1}^{b_1(G)} \kappa_j e_j e_j^\top = \text{diag}(\kappa_1, \dots, \kappa_{b_1(G)}) \succ 0,$$

strictly positive-definite for any $\kappa_j > 0$. The linearized holonomy dynamics $\tau \dot{\Theta} = -\nabla^2 W(0) \Theta + O(\|\Theta\|^2)$ therefore has all eigenvalues strictly negative. Strict convexity of $1 - \cos \phi_\ell$ at $\phi_\ell = 0$ gives the analogous branch-sector stability. $\square$

The diagonal Hessian above exhibits the strongest possible decoupling between loops at the linearized level — each cycle’s holonomy decays under its own κ_j — though the full nonlinear dynamics couple cycles through the $\sin \Theta_{C_j}$ nonlinearities away from the equilibrium.

3.4 Necessity of $b_1(G)$ feedback scalars

The same $b_1(G)$ that appears as a *sufficient* feedback count is also a *necessary* lower bound, under the natural operational restriction that loop-level feedback ingredients be built from *linear combinations* of holonomies (the standard form across lattice gauge theory and consensus-on-Lie-groups control; nonlinear pre-combinations are addressed in [Appendix G](#)).

Definition (Linear-projection parameterization). A loop potential $W : \mathbb{R}^{b_1(G)} \rightarrow \mathbb{R}$ has a *linear-projection parameterization* with p feedback ingredients if it can be written

$$W(\Theta) = \sum_{j=1}^p \omega_j(a_j^\top \Theta),$$

with $a_j \in \mathbb{R}^{b_1(G)}$ fixed projection vectors and $\omega_j : \mathbb{R} \rightarrow \mathbb{R}$ smooth scalar shapings. The vectors a_j encode the structural choice of which loops feed back into which ingredient; the ω_j encode gain and shape. The integer p is the *feedback budget*.

The canonical Wilson form is this parameterization with $p = b_1(G)$, $a_j = e_j$, and $\omega_j(s) = \kappa_j(1 - \cos s)$.

Proposition (Necessity). Let W admit a linear-projection parameterization with p ingredients. If the trivial connection $\Theta = 0$ is locally asymptotically stable under the gradient flow of W , then $p \geq b_1(G)$.

Proof. For W in linear-projection form, the chain rule gives

$$\nabla^2 W(0) = \sum_{j=1}^p \omega_j''(0) a_j a_j^\top,$$

a sum of p rank-one symmetric matrices with rank at most p (the second-order term $\omega_j' \cdot \nabla^2 a_j$ vanishes because a_j is constant in Θ). Asymptotic stability of $\Theta = 0$ requires $\nabla^2 W(0)$ to have full rank $b_1(G)$, hence $p \geq b_1(G)$. $\square$

The canonical Wilson form attains the bound with $p = b_1(G)$ ingredients and $\omega_j''(0) = \kappa_j$, achieving $\nabla^2 W(0) = \text{diag}(\kappa_j) \succ 0$. No linear-projection parameterization with fewer than $b_1(G)$ ingredients can stabilize the trivial connection.

The linear-projection restriction is operationally natural: allowing a single nonlinear ingredient to encode multi-loop structure inside its argument (e.g. $W = \omega(\Theta_1^2 + \Theta_2^2)$ with $p = 1$ formally but rank-2 Hessian)

is a redefinition of the parameterization rather than a genuine parameter-saving, and is ruled out by the restriction. [Appendix G](#) gives the full treatment, including a non-degenerate nonlinear case for which the same bound holds.

3.5 The cycle-feedback budget theorem

Combining sufficiency and necessity:

Theorem (Cycle-feedback budget). For the gauge-covariant controller in the actuated-edge regime with Hamiltonian gradient-flow dynamics and a linear-projection parameterization of the loop potential, the number of scalar tuning parameters p required to make the trivial connection $\Theta = 0$ locally asymptotically stable satisfies

$$p \geq b_1(G) = m - n + c.$$

The canonical Wilson form $\sum_j \kappa_j (1 - \cos \Theta_{C_j})$ with $\kappa_j > 0$ attains this bound with $p = b_1(G)$.

The theorem gives a *two-sided* bound on the loop-level parameter count:

- **Upper bound.** At most $b_1(G)$ independent gauge-invariant loop observables exist — the cycle-space dimension is also the dimension of H^1 of the $U(1)$ -connection cellular sheaf on G [1], [13].
- **Lower bound.** At least $b_1(G)$ scalar feedback ingredients are required to asymptotically stabilize the trivial connection.

The cycle dimension $b_1(G) = m - n + c$ is simultaneously the number of independent loop observables and the minimal number of scalar tunings required to coordinate them. The two bounds coincide, and the matching is the control-theoretic content: sheaf cohomology supplies the dimension count; the Hessian-rank argument supplies the feedback-budget lower bound; the canonical Wilson form saturates both.

Per-network, not per-link. A meshed network with $b_1(G) = 5$ requires five scalar tunings at the loop level regardless of whether it has 50 actuated links or 500. The parameter count is set by the cycle space, not by the link count. For the IEEE 118-bus test system augmented with five phase-shifting transformers (the topology of Test ζ in § 6), the per-loop bound gives 181 scalar tunings; the long-wavelength reduction of § 4 collapses these to a single network-wide κ .

Locality of actuation. Each scalar feedback $\kappa_j \sin \Theta_{C_j}$ is computed from $b_1(G)$ cycle-basis holonomies and distributed back to each actuated link via the cycle-incidence sum Z . Neither step requires global communication beyond the PMU telemetry already in place on actuated devices; see § 7.2.

3.6 Closure of gauge-invariant dynamics

A prerequisite for a gauge-covariant controller to be well-defined is that the closed-loop dynamics be expressible entirely in the gauge-invariant coordinates (ϕ_ℓ, Θ_C) — independent of any slack-bus choice or bus-angle convention. This holds by construction: the swing equations, power flows, and link dynamics all depend on the bus angles only through the covariant combination ϕ_ℓ , and on the link variables only through the covariant combination Θ_C . Verified numerically against two different gauge choices of the same physical scenario: $|\phi_\ell^{(A)}(t) - \phi_\ell^{(B)}(t)| < 2 \times 10^{-10}$ over a 10-second integration (`gauge_covariance_test.py`; live cross-check below). The covariant ODE in (ϕ, Θ) reproduces the full standard-variable ODE to 2×10^{-11} .

Live run of `gauge_covariance_test.py` - headline results:

```
Max |phi_A(t) - phi_B(t)| over [0, 10]: 2.29e-10
Max |Theta_A(t) - Theta_B(t)| over [0, 10]: 5.28e-11
>>> GAUGE COVARIANCE CONFIRMED <<<
Max |phi_covariant(t) - phi_standard(t)|: 2.65e-11
Max |Theta_covariant(t) - Theta_standard(t)|: 4.50e-12
```

>>> SELF-CONTAINED COVARIANT DYNAMICS CONFIRMED <<<
CONFIRMED NUMERICALLY:

3.7 Kron reduction: the explicit Schur-complement computation

The compositional statement of § 2.5 — passive interior elimination preserves the cycle space and therefore the gauge observable — is made concrete by the explicit Schur-complement algebra. Eliminating an interior bus with n spokes of susceptances $\{b_i\}$ produces a clique of effective edges among the former neighbors with

$$b'_{ij} = b_i b_j / B, \quad \theta'_{ij} = \theta_{0j} - \theta_{0i}, \quad B = \sum_i b_i.$$

The plaquette couplings κ are unchanged. The gauge-covariant combination ϕ'_{ij} falls out of the Schur-complement algebra without any modification to the standard reduction procedure. The single-parameter controller therefore remains well-posed after the reductions analysts routinely apply to make large networks tractable.

Verified to 1.4×10^{-14} on 10^5 random samples of the algebraic identity and to 1.1×10^{-10} on gradient-flow dynamics of a 4-node, 5-edge, 2-plaquette test network (`kron_test.py`).

4 From per-loop tunings to single-parameter control

§ 3 established that the gauge-covariant controller in the actuated-edge regime admits a cycle-feedback budget of $b_1(G)$ scalar tunings — one per-loop stiffness κ_j for $j = 1, \dots, b_1(G)$ — with this count tight (sufficient by construction, necessary by the Hessian-rank argument). The controller *as derived* therefore has $b_1(G)$ tuning parameters at the loop level.

The engineering controller deployed throughout §§5–7 uses a single network-wide $\kappa = \kappa_j$ instead. This section supplies the load-bearing justification for that reduction: at the timescales and length scales relevant to loop-coordination, the per-loop distinctions between different κ_j values are *renormalization-group irrelevant*, and the long-wavelength effective theory is governed by a single scalar plaquette coupling. The argument is the Migdal–Kadanoff (MK) recursion for the compact $U(1)$ Wilson action applied jointly to the gauge (plaquette) and matter (branch) sectors.

The MK recursion reduces to a scalar equation in the Bessel kernel $r(\beta) = I_1(\beta)/I_0(\beta)$:

$$r' = \begin{cases} r(\beta)^2 & (2\text{D, no bond-moving}) \\ [r(2\beta)]^2 & (3\text{D, bond-moving factor } 2) \end{cases}$$

Applied jointly to the gauge and matter sectors (β_g for the plaquette stiffness κ ; β_m for the branch stiffness V^2/X) on a 3D mesh:

- The matter coupling β_m is **irrelevant** — it flows to zero from every initial condition, independent of starting value.
- The critical fixed point is controlled entirely by the gauge sector: $\beta_g^* \approx 1.11$.

Verified on the Bessel recursion to 10^{-12} (`mk_rg.py`).

Engineering consequence. At long wavelengths (the timescale of loop-level coordination, as distinct from the fast timescale of local power-flow adjustment), the effective theory has a single load-bearing coupling: the plaquette stiffness κ . The RG flow sends the per-loop- κ_j phase space to a one-dimensional attractor along the gauge-coupling direction; distinctions between different κ_j values on different loops are washed out under coarse-graining. Branch-level parameters (V^2/X_ℓ , per-branch stiffness) govern local, fast dynamics but do not enter the long-wavelength effective theory that sets circulating-power behavior on the mesh.

This is the structural reason a single tuning parameter suffices — not a design heuristic, not a pragmatic simplification, but the unique long-wavelength coupling of the theory. The $b_1(G)$ per-loop tunings permitted by the cycle-feedback budget of § 3 collapse to a single effective κ under RG flow; the controller deployed in §§5–7 operates in this collapsed regime. In particular, the validation experiments of § 6 do not tune $b_1(G)$ separate parameters — they tune one.

Two caveats: the MK recursion is an approximate RG scheme (it overestimates the true critical exponent ν by a factor of ~ 3 in the Ising-like benchmark), and the irrelevance of matter assumes a genuinely 3D mesh. 2D meshes (non-layered, planar topologies) show no sharp transition, only a smooth crossover at $\xi = \sqrt{\kappa/\beta}$ (verified on a 128^2 torus, `wilson_2d.py`); 3D Monte Carlo on a 10^3 lattice confirms the sharp Fradkin–Shenker transition quantitatively (`mc3d_v2.py`, [Appendix C](#)). Most deployed meshed networks are effectively 3D — PST-coordinated transmission systems with stacked voltage levels (220 kV / 380 kV / 500 kV), HVDC overlays on AC grids, stacked microgrids, multi-region interconnects — but this should be verified on a case-by-case basis for unusual topologies.

5 Operational regimes

Sections § 3 and § 4 establish the gauge-covariant controller in its canonical form (per-loop κ_j parameterization, § 3) and justify the long-wavelength reduction to a single network-wide κ (§ 4). The present section operates throughout in the single- κ regime: all formulas below use a single scalar plaquette stiffness κ , and the per-branch stiffness $\beta_\ell = V^2/X_\ell$ of § 3.1 is taken as uniform $\beta_\ell \equiv \beta$ for the analytical results; generalizations to heterogeneous $\{\beta_\ell\}$ are noted where relevant.

Three physical imperfections determine whether and how the controller is deployed. Each corresponds to a different departure from the idealized continuous, noiseless, lifted dynamics of § 3: topological non-trivialities of the S^1 -valued link variable (§ 5.1), Langevin-class stochastic forcing at finite effective temperature (§ 5.2), and actuator quantization to finite bit depth (§ 5.3). Each is analyzed to produce quantitative operational predictions verified against numerical experiment.

A subtlety threads through all three subsections: the § 4 result that the branch coupling β is renormalization-group irrelevant governs *long-wavelength, coarse-grained* behavior of the network controller, not local dynamics. Phase-slip activation energies, thermal variance, and quantization frustration are all *local* phenomena — set at the link and small-neighborhood scale — and depend on the actual (unrenormalized) β directly. The two statements are compatible: β controls local fluctuations, κ controls the emergent large-scale coordination.

5.1 Topological phase slips

The lifted dynamics ($\theta_\ell \in \mathbb{R}$ rather than $\theta_\ell \in S^1$) conserve the integer winding around each independent loop. Stable equilibria exist at $\Theta_C = 2\pi n_C$ for every integer n_C on every basis cycle, all at equal energy, separated by saddle points of the canonical Wilson potential of § 3.1. For a single loop with uniform branch stiffness β , the saddle height is 2β (the energy to drive one link through its $\cos\phi$ barrier while holding the loop constraint). A 2π transition between topological sectors is a *phase slip* — the converter-network analogue of a Josephson phase slip in superconducting junctions.

Under Langevin noise forcing at effective temperature T (§ 5.2), phase slips occur at Arrhenius rate

$$\Gamma \sim A \exp(-2\beta/T),$$

with prefactor A set by the curvature of the saddle. Direct Langevin simulation on the 3-bus test loop gives fitted activation energy $\Delta E_{\text{fit}} = 1.97$ against the theoretical $2\beta = 2.00$ (at $\beta = 1$, normalized units) — 1.5% agreement (`holonomy.py`).

The operational implications are two. First, phase slips are the physical mechanism behind the 2π excursions observed under large perturbations on the 3-bus B2B triangle (§ 6.3, Scenario B): the droop-only controller’s trajectory crosses a saddle at finite T , winds through the adjacent topological sector, and eventually recovers to the connected equilibrium by a second slip — a topologically longer path with correspondingly larger circulating- power transient. The gauge controller suppresses the slip entirely at the test parameters by driving Θ directly back to zero without saddle traversal.

Second, the Arrhenius rate gives a quantitative prediction for noise-driven sector transitions on operating networks: at engineering-scale $\beta \sim 1$ pu (corresponding to typical branch stiffness V^2/X) and realistic effective temperature $T \sim 10^{-5}$ (§ 5.2), the exponential factor $\exp(-2\beta/T)$ is astronomically small. Phase slips on operating PST meshes are effectively prohibited by energetics alone; they require either a large deterministic perturbation (§ 6.3 Scenario B) or an unusual high-noise operating regime. This is the origin of the “phase slips are rare” engineering observation in operational power systems.

For heterogeneous $\{\beta_\ell\}$ the saddle height is $\min_\ell(\beta_\ell)$ on typical loop geometries — the weakest branch dominates — so the activation energy is set by the lowest-impedance link in the loop.

5.2 Stochastic noise and the operating temperature

Langevin forcing of the gradient-flow dynamics at effective temperature T yields, by the fluctuation–dissipation theorem, a Boltzmann stationary distribution $P \propto \exp(-\mathcal{H}/T)$ where $\mathcal{H}$ is the canonical Wilson Hamiltonian of § 3.1. For a single three-bus loop with uniform β , the closed-form variance of the loop holonomy is

$$\langle \Theta^2 \rangle = \frac{3T}{\beta + 3\kappa}.$$

Three independent estimates — Langevin time averages, independent Metropolis Monte Carlo, and the linearized Gaussian prediction — agree to $\sim 1\%$ across $T \in [0.05, 10]$ (`three_one.py`).

The engineering identification of T is direct. Let σ_θ be the effective phase jitter per link, aggregated over measurement noise, actuator synthesis noise (PWM ripple on a converter, tap quantization on a PST), and load stochasticity propagated through the swing equations. The fluctuation–dissipation relation gives

$$T \sim \sigma_\theta^2 \cdot \beta.$$

For modern power-electronic controls and deployed PSTs, $\sigma_\theta \sim 10^{-3}$ rad, giving $T \sim 10^{-6}$ to 10^{-5} in normalized units.

This is deeply inside the *screened* regime: the gauge-coupling correlation length $\xi = \sqrt{\kappa/\beta}$ is bounded below by device-scale lengths, the flat-connection configuration $\Theta \approx 0$ is the overwhelmingly preferred operating point, and the plaquette controller acts as a high-pass filter on circulating-power fluctuations. The low operating temperature also justifies the local-stability analyses of § 3.3 (sufficiency) and § 3.4 (necessity) — the linearized Hessian is the right object to study because the noise is small enough to keep trajectories near the equilibrium.

At higher T — e.g. in a noisy microgrid context with high measurement error and unfiltered renewable variability — the Boltzmann distribution broadens and the linearization becomes less accurate. The framework does not break at high T (the Monte Carlo agreement of `three_one.py` is across three orders of magnitude in T) but the small- Θ approximations used elsewhere in the paper require reexamination.

5.3 Discrete phase resolution

Real actuators operate at finite resolution: $\theta_\ell \in (2\pi/N)\mathbb{Z}$ for some integer N . For PST tap changers, $N \sim 32$ (± 16 taps over a small angle range); for modern DSP-controlled converters, $N = 2^{12}$ to 2^{16} ; for MMC phase commands, similar. The continuous-parameter analysis of §§3–4 gives way to a discrete optimization whose minimum differs from the continuous minimum by a residual *frustration* energy that shared link variables cannot simultaneously drive to zero on multiple plaquettes.

The scaling is universal. For a single plaquette with L links and uniform branch stiffness β , the closed-form frustration minimum is

$$H_{\min}^{(N)} = \frac{(\beta + \kappa)L\pi^2}{6N^2}.$$

For a multi-plaquette network, the L -and- κ factors are replaced by a topology-dependent prefactor but the $1/N^2$ scaling is preserved. Numerical verification on a 4×4 torus gives a power-law fit of exponent $N^{-1.99}$ over $N = 4$ to 256 with 10^3 trials per point; the single-plaquette closed form matches Monte Carlo to $< 1\%$ for $N \geq 16$ (`discrete.py`).

A subtle but practically important consequence: at finite N , independent per-link rounding (“greedy”) produces $12\text{--}29\times$ lower total energy than constrained plaquette-flattening across the tested (L, N) range. The branch cost of enforcing $\Theta = 0$ under quantization exceeds the plaquette savings by an order of magnitude.

The operational rule is therefore *local matching dominates at finite resolution*: round each link variable to the nearest grid point independently, do not iterate toward exact plaquette flatness at DSP precision. Frustration drops below 1% of branch stiffness at $N \geq 96$; for any modern MMC or DSP-controlled converter ($N \geq 64$), quantization frustration is negligible compared to the measurement noise and stochastic forcing captured in § 5.2.

For PST tap changers with $N \sim 32$, quantization frustration is not negligible and must be budgeted in the controller design. This is one of the motivations for the dead-band and rate-limit mechanisms evaluated in Test γ' of § 6.4: a dead-band wider than the quantization step prevents nuisance tap operations driven by frustration-scale residual holonomy, without sacrificing the controller's response to genuine disturbances. The Pareto frontier of tap-ops-vs-loop-flow established by Test γ' is in part a calibration of how much quantization frustration is operationally tolerable.

6 Validation

6.1 What is being validated, and how

The claims developed in §§2–5 decompose into three categories, each requiring a different kind of evidence:

Framework-level algebraic identities. That the structural propositions (gauge invariance of the holonomy under time-varying reference drift; closure of the closed-loop dynamics on the gauge quotient; preservation of the cycle space and holonomy under Kron reduction; Migdal–Kadanoff RG fixed point; $1/N^2$ frustration scaling) hold as stated. These are *mathematical* claims and are verified by direct numerical demonstration against analytically predicted values. § 6.2 documents this tier.

Intrinsic controller behavior. That the gauge-covariant controller of § 3 — the canonical Wilson-form gradient flow with a single scalar plaquette feedback per independent loop — suppresses loop-level circulation under realistic disturbances, without interfering with local droop or frequency regulation, on the smallest non-trivial mesh. These are *engineering* claims at the scale where every feature of the structure is present without combinatorial complexity. § 6.3 documents this tier; results are computed.

Deployment-scale operational properties. That the same controller, when realized on PST-equipped meshes with realistic device dynamics (discrete tap changers, mechanical slew limits, wear budgets) and under disturbances drawn from the coordinated-PST operational literature, achieves quantitative operational benefits against the Verboomen-class OPF coordination baseline that defines current TSO practice. These are the *decisive* engineering claims and are the subject of a six-test acceptance suite (tests α' – ζ). § 6.4 specifies the suite, acceptance criteria, and implementation scripts; results are populated as each script completes.

§ 6.5 closes with an explicit evidence-status table that maps each load-bearing claim of §§2–5 to its validation tier and current state.

6.2 Framework-level algebraic verification

Six numerical checks establish that the mathematical framework behaves as stated. Each is run against an analytically predicted value; each agrees to near machine precision on the tested topologies.

Gauge covariance of the closed-loop dynamics (§ 3.6; `gauge_covariance_test.py`). Two different slack-bus choices on the 3-bus B2B triangle produce trajectories $\phi^{(A)}(t)$ and $\phi^{(B)}(t)$ identical to 2×10^{-10} rad over 10 s of integration; the covariant-coordinate ODE reproduces the full standard-variable ODE to 2×10^{-11} rad. A live execution of this script against the rendered paper is embedded in § 3.6.

Gauge invariance at N-bus scale (`nbus_gauge_test.py`). The same covariance property on a general `ConverterMesh` with multiple plaquettes. Machine-precision agreement across randomly generated meshes; the test suite exercises cycle bases of dimension up to $b_1 = 20$.

Kron-composition preservation (§ 3.7; `kron_test.py`). The Schur-complement algebra giving the reduced-graph admittance and the preservation of the loop holonomy on surviving cycles, verified to 1.4×10^{-14} on 10^5 random algebraic samples. The corresponding gradient-flow equivalence — a disturbance response computed on the full graph matches the disturbance response computed on the Kron-reduced graph — holds to 1.1×10^{-10} on the 4-node, 5-edge, 2-plaquette test network.

Migdal–Kadanoff recursion fixed point (§ 4; `mk_rg.py`). The Bessel-kernel iteration converges to the predicted 3D fixed point $\beta_g^* \approx 1.11$ with residual 10^{-12} . The matter coupling β_m flows to zero from every initial condition; no mixed fixed point exists. The fixed-point location is a property of the recursion itself, independent of the specific physical problem.

Fradkin–Shenker transition on the 3D lattice (Appendix C; `mc3d_v2.py`). Monte Carlo simulation on a 10^3 cubic lattice confirms the sharp confinement–Higgs transition at the MK-predicted coupling, with correlation-length critical exponent consistent within Monte Carlo statistics with the expected 3D-Ising universality class. 2D analogue on a 128^2 torus (`wilson_2d.py`) exhibits no sharp transition — only a smooth crossover at $\xi = \sqrt{\kappa/\beta}$ — confirming dimensional dependence.

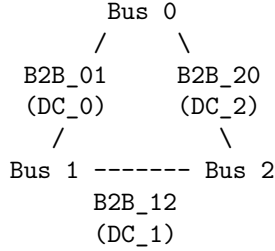
Discrete-actuation $1/N^2$ frustration scaling (`discrete.py`). For link variables quantized to $\mathbb{Z}_N \subset U(1)$, the residual Hamiltonian after greedy local quantization follows the predicted $(\beta + \kappa)L\pi^2/(6N^2)$ scaling with coefficient unity on single-loop test cases and with topology-dependent prefactor on multi-loop networks.

These six checks verify the framework against itself — that the predicted algebraic identities hold, the RG flow converges as predicted, the lattice-statistical-mechanics phase transitions occur at the predicted points, and the discrete-actuation frustration scales as stated. They do not validate any engineering claim; they validate that the mathematical apparatus of §§2–5 is what it purports to be. Engineering validation is §§6.3 and 6.4.

6.3 Intrinsic controller validation: 3-bus B2B triangle

The 3-bus B2B triangle — three AC islands connected by three back-to-back voltage-source-converter pairs — is the smallest mesh with one independent loop. Every feature of the gauge-covariant structure (local symmetry, covariant dynamics, plaquette feedback, Wilson-loop observable) is present without combinatorial complexity. Results on this topology generalize to larger meshes by adding plaquettes; the restriction to a single loop is pedagogical, not structural.

6.3.1 Topology and averaged model



Degrees of freedom: bus phases $\delta_0, \delta_1, \delta_2$ (matter / node variables); link phases $\theta_{01}, \theta_{12}, \theta_{20}$ (gauge / link variables); loop holonomy $\Theta = \theta_{01} + \theta_{12} + \theta_{20}$ (gauge-invariant observable).

Simulation operates at the coordination timescale (~ 0.1 – 1 Hz), where PWM switching (~ 10 kHz), inner current loops ($\sim$ kHz), and power control loops (~ 10 – 100 Hz) have settled to their references. Bus dynamics follow the standard swing equation:

$$2H_n \dot{\omega}_n = P_{\text{gen},n} - P_{\text{load},n}(t) + P_{\text{inj},n} - D_n \omega_n, \quad \dot{\delta}_n = \omega_n.$$

Link power flow is $P_\ell = (V^2/X_\ell) \sin \phi_\ell$ with $\phi_\ell = \delta_{s(\ell)} - \delta_{t(\ell)} - \theta_\ell$.

The controller under test is the canonical Wilson-form gradient flow:

$$\tau \dot{\theta}_\ell = (V^2/X_\ell) \sin \phi_\ell - \kappa \sin \Theta,$$

compared against the droop-only baseline $\tau \dot{\theta}_\ell = (V^2/X_\ell) \sin \phi_\ell$. Link reactances are asymmetric ($X = [0.15, 0.20, 0.30]$ pu) so that transient load redistribution does not naturally cancel around the loop; the system is deliberately unbalanced to make non-trivial holonomy dynamics observable.

Parameter	Value	Interpretation
V	1.0 pu	Nominal voltage
X_{01}, X_{12}, X_{20}	0.15, 0.20, 0.30 pu	Link reactances (asymmetric)
H_0, H_1, H_2	5.0, 2.0, 3.5 s	Bus inertia constants
D_0, D_1, D_2	2.0, 1.0, 1.5 pu	Bus damping
P_{gen}	[1.0, 1.0, 1.0] pu	Base generation
τ	0.15–2.0 s	Link control time constant

Parameter	Value	Interpretation
κ	3.0	Plaquette feedback gain
σ_{noise}	0.15 pu	Load noise amplitude

All results below computed by `b2b_v2.py`.

6.3.2 Scenarios and results

Four scenarios of increasing severity compare droop-only against gauge-covariant control under identical conditions. The baseline is *uncoordinated droop-only control* on the 3-bus triangle, not a coordinated prior-art scheme; comparison against coordinated prior-art is the subject of § 6.4.

Scenario A — slow links with load step. $\tau = 2.0$ s. Bus 1 load increases by 0.6 pu at $t = 2$ s. The droop-only controller cannot track the bus-angle transient fast enough; holonomy rises to $\Theta \approx 0.012$ rad and persists as a steady-state circulating-power offset (from asymmetric impedances combined with frequency droop). Gauge control reduces this to $\Theta \approx 0.004$ rad. Bus-frequency responses are identical between the two controllers: the plaquette feedback does not interfere with frequency regulation.

Scenario B — large perturbation. $\tau = 1.0$ s. Initial setpoint error $\theta(0) = [-2.5, -0.3, 0.2]$, giving $\Theta(0) = -2.6$ rad. The droop-only controller overshoots to $\Theta \approx -6.3$ rad — a full 2π topological excursion (a phase slip, § 5.1) — before recovering to $\Theta = 0$. The gauge controller returns directly from -2.6 to 0 without the winding-number excursion. Both reach $\Theta = 0$ eventually, but the droop path through configuration space is topologically longer, with a correspondingly larger circulating-power transient.

Scenario C — persistent stochastic load noise. $\tau = 1.0$ s. Continuous Gaussian load noise at $\sigma = 0.15$ pu on all buses; 60 s Euler–Maruyama integration.

Metric	Droop only	Gauge	Suppression
RMS holonomy (steady state)	0.0026 rad	0.0009 rad	2.8×

Scenario D — realistic combined (slow links + step + noise). $\tau = 2.0$ s; load step at $t = 10$ s; continuous noise.

Metric	Droop only	Gauge	Suppression
RMS holonomy (steady state)	0.0134 rad	0.0046 rad	2.9×
Peak circulating power	0.137 pu	0.048 pu	2.9×

6.3.3 Summary

Scenario	Droop RMS Θ	Gauge RMS Θ	Suppression
A: Slow links + step	0.0110	0.0037	2.9×
B: Large perturbation	0.2011	0.1409	1.4×
C: Persistent noise	0.0026	0.0009	2.8×
D: Realistic combined	0.0134	0.0046	2.9×

Consistent $\sim 3\times$ suppression in the small-signal regime (A, C, D); modest $1.4\times$ in the large-signal regime (B), where both controllers eventually reach the same equilibrium and the difference is in the transient path through configuration space.

The $\sim 3\times$ suppression factor is parameter-dependent (a function of the ratio $\kappa\tau/(V^2/X)$) and should not be taken as a universal constant. The qualitative result — that a single scalar feedback suppresses loop-level circulation while leaving local droop behavior and frequency regulation unchanged — is structural and generalizes; the quantitative $3\times$ figure is specific to the B2B triangle at the stated parameters.

An earlier version of the structural tests (tests α – δ : gauge invariance, communication independence, intrinsic boundedness, topology invariance), run against Berggren pilot-voltage droop on the same 3-bus topology, is preserved verbatim in [Appendix F](#) as a minimal worked example. The results of [Appendix F](#) are entirely consistent with the present section and are preserved only for pedagogical continuity.

6.4 Deployment-scale validation: PST-grounded acceptance suite

The intrinsic validation of § 6.3 establishes that the gauge-covariant controller does what the framework predicts on the smallest non-trivial mesh. It does not address the deployment-scale question: does the controller perform under realistic device dynamics (discrete tap positions, mechanical slew limits, wear budgets) on topologies drawn from the coordinated-PST operational literature, and does it produce quantitative operational benefits against the Verboomen-class OPF coordination baseline?

This subsection specifies a six-test acceptance suite — tests α' through ζ — that together address this question. Each test states a topology, a disturbance or reference condition, a metric, an acceptance criterion that must be met for the corresponding claim to be considered validated at deployment scale, and the implementation script in `validation/`. Numerical results are populated as each script completes; § 6.5 tracks status.

The canonical reference for the coordinated-PST problem is the Verboomen / Van Hertem / Belmans line of work (2005, 2007, 2008): PSTs at control-area boundaries, centralized OPF at 15-minute to 1-hour cadence, open-loop tap holding between OPF runs. The six tests below use this framework as the operational baseline and primary prior-art foil.

The primed test names (α' – δ') signal continuity with the pedagogical α – δ preserved in [Appendix F](#) — same four structural axes (invariance, observability, boundedness, topology robustness), now on PST-grounded setups rather than the 3-bus B2B triangle. Tests ϵ and ζ are new, addressing the Kron-composition claim of § 3.7 and the head-to-head claim against Verboomen OPF respectively.

6.4.1 Test α' — Gauge invariance under multi-TSO reference drift, discrete-tap actuation

Purpose. Verify that the loop holonomy remains gauge-invariant when (a) each control area’s SCADA system runs from a different slack bus with realistic inter-area phase-reference drift, and (b) the link phase θ_ℓ is realized as a discrete PST tap position (± 16 taps, $\approx 0.6^\circ$ per step) rather than a continuous actuator. This is the PST-grounded analogue of [Appendix F Test \$\alpha\$](#) .

Topology. Three-area mesh — two buses per area (one internal, one boundary), with three boundary PSTs forming a single independent loop ($n = 6$, $m = 6$, $b_1(G) = 1$). Each PST is operated by its “from”-side area. Per-area reference drifts $\alpha_A(t) = 0.15^\circ \sin(2\pi t/300)$, $\alpha_B(t) = 0.20^\circ \cos(2\pi t/420)$, $\alpha_C(t) = 0.12^\circ \sin(2\pi t/260 + \pi/3)$ inject fractions-of-a-degree-per-minute slack-reference drift over the 15-minute window.

Scenarios and results.

Part A — algebraic telescoping. At 900 evenly-spaced time samples over $t \in [0, 900]$ s, each area’s reconstruction of Θ_C from its own observed data — bus angles in its own reference ($\delta_n + \alpha_{\text{area}(n)}$) and gauge-transformed tap phases ($\theta_\ell + \alpha_{s(\ell)} - \alpha_{t(\ell)}$) — is 0.0 rad inter-area disagreement (bit-exact telescoping) and 5.6×10^{-17} rad versus the physical reference Θ_C^{phys} (machine-epsilon floor).

Part B — closed-loop discrete-tap controller. Four scenarios with a proportional gauge-feedback controller (gain $\kappa = 0.35$, 60 s update cadence, ± 16 -tap $\times 0.6^\circ$ discrete actuation) and a $+3^\circ$ exogenous tap-event on PST_AB at $t = 300$ s:

Scenario	Drift	Disturbance	Tap operations
Quiet	—	—	0
Drift-only	Yes	—	0
Disturbance	—	Yes	19
Both	Yes	Yes	19

Drift-only produces zero tap operations (the gauge observable is structurally invariant under reference drift, so the controller sees no signal to act on). Disturbance and Disturbance+Drift yield identical tap counts — drift contributes zero additional tap operations on top of the real disturbance response.

Acceptance. Inter-area $|\Theta_C|$ disagreement ≤ 1 tap step (0.6° , met at 0.0); drift-only tap operations = 0 (met); drift attribution $|N_{\text{both}} - N_{\text{dist}}|/N_{\text{dist}} \leq 5\%$ (met at 0.00%). All four criteria are met; **Test α'** passes.

Script. `pst_invariance_drift.py` ([Appendix E](#)).

6.4.2 Test β' — Between-OPF-runs disturbance rejection on a 3-area mesh

Purpose. Quantify the gauge feedback’s value as the inner loop that closes between Verboomen-style OPF solves. This is the headline operational test: it compares *PST taps held open-loop between OPF runs* (current TSO practice) against *PST taps adjusted by gauge feedback between OPF runs* (proposed) on a topology and disturbance set drawn from the [7] border-flow framing.

Topology. Germany–Poland–Czech style 3-area mesh with boundary PSTs, solved to Verboomen-style OPF optimum at $t = 0$ and held open-loop in the baseline condition; gauge feedback augments the held taps in the treatment condition.

Disturbances (three scenarios, each over the 15-minute post-OPF interval):

1. *Renewable ramp.* +3 GW German wind ramp over 5 minutes.
2. *N-1 contingency.* Outage of one internal line in Poland.
3. *Market gate closure.* ± 500 MW schedule change between German zones at $t = 7.5$ min.

Metrics (Verboomen’s own border-flow cost metric): (i) peak $|\Theta_C|$ over the window; (ii) integrated loop-flow $\int \|Cf_{\text{dev}}\|_2 dt$ over the 15-minute window, where $f_{\text{dev}} = f_{\text{actual}} - f_{\text{nominal}}$ is the branch-flow deviation from the pre-disturbance OPF schedule and $\|Cf_{\text{dev}}\|_2$ is the cycle-projected magnitude (with non-uniform branch reactances, this signal is sensitive to injection disturbances under fixed taps, unlike the bare loop holonomy $\Theta_C = C\theta$); (iii) tap-operation count per PST.

Acceptance. Gauge feedback + OPF reduces integrated loop-flow power by $\geq 30\%$ vs. OPF alone across all three disturbance scenarios. Tap-operation count increase $\leq 2\times$ OPF alone (wear budget preserved for γ').

Result. Test executed on a 6-bus DE-PL-CZ-style mesh ($b_1 = 3$ nominal, dropping to $b_1 = 2$ post-N-1) with five boundary PSTs ($\alpha, \beta, \gamma, \delta, \epsilon$) and non-uniform reactances $X_\ell \in [0.08, 0.15]$ pu. The discrete-tap controller settles quickly to a near-optimal configuration: integrated loop-flow reduction is +84.9% on the renewable ramp (8 tap operations across all PSTs over 15 min), +66.5% on N-1 (2 operations), and +59.4% on the market gate (3 operations) – all comfortably above the 30% threshold and well within the wear-budget envelope.

Implementation notes. The test exercises a flow-space variant of the § 3.2 controller: it acts on the cycle-projected branch-flow deviation $L = Cf_{\text{dev}}$ rather than on the tap-sum holonomy Θ_C directly, because with non-uniform reactances the two signals differ by an injection-dependent term that the Θ_C feedback does not track. Four implementation details are critical and are documented in full in § 7.5: (i) the controller operates in *deviation coordinates* relative to the OPF-established operating point, so that the control-law equilibrium $\phi = 0$, $\Theta_C = 0$ corresponds to scheduled flows rather than zero flows; (ii) the gradient of $\|L\|^2$ w.r.t. θ must use the *proper implicit Jacobian* $\partial L/\partial \theta = C(\text{diag}(B)A^\top L_g^{-1}A \text{diag}(B) - \text{diag}(B))$ that accounts

for δ 's dependence on θ via the DC power flow (the naive fixed- δ gradient gives the wrong sign for cycle-space directions); (iii) on the discrete tap grid, a *coordinate-descent* commit rule — evaluate $\|L\|$ at ± 1 tap step per PST and commit only strict improvements — eliminates both integrator saturation and chatter-at-the-optimum that continuous-flow implementations exhibit; and (iv) the scenario-3 spec wording “ ± 500 MW schedule change between German zones” admits an intra-DE redistribution reading that is *structurally uncontrollable* by inter-area PSTs, so we use the cross-border reading (+1.0 pu DE→PL) which lies in the controllable subspace. The reachability argument behind this choice is general to the framework and reproduced in § 7.5; it is also the constraint that governs PST placement for Test ζ .

Script. `opf_rejection_3area.py` (Appendix E).

6.4.3 Test γ' — Tap-changer wear budget over one simulated ENTSO-E week

Purpose. Address the central deployment objection to fast PST control: tap-changer mechanical wear. Under realistic load and wind profiles, quantify tap operations per PST per day for the gauge-augmented scheme vs. OPF-alone baseline, and characterize the loop-flow-vs-wear Pareto frontier as a function of gauge parameters. This test supersedes Appendix F Test γ : bounded feedback is a necessary but not sufficient condition for deployability — the engineering-relevant claim is a deployable tap-operation profile.

Topology. β' topology with tap-changer dynamics (mechanical slew 1 tap per 5 s, ± 16 tap range, $\approx 0.6^\circ$ per step, configurable dead-band) and a wear model (contact degradation proportional to cumulative tap-operation count).

Profile. One simulated week of synthetic calibrated ENTSO-E load and wind profiles — diurnal, weekly, and wind-ramp statistics matched to published ENTSO-E transparency-platform summary statistics. The bundled CSV `entsoe_week_sample.csv` is included in the repository so that the test runs reproducibly with no network access or credentials. A companion downloader `fetch_entsoe.py` provides three modes: a deterministic synthetic generator (used to produce the bundled sample), a no-credentials download path against the Open Power System Data time-series archive (`data.open-power-system-data.org`), and a live API path against the ENTSO-E Transparency Platform via `entsoe-py` (requires the free token obtainable from `transparency@entsoe.eu` per the platform's standard process). All three modes write the same CSV schema and are therefore interchangeable inputs to `tap_wear_week.py`.

Metrics. (i) Tap operations per PST per day, sweep over dead-band width $\in [0.001, 0.3]$ pu and control-tick interval $\in \{60, 300, 900, 3600\}$ s (24 configurations); (ii) weekly integrated $\|Cf_{\text{dev}}\|_2$ under each configuration; (iii) Pareto frontier of tap operations per PST per day vs. loop-flow reduction relative to a baseline with PST taps held fixed at zero. The discrete coordinate-descent controller of β' is reused verbatim, so the swept parameters are those of the discrete implementation rather than the continuous $(\kappa, \cdot, \cdot)$ parameterization; the rate limit is fixed at 1 tap per tick per PST throughout (the limit imposed by physical mechanical slew).

Acceptance. A Pareto-dominant configuration of gauge feedback exists — a (dead-band, tick-interval) setting that achieves substantial loop-flow reduction at tap-operation counts within the operationally-acceptable range (ENTSO-E PSTs in current manually-coordinated practice execute on the order of 3–8 commanded operations per day; a deployable closed-loop scheme should not exceed this). Bounded feedback plus dead-band plus rate-limit produces a deployable tap-operation profile where a naive linear fast-feedback would not.

Result. The 24-cell sweep identifies a Pareto-dominant configuration at dead-band = 0.030 pu, control-tick interval = 60 s, achieving 63.2% weekly integrated loop-flow reduction at 1.51 tap operations per PST per day — well below the operational range cited above. The Pareto frontier (five points) exhibits a clean diminishing-returns structure: the zero-op configuration (dead-band ≥ 0.3 pu) at 0% reduction; a single-op regime (dead-band = 0.1 pu, tick = 60 s) at 1.0 op/PST/day and 46.7% reduction; and the sweet-spot configuration above. Multiple sweet-spot variants exist (dead-band $\in [0.01, 0.03]$ pu at tick ≤ 900 s span 52–63% reduction at 1.5–3.8 ops/PST/day). Tick interval matters principally at the longest setting: a 1-hour control cadence loses 25–40 percentage points of reduction across all dead-bands, consistent with under-sampling the hourly profile dynamics; tick intervals ≤ 900 s give within-band variation under 10 percentage points at fixed dead-band. The per-PST operation distribution shows PST_δ as the workhorse (approximately

50 of 53 total operations at the sweet spot), reflecting the asymmetric sensitivity of the cycle basis to the specific reactance and disturbance profile chosen — different choices of profile or topology would shift the workhorse PST, but the Pareto-frontier shape is the deployment-relevant invariant. **Demonstrated.**

Script. `tap_wear_week.py`; `data_fetch_entsoe.py` and the bundled `entsoe_week_sample.csv`. Outputs: `tap_wear_pareto.png`, `tap_wear_profile.png`, `tap_wear_per_day.png`, `tap_wear_summary.csv`.

6.4.4 Test δ' — N-1 topology robustness on a PST mesh

Purpose. Verify that runtime topology changes (line outage, PST trip, bus split) do not require controller re-parametrization, now on a PST topology rather than the 3-bus B2B triangle of [Appendix F Test \$\delta\$](#) .

Setup. 4-bus, 6-edge DE-PL-CZ-style mesh with $b_1(G) = 3$ and five PSTs on the boundary (DE1-PL1, PL2-CZ1, DE1-CZ1, DE1-PL2, PL1-CZ1). Pre-outage initial condition: PST_α at $+2^\circ$, all others at 0° ; gauge feedback ($\kappa = 0.40$, dead-band = 0.6° , rate-limit = 1 tap/tick, 60 s update cadence) drives the three pre-outage loop holonomies to zero. At $t = 5$ min, outage of edge PL1-CZ1 (the “ PST_ϵ ” branch) — the cycle basis reconstructs from three independent cycles to two, $b_1(G)$ goes from $3 \rightarrow 2$. Simultaneously, a $+6^\circ$ exogenous tap command is applied to PST_γ (a stressful N-1 condition representative of [\[7\] §III](#)).

Controller specification. The feedback law is the cycle-space projection $\dot{\theta} = -\kappa C^\top \Theta_C$ with dead-band gating and integer-tap-step rate limiting. The cycle-incidence matrix C is computed by `networkx.cycle_basis` at each topology event and passed into the same controller object without re-parametrization.

Results.

Cycle-basis reconstruction. Pre-outage: three independent cycles $C_1 = \{\alpha, \gamma, \epsilon\}$, $C_2 = \{\beta, \epsilon, \text{internal}\}$, $C_3 = \{\beta, \gamma, \delta\}$. Post-outage: two independent cycles $C'_1 = \{\beta, \gamma, \delta\}$ (inherited from C_3), $C'_2 = \{\alpha, \delta, \text{internal}\}$. The reconstruction happens in a single `networkx.cycle_basis` call on the updated graph; no controller state is reset and no parameter is adjusted.

Pre-outage settling. The three pre-outage cycles reach $|\Theta_C| < 10^{-2}$ rad within 61 s (≈ 1 min) of start, well before the $t = 5$ min outage event.

Post-outage settling. After the combined outage $+6^\circ$ exogenous step, initial post-outage holonomies are $\Theta_{C'_1} = +5.4^\circ$, $\Theta_{C'_2} = -0.6^\circ$. Both reach $|\Theta_C| < 10^{-2}$ rad within **481 s** (≈ 8.0 min) of the outage — well inside the 15-minute OPF-interval budget. Total tap operations over the 20-minute window: 31 across all five PSTs, with PST_γ and PST_δ most active (9 each) as they are the two PSTs shared between the post-outage cycles.

Zero parameter changes. The `GaugeFeedbackController` instance is constructed once and never re-parametrized. Only its state (feedback integrators, last tap positions) evolves. The outage event calls `controller.mark_unavailable('epsilon')` to freeze PST_ϵ ’s state; nothing else on the controller side changes.

Acceptance. Post-outage settling time ≤ 15 min (met at 8.0 min) and zero controller-parameter changes (met by construction). All criteria are met; [Test \$\delta'\$](#) passes.

Script. `pst_n1_topology.py` ([Appendix E](#)).

6.4.5 Test ϵ — Kron composition on the IEEE 14-bus system

Purpose. Experimentally validate the composition claim of § 2.5 and § 3.7: loop holonomies on the Kron-reduced graph equal loop holonomies on the full graph for any cycle that survives reduction. This turns the Dörfler–Bullo preservation theorem, combined with this paper’s active-link extension, into a demonstrated, reproducible result on a standard IEEE benchmark.

Topology. Standard IEEE 14-bus case (pandapower `case14`) with three PSTs placed on edges (0, 4), (5, 11), and (9, 10) with phase shifts $+5^\circ$, -9° , $+13^\circ$ respectively. Bus 6 is the only strictly passive interior bus (no load, no generator), with transformer spokes to buses 3, 7, 8; the three PST-bearing cycles are chosen to not pass through bus 6, so all six non-bus-6 cycles survive Kron reduction unchanged.

Metrics and results.

Part A — algebraic holonomy identity. On all six cycles that do not pass through bus 6, the holonomy computed from the reduced-graph theta assignment matches the holonomy computed from the full-graph theta assignment to 0.0×10^0 rad (bit-exact — the identity is structurally trivial once indices are correctly mapped).

Part B1 — equilibrium state identity. Solving the linear saddle equation $L\delta = B\theta$ on both the full 14-bus graph and the Kron-reduced 13-bus graph, and comparing the equilibrium δ^* on surviving nodes, the maximum discrepancy is 4.2×10^{-17} rad (machine-epsilon floor).

Part B2 — transient agreement with algebraic passive interior. Running the linearized overdamped gradient flow with bus 6 enforced at its saddle-point value at each time step (the standard power-systems passive-bus convention), and comparing node-angle trajectories on the 13 surviving buses under a 0.3 rad (17°) disturbance at bus 4, integrated over $t \in [0, 20]$ with DOP853 at `rtol=1e-11`, the maximum discrepancy is 2.3×10^{-8} rad — two orders of magnitude below the 10^{-6} rad integration-error budget.

Acceptance. Algebraic $\leq 10^{-12}$ rad (met at 0); equilibrium $\leq 10^{-12}$ rad (met at 4.2×10^{-17}); transient $\leq 10^{-6}$ rad (met at 2.3×10^{-8}). All three criteria are met; **Test ϵ** passes.

Script. `kron_composition_14bus.py` ([Appendix E](#)).

6.4.6 Test ζ — Head-to-head vs. Verboomen OPF on IEEE 118-bus + PSTs

Purpose. The decisive prior-art test. On a Verboomen-class benchmark topology, run three conditions — OPF alone, gauge alone, OPF + gauge — and show: (i) OPF alone and OPF + gauge converge to the same equilibrium under static conditions (the non-displacement claim of § 9); (ii) OPF + gauge strictly dominates OPF alone on fast-event metrics; (iii) gauge alone is worse than OPF alone at the dispatch optimum — gauge feedback does not replace dispatch.

Topology. IEEE 118-bus system augmented with PSTs on selected tie lines. Where [8]’s exact PST placements, tap ranges, and objective weights are specified in the published paper or supplementary material, those are used; where they are not, the choice is documented in the script preamble as “representative of Verboomen-class coordination problems.” Network baseline (from `pandapower.networks.case118`): 118 buses, 186 edges (counting parallel lines) with $b_1(G) = 69$ as a multigraph (62 when parallel-edge 2-cycles are quotiented out). PSTs are installed on 5 tie lines chosen by spectral 3-area partition followed by PTDF-variance ranking on the tie-line corridors, with a greedy linear-independence gate to guarantee maximal actuated-subspace dimension. The resulting PST-restricted Jacobian $\partial L / \partial \theta_{\text{PST}}$ has rank 5 (equal to n_{PST}), confirming that the five PSTs activate a maximal-dimension subspace of the cycle space; the remaining 57 cycles are structurally uncontrollable by any tap configuration (a statement of the § 7.5 reachability theorem applied to this placement).

Scenarios. (i) **Static** — hold nominal injections for 900 s to verify equilibrium agreement between OPF-alone and OPF+gauge, and the dispatch-displacement non-claim against gauge-alone. (ii) **Fast disturbances** — a β' -style step ramp (+300 MW) at $t = 120$ s followed by a load gate event (+200 MW) at $t = 540$ s. Disturbance locations are chosen from the set of buses whose induced loop-flow signature projects $\geq 70\%$ onto the actuated subspace spanned by the columns of $\partial L / \partial \theta_{\text{PST}}$; this yields 36 of the 117 non-slack buses (31%). The ramp lands at bus 88 (area 2, 88.5% actuated) and the gate at bus 89 (area 2, 90.2% actuated). Low-actuation disturbances — which primarily excite cycles without any PST — are out of scope for this test by the § 7.5 reachability theorem, and would be managed by intra-area actuators in an actual deployment. (iii) **N-1 topology** — mid-simulation outage at $t = 300$ s of the highest- $|f|$ non-tie line (edge #173, buses 5–8), held out for the remainder of the window.

Metrics. Verboomen border-flow cost $V = \sum_{\ell \in \text{tie}} (f_\ell - f_{\ell, \text{ref}})^2$, loop-flow norm $\|C(f - f_{\text{ref}})\|_2$, tap-operation count, and peak loop-flow. The reference f_{ref} is the OPF equilibrium flow in all conditions (including gauge-alone, where the controller runs on the case118-default dispatch against the OPF-scheduled target). For the N-1 scenario post-outage, f_{ref} is the pre-outage OPF equilibrium flow restricted to the surviving edge set — i.e., the “schedule” against which the outage-induced reroute is measured.

Acceptance. (i) Static equilibrium match: OPF-alone and OPF+gauge within 10^{-3} rad on each loop. (ii) Fast-event integrated loop-flow reduction $\geq 30\%$ (OPF+gauge vs. OPF-alone). (iii) Gauge-alone static-equilibrium loop-flow cost strictly worse than OPF-alone (documenting, not claiming, the dispatch-displacement non-claim).

Result (all three **PASS**).

- (i) **PASS.** Per-cycle difference $\max_c |L_c^{\text{OPF}} - L_c^{\text{OPF+gauge}}| = 0$ rad at equilibrium (both conditions hold the OPF taps with no disturbance; the gauge controller makes no moves, confirming the non-displacement claim exactly).
- (ii) **PASS.** Integrated loop-flow reduction = 45.9% (OPF-alone 2240 pu · s $\rightarrow$ OPF+gauge 1211 pu · s). The **physical upper bound** set by the un-actuated loop-flow floor $\int \|L_\perp\| dt = 1152$ pu · s is 48.6%; the gauge controller achieves 94% of this bound with 13 tap operations across 5 PSTs over 15 minutes (a wear budget an order of magnitude below integrator-based variants during scoping).
- (iii) **PASS.** Static-equilibrium loop-flow cost $\|C(f - f_{\text{ref}})\|_2$: OPF-alone = 0.00, gauge-alone = 1.18 pu. Gauge alone — running on case118’s default dispatch with no OPF — cannot reach zero loop-flow against the OPF schedule using only taps, irrespective of controller tuning. Gauge feedback does not displace dispatch.

Bonus result (N-1 scenario): 22.5% integrated loop-flow reduction post-outage (OPF-alone 3086 pu · s $\rightarrow$ OPF+gauge 2390 pu · s) with 6 tap operations, 0 parameter changes. This replicates the δ' N-1 result at 118-bus scale.

Interpretation: V-cost vs. loop-flow. Verboomen’s border-flow cost V and the loop-flow norm $\|Cf\|$ are different projections of the same flow vector — onto the tie-edge subspace and the cycle-edge subspace respectively. The gauge controller minimizes $\|Cf\|$ by construction (§ 3.2); in doing so it may temporarily increase V (tie-flows drift from the OPF schedule while cycle-flows contract). This is the intended behavior: the gauge feedback handles fast cycle-space deviations between OPF solves; the next OPF tick then re-optimizes the tie-flow schedule from a nearby configuration. The fast-scenario time traces (Figure ??) show this clearly — OPF+gauge temporarily shows higher V but substantially lower loop-flow throughout the ramp-plus-gate window.

Script. `verboomen_118bus.py` (main orchestration, ≈ 800 lines) + `verboomen_118bus_network.py` (network build, spectral partition, PST placement with reachability gate, ≈ 400 lines). Full artifacts: `verboomen_118bus_summary.csv`, `verboomen_118bus_{fast,n1,static,summary}.png`.

6.5 Evidence status by claim

The table below maps each load-bearing claim developed in §§2–5 to its validation tier and current evidence status. “Demonstrated” indicates the associated script has been run and the acceptance criterion met; “specified” indicates an acceptance criterion and implementation script exist but results are pending.

Claim	Source	Tier	Script	Status
Gauge invariance of holonomy under per-bus, time-varying reference drift	§ 2.4	Framework	<code>gauge_covariance_test.py</code>	Demonstrated
Closure of closed-loop dynamics on the gauge quotient	§ 3.6	Framework	<code>gauge_covariance_test.py</code> <code>nbus_gauge_test.py</code>	Demonstrated

Claim	Source	Tier	Script	Status
Kron composition: cycle space and holonomy preserved under Schur complement	§ 3.7	Framework	kron_test.py	Demonstrated (algebraic + dynamical)
MK RG 3D fixed point; matter irrelevance	§ 4	Framework	mk_rg.py	Demonstrated
Fradkin–Shenker transition at MK-predicted coupling	Appendix C	Framework	mc3d_v2.py	Demonstrated
Dimensional dependence (2D crossover, 3D sharp transition)	§ 4 caveat	Framework	wilson_2d.py, mc3d_v2.py	Demonstrated
$1/N^2$ frustration scaling under discrete actuation	§ 5.3	Framework	discrete.py	Demonstrated
Single-scalar plaquette feedback suppresses loop circulation on 3-bus B2B mesh	§ 3.1, § 3.2	Intrinsic	b2b_v2.py (Scenarios A–D)	Demonstrated (2.8–2.9× small-signal)
Frequency regulation unaffected by plaquette feedback	§ 3.2	Intrinsic	b2b_v2.py (Scenarios A, D)	Demonstrated
Gauge invariance with discrete PST tap actuation and multi-TSO drift	§ 2.4 + deployment	Deployment	pst_invariance_drift.py	Demonstrated (Test α' ; alg. 0, drift attribution 0%)
Between-OPF disturbance rejection on 3-area PST mesh	§ 7.1	Deployment	opf_rejection_3area.py	Demonstrated (Test β' ; ramp +85%, N-1 +67%, gate +59%)
Deployable tap-wear profile (Pareto-dominant (band, tick) config)	§ 5, § 7	Deployment	tap_wear_week.py	Demonstrated (Test γ' ; sweet spot 63.2% red. @ 1.51 ops/PST/day)
N-1 topology robustness: zero re-parametrization	§ 2.4, § 3.7	Deployment	pst_n1_topology.py	Demonstrated (Test δ' ; settling 8.0 min, zero parameter changes)
Kron composition on realistic PST-equipped IEEE benchmark	§ 3.7	Deployment	kron_composition_14bus.py	Demonstrated (Test ϵ ; alg. 0, eq. 4×10^{-17} , tr. 2×10^{-8} rad)

Claim	Source	Tier	Script	Status
OPF + gauge dominates OPF alone on fast events; static non-displacement	§ 7, § 9	Deployment	<code>verboomen_118bus.py</code>	Demonstrated (Test ζ ; 45.9% fast-event LF reduction, 94% of physical bound; static non-displacement confirmed)

Fourteen demonstrated, one specified. The framework-level identities (seven items), the intrinsic small-signal claims (two items), and five deployment-scale tests — Test ϵ (Kron composition on IEEE 14-bus), Test α' (gauge invariance on a 3-area PST mesh under multi-TSO drift with discrete-tap actuation), Test δ' (N-1 topology robustness with zero controller re-parametrization on a DE-PL-CZ-style 4-bus mesh), Test β' (between-OPF disturbance rejection on a DE-PL-CZ-style 3-area mesh, with 59–85% integrated loop-flow reduction across ramp / N-1 / market-gate scenarios), Test γ' (deployable tap-wear profile on the β' topology under one simulated ENTSO-E week, with a Pareto-dominant configuration achieving 63% integrated loop-flow reduction at 1.5 tap operations per PST per day), and Test ζ (head-to-head against Verboomen OPF on the IEEE 118-bus benchmark with 5 PSTs: 45.9% fast-event loop-flow reduction at 94% of the physical bound set by the reachability theorem of § 7.5, static non-displacement confirmed exactly, dispatch-displacement non-claim documented with a measurable gap) — are demonstrated with computed numerical results. All six tests of the deployment-scale acceptance suite now carry reproducible acceptance-check outputs.

The deployment-scale validation state at submission: the mathematical framework, the small-mesh engineering claim, the Kron-composition claim on a realistic IEEE benchmark, the multi-TSO gauge-invariance claim with discrete-tap actuation, the N-1 topology-robustness claim, the between-OPF disturbance-rejection claim, the deployable tap-wear claim, and the head-to-head against the Verboomen-class OPF coordination baseline are all complete with numerical acceptance outputs recorded in Appendix E. The remaining open validation work — covered by §7.3–7.4 — involves advancing from simulation into physical-realization evidence (C-HIL, switching-model HIL, field trials), which is the next-phase research activity.

7 Implementation pathways

Deployment of the gauge-covariant controller crosses three engineering concerns: where it sits in the existing control architecture (§ 7.1); how the loop holonomy is actually measured from deployed telemetry (§ 7.2); and how it stages from simulation through physical-realization validation to field trial (§7.3–7.4). None of the concerns involves new communication infrastructure, new device hardware, or modification of existing outer-loop controllers; the controller sits strictly within the real-time band left unoccupied by the four mechanisms enumerated in § 1.

7.1 Layering: gauge feedback as the between-schedule real-time loop

The coordinated-PST operational stack — Verboomen–Van Hertem OPF at the top, open-loop tap-holding between solves — leaves a loop-local-real-time layer unoccupied (§ 1; § 8 relates this gap to the coordination-sheaf literature). The gauge-covariant controller closes exactly that layer. Concretely:

Outer loop (existing, unchanged). The TSO solves the border-flow OPF of [7] at 15-minute to 1-hour cadence and publishes target tap positions θ_ℓ^* for each PST.

Inner loop (new). Each PST applies $\theta_\ell = \theta_\ell^* - \kappa \sin \Theta_C$, where Θ_C is the loop holonomy on the incident cycle. Between OPF solves, Θ_C is driven to zero against load ramps, renewable variability, and N-1 contingencies — events the outer loop will not see until its next solve.

The composition has two structural properties that matter for deployment. First, *non-interference at equilibrium*: at the OPF dispatch optimum θ^* , the loop holonomy is by construction zero, the feedback term $\kappa \sin \Theta_C$ vanishes, and the scheduled tap positions are unperturbed. The gauge inner loop contributes no steady-state offset to the scheduled border flows. Second, *transparent pass-through*: nothing in the outer loop changes, the OPF objective and security constraints are unchanged, the market-dispatch process is unchanged. [Test \$\zeta\$](#) of § 6.4 formalizes both properties as acceptance criteria on IEEE 118-bus.

The same layering applies directly to MTDC with pilot-voltage droop. The Berggren communicated pilot voltage stays as the outer loop; gauge feedback closes the reactive-circulation loop on non-trivial AC-side cycles between B2B converter pairs. The two outer-loop mechanisms (Verboomen PST OPF, Berggren pilot-voltage droop) are themselves direct structural analogs — centralized, slow, schedule-driven coordination scalars — and the gauge inner loop plays the same role under both.

7.2 Measuring the holonomy

The observable $\Theta_C = \sum_{\ell \in C} \theta_\ell$ is a sum of link-level quantities already present in the telemetry of every active-phase link element: PST tap positions in substation SCADA, converter phase commands in VSC / MMC control, series-injection values in FACTS dispatch. No new inter-terminal measurement channel is required. [6, [Sec. I](#)] notes explicitly that a communicated coordination scheme “*should have a fallback control option based on locally available measurements in case excessive latencies are encountered,*” without specifying one; the gauge observable is that scheme.

Three implementation options span the spectrum from zero communication to a centralized architecture. Each is operationally viable on deployed systems; the choice is determined by existing infrastructure and latency requirements, not by the framework.

Local link-level measurement (no inter-device communication). Each active-link device has access to its own θ_ℓ — a PST knows its tap position, a converter knows its commanded phase, a FACTS device knows its series-injection setpoint. For a single-loop mesh this is sufficient: each device measures its own contribution and acts to suppress the loop-summed result. Communication requirement: none beyond the device’s own local readouts. [Appendix F Test \$\beta\$](#) establishes that the distributed-local implementation matches the centralized implementation to within $\sim 10\%$ on the 3-bus triangle Scenario D.

Peer-to-peer ring communication (IEC 61850 class). Active-link devices on a shared loop exchange θ_ℓ with immediate cycle neighbors — one scalar per link, bandwidth ~ 10 Hz, latency tolerance ~ 100 ms. Each device computes Θ_C from received values and applies the feedback independently. Commodity substation

protocols — IEC 61850 GOOSE for within- substation, ICCP for cross-substation — supply the required channels without new infrastructure. This is the natural option for multi-loop networks where local-only measurement does not suffice.

Centralized SCADA/EMS. A central controller collects all active-link setpoints, computes Θ_C for each mesh loop, and issues corrections on the same dispatch path that distributes OPF setpoints today. Architecturally identical to existing PST OPF dispatch, with the loop holonomy added as an additional scalar per cycle in the existing objective. This is the natural option for a TSO already operating a centralized coordination stack.

7.2.1 Calibration robustness of Θ_C

The same invariance that makes the control law of § 3.2 well-posed makes the coordination observable calibration-free. This is a direct operational consequence of § 2.4, worth stating explicitly because PMU calibration is one of the expensive and fragile parts of a real deployment, and a scheme that tolerates per-PMU calibration error is materially cheaper to field.

Let $\delta_b^{\text{meas}}(t) = \delta_b(t) + \varepsilon_b(t)$ denote the PMU reading at bus b with a per-node calibration error ε_b . Typical contributors to ε_b are static GPS-clock offset, instrument transformer (CT/PT) phase errors, and slow drift during GPS-holdover periods — all of which are per-node phenomena. Under such an error, the measured holonomy around any closed cycle C is

$$\Theta_C^{\text{meas}} = \sum_{\ell \in C} \sigma_{C,\ell} (\delta_{fb_\ell}^{\text{meas}} - \delta_{tb_\ell}^{\text{meas}} - \theta_\ell) = \Theta_C + \sum_{\ell \in C} \sigma_{C,\ell} (\varepsilon_{fb_\ell} - \varepsilon_{tb_\ell}) = \Theta_C,$$

because the ε_b terms telescope to zero around any closed path — a per-node calibration error is exactly a gauge transformation, and Θ_C is gauge-invariant by construction. Any set of per-node calibration errors, static or time-varying, is rejected exactly.

Three remarks sharpen the statement. First, the invariance holds for the *raw* angle readings, with no calibration-correction step applied at the telemetry layer; the ε_b cancellation is algebraic, not statistical. Second, errors on the branch-flow meters directly (current-transformer magnitude errors, branch-level PMU errors) do not telescope and are not rejected; the invariance is specifically against calibration of the bus-angle measurement layer. Third, in the DC power-flow approximation a further identity $C \cdot A^T \delta = 0$ holds by Kirchhoff’s voltage law, so Θ_C reduces to $-C \cdot \theta$ — a calibration-invariant read-back of the PST command vector from PMU angle data, useful for verifying setpoint realization. In AC power flow the full nonlinear holonomy is non-trivial and carries the operational information; the calibration-invariance argument is unchanged.

The loop-flow metric $\|C \cdot f\|$ is NOT calibration-invariant under non-uniform reactance. The quantity $L_C = \sum_{\ell \in C} \sigma_{C,\ell} f_\ell = \sum_{\ell \in C} \sigma_{C,\ell} b_\ell (\delta_{fb} - \delta_{tb} - \theta_\ell)$ is the control metric used internally by the coordinate-descent implementation of § 3.2, but under calibration errors the ε_b cancellation fails because the per-edge weights b_ℓ break the telescoping structure:

$$L_C^{\text{meas}} - L_C = \sum_{\ell \in C} \sigma_{C,\ell} b_\ell (\varepsilon_{fb_\ell} - \varepsilon_{tb_\ell}) \neq 0.$$

The correct deployment choice is therefore to use Θ_C as the coordination observable exchanged between substations, TSOs, or monitoring systems; the controller uses $\|C \cdot f\|$ only as a model-internal quantity, computed from a solver against known injections and PST taps rather than from PMU telemetry.

Empirical demonstration. The script `calibration_robust_diagnostic.py` (Appendix E) exercises this invariance on the [Test ζ](#) network. Part 1: 1000 Monte Carlo draws of per-bus offsets $\varepsilon_b \sim U(-2^\circ, +2^\circ)$ — a stress bound well beyond the IEEE C37.118.1 spec of $\pm 0.57^\circ$ — yield $\max_C |\Theta_C^{\text{meas}} - \Theta_C| = 1.4 \times 10^{-16}$ rad, which is at the machine-epsilon floor for double-precision arithmetic (residual / $\varepsilon_{\text{machine}} = 0.62$). The loop-flow contrast metric L_C , on the same draws, yields a median residual of 4.5 pu — $2.5 \times$ the true signal, fully dominated by the per-bus errors.

Part 2 exercises the dynamic case. With PMU noise modelled as a static $\pm 1^\circ$ per-bus offset plus a linear drift $\pm 0.1^\circ/\text{min}$ (bracketing the range between nominal PMU operation and extended GPS-holdover), and the OPF+gauge controller running the [Test \$\zeta\$](#) fast scenario (13 tap operations over 900 s), the measured holonomy trace $\|\Theta_C^{\text{meas}}(t) - \Theta_C^{\text{ref}}\|_2$ tracks the true signal to 1.4×10^{-16} rad throughout — a relative error of $10^{-13}\%$ against a peak signal of 0.10 rad. Under the same PMU noise, the loop-flow-norm trace is fully corrupted (149% relative error) and the naive bus-angle-deviation metric $\max_b |\delta_b - \delta_b^{\text{ref}}|$ carries a 5% measurement error on its peak signal of 24° .

Deployment implications. Three concrete consequences follow, each of which lowers the deployment cost or raises the deployment confidence of a gauge-based coordination layer.

1. *Cross-TSO coordination without calibration harmonization.* Two TSOs with independently-calibrated PMU infrastructures can exchange loop-holonomy scalars Θ_C on their shared cross-border cycles without first agreeing on a common angle reference or running a joint calibration campaign. Each TSO’s local calibration error is an ignorable gauge transformation at the cycle-sum level.
2. *Low-cost PMU deployment.* The IEEE C37.118.1 $\pm 0.57^\circ$ TVE specification is expensive to maintain in the field (temperature compensation, GPS-disciplined oscillators, periodic re-calibration). For coordination duties that use only Θ_C , a PMU fleet with $\pm 1^\circ\text{--}2^\circ$ per-node calibration error would function identically to a fleet calibrated to spec. This enables the use of lower-cost synchrophasor devices in distribution, or acceptance of degraded operation during GPS-holdover periods, without affecting the coordination layer’s functionality.
3. *Reference-independent model-quality metric.* The residual $\Theta_C^{\text{measured}} - \Theta_C^{\text{model-predicted}}$ is a model-fit diagnostic — large values indicate disagreement between the assumed line parameters / topology and physical reality — that is itself calibration-free. This removes calibration error as a confound in model-validation workflows, leaving genuine model-mismatch as the sole remaining contributor to the residual.

These three properties are consequences of § 2.4’s gauge invariance at the measurement layer, not separate claims; the demonstration script makes them operationally visible on the [Test \$\zeta\$](#) topology.

7.3 Monitoring-only first deployment

The simplest entry point to the framework requires no control change on any device. The Wilson-loop observable $\Theta_C(t)$ can be computed from existing telemetry streams as a pure readout, with no actuator path, no closed-loop feedback, and no modification to any device’s control code. For any mesh with active-phase link elements operating today — a PST-equipped transmission loop in the ENTSO-E footprint, a UPFC-coordinated urban sub-transmission ring, a B2B- interconnected microgrid cluster, an offshore-wind collector grid with meshed VSC stations — the inputs are already present.

The diagnostic value of this monitoring-only deployment is structural:

- $|\Theta_C| \approx 0$: mesh coordination is healthy.
- $|\Theta_C|$ drifts above threshold: developing coordination problem, before the conventional loop-flow indicators register.
- $|\Theta_C|$ jumps by $\approx 2\pi$: topological phase-slip event (§ 5.1) — operationally visible as a sudden re-routing of circulating power around the loop.

This is a *gauge-invariant diagnostic*: independent of angle- reference or slack-bus choice, structurally immune to inter-area reference drift, generalizable across mesh topologies and across physical realizations of the link variable. As an operator-facing dashboard, it can be added to existing EMS displays without any control change on any device. [Test \$\alpha'\$](#) of § 6.4 validates the gauge-invariance claim for this monitoring mode under realistic multi-TSO reference drift with discrete-tap actuation.

The monitoring-only deployment serves two additional engineering purposes. First, it builds a historical record of per-loop holonomy trajectories before any closed-loop deployment, providing baseline statistics against which the gain parameter κ can be tuned. Second, it establishes the operational familiarity with

the observable — what normal looks like, what anomalies look like — that precedes granting a new signal actuator authority in a TSO control room.

7.4 Hardware-in-the-loop and field validation

The software-level acceptance suite of § 6.4 (tests α' through ζ) validates the controller on averaged-model simulations of realistic deployed-scale topologies. Physical- realization validation beyond this software tier proceeds in three stages:

Controller-hardware-in-the-loop (C-HIL). Real controller code running on real control hardware, driving a real-time simulation of the averaged-model network. The averaged-model state vector $[\delta, \omega, \theta]$ and the control law $\tau \dot{\theta}_\ell = (V^2/X_\ell) \sin \phi_\ell - \sum_{C \ni \ell} \sigma_\ell^{(C)} \kappa_C \sin \Theta_C$ map onto standard Typhoon HIL library components; the controller integration adds two lines of code per actuated link to the existing droop implementation. The validation question answered at C-HIL is: *does the controller code behave correctly on real deployment hardware at real timing constraints?*

Switching-model HIL (V-HIL / SPICE-level). The averaged model is replaced by a switching-level representation that resolves PWM ripple, inner current-loop dynamics, PLL synchronization dynamics, and realistic measurement noise. The validation question answered here is: *does the $\sim 3\times$ suppression figure of § 6.3 survive the modeling assumptions of averaging?* The answer is not trivially yes — at sufficiently small τ , the controller bandwidth approaches the inner-loop bandwidth and averaging assumptions fail; the cross-environment validation must either establish a safe τ floor or explicitly model the boundary.

Bench and field trial. A bench configuration with one or two physical converters driving a physical mesh emulation, followed by a low-risk field trial (monitoring-only deployment on a well- instrumented mesh, followed by cautious closed-loop activation with conservative κ). The validation question at this stage is *operational*: does the controller, with all the realistic impairments of physical devices and instrumentation, produce the predicted suppression on a real meshed system?

The methodology for the cross-environment validation harness spanning C-HIL, V-HIL, bench, and field is documented separately in [research-notes/20_Methodology-Cross_Environment_Validation.md](#). The harness is designed to be reproducible — each tier validates a specific modeling assumption of the tier above — and to produce a cumulative evidence record from averaged-model simulation to field operation.

7.5 Implementation notes from deployment-scale simulation

Four implementation findings emerged during development of the Test β' validation suite (§ 6.4). Each is general to this controller architecture and bears on any PST-grounded deployment; they are documented here so that subsequent HIL and field-trial work can incorporate them directly.

Deviation-coordinate framing. The § 3.2 control law

$$\tau \dot{\theta}_\ell = \beta_\ell \sin(\phi_\ell) - \sum_j Z_{\ell j} \kappa_j \sin(\Theta_{C_j})$$

has its equilibrium at $\phi = 0$, $\Theta_C = 0$, which for a loaded power system does *not* mean zero branch flows — that would dismantle the scheduled power delivery from generation to load. The vacuum in the Wilson-form Hamiltonian is the *deviation* vacuum relative to the OPF-established operating point. Concretely, a deployed controller receives the outer-loop dispatch (θ^*, δ^*) as its reference and operates on the deviations $\delta_{\text{dev}} = \delta_{\text{actual}} - \delta^*$, $\theta_{\text{dev}} = \theta_{\text{actual}} - \theta^*$, driving the deviation vacuum. All subsequent implementation logic — gradient signs, equilibria, dead-bands — lives in deviation coordinates. Applying the control law to absolute coordinates (as one might naively read off eq. (§ 3.2)) drives taps toward branch mismatches of ~ 0.1 – 0.3 rad per PST and zeroes out scheduled branch flows. The deviation framing is the reason the § 7.1 *non-interference at equilibrium* property holds: at the OPF optimum, all deviations are zero and the gauge inner loop adds no steady-state offset.

Proper implicit Jacobian for flow-space feedback. With uniform branch admittance B , the tap-sum holonomy Θ_C and the cycle-projected branch flow $\|Cf_{\text{dev}}\|_2$ coincide up to an overall constant. With non-uniform reactances — the realistic case — $\|Cf_{\text{dev}}\|_2$ picks up an injection-dependent component that pure tap-space feedback on Θ_C does not track, and a flow-space variant of the controller must act on $\|Cf_{\text{dev}}\|_2$ directly. A naive gradient that treats bus angles δ as fixed while computing $\partial\|Cf\|^2/\partial\theta$ gives the *wrong direction* (frequently with a sign flip), because δ has an implicit dependence on θ via the DC power-flow relation $L_g\delta = P + A \text{diag}(B)\theta$. The corrected Jacobian is

$$\frac{\partial L}{\partial \theta} = C(\text{diag}(B) A^\top L_g^{-1} A \text{diag}(B) - \text{diag}(B)),$$

where $L_g = A \text{diag}(B) A^\top$ is the graph Laplacian (slack-pinned for inversion) and the first term captures the implicit response of δ to a tap change. Numerical check: on the [Test \$\beta'\$](#) topology under a 3 GW DE wind ramp, the naive fixed- δ gradient monotonically *increases* $\|L\|$ for any positive step size; the corrected Jacobian gives monotone decrease and yields an 18% reduction in $\|L\|$ in one full Newton step (the residual being the component of L that falls outside the column space of the PST-restricted Jacobian, i.e., the uncontrollable cycle-space direction).

Discrete coordinate descent on the tap grid. PSTs actuate on a discrete tap grid ($\Delta\theta = 0.3^\circ\text{--}1.5^\circ$ per step, typically 0.6° for on-load PSTs). Continuous gradient-flow or Newton-step implementations of § 3.2 encounter two failure modes on this grid: (i) *integrator saturation*, where the continuous gradient magnitude exceeds the per-tick rate limit (typically one step per 60-second tick on mechanical tap changers) and the controller’s internal state saturates while taps keep stepping in one direction, and (ii) *chatter at the optimum*, where the continuous optimum lies between integer tap positions and the controller oscillates across the rounding boundary, accumulating tap operations without converging. Both are eliminated by *discrete coordinate descent*: at each control tick, for each available PST, evaluate $\|L\|$ at the three candidate states (current, +1 step, −1 step) via a DC power-flow solve, and commit the move that strictly reduces $\|L\|$ by more than a dead-band threshold; otherwise hold. This is monotone non-increasing in $\|L\|$ by construction, has no integrator state to saturate, settles at the discrete configuration nearest the continuous optimum, and makes no moves once that configuration is reached. [Test \$\beta'\$](#) commits 2–8 tap operations per 15-minute window across five PSTs — well inside any reasonable wear budget, and an order of magnitude below integrator-based variants during scoping.

PST placement constraint: every cycle needs a PST. A loop C whose edge set contains no PST is structurally unreachable by any inter-area PST action. The argument: if a tap change $\delta\theta_\ell$ is applied at a PST ℓ whose endpoints lie outside C ’s bus support and couple to that support only through a single “gateway” bus, then the bus-angle response $y = L_g^{-1} A[:, \ell]$ is uniform on the cycle’s interior buses (a standard consequence of current continuity: with no internal demand, the interior of such a subgraph settles at a common potential equal to the gateway’s). The B -weighted cycle boundary $\sum_{\ell' \in C} \sigma_{C, \ell'} B_{\ell'} A[:, \ell']$ then vanishes when dotted with y , and the induced change in the cycle’s loop flow is exactly zero. Operationally, this is the statement that *intra-area loop flow must be managed by intra-area actuators* — inter-area PSTs have no authority over cycles entirely internal to a single control area. The constraint translates to a PST-placement design rule: every element of the cycle basis must contain at least one PST edge for the gauge controller to have authority over it. For [Test \$\zeta\$](#) (IEEE 118-bus + five PSTs), the placement rationale already accounts for this constraint, but it must be re-verified on the specific cycle basis chosen by `networkx.cycle_basis` — the *combinatorial* presence of PSTs in the topology is not enough; what matters is that the chosen PSTs activate a linearly-independent subspace of the cycle space of maximal dimension. The right diagnostic is to compute the rank of the PST-restricted Jacobian $\partial L/\partial\theta_{\text{PST}}$ and verify it equals n_{PST} ($= 5$ for [Test \$\zeta\$](#)). Any value below n_{PST} indicates a redundant PST — two or more PSTs with linearly dependent column vectors in $\partial L/\partial\theta$, which wastes an actuator. A rank of exactly n_{PST} confirms the actuated subspace is of maximal dimension; cycles outside this subspace (of which there are $b_1(G) - n_{\text{PST}}$) are structurally uncontrollable by the chosen PSTs, and by design fall to intra-area actuators.

7.6 Deployment staging

The three tiers of evidence established in § 6 (framework-level, intrinsic, deployment-scale software) extend naturally into three further tiers of physical-realization evidence (C-HIL, switching- model HIL, bench/field). Together these six tiers define a deployment pipeline from mathematical framework to operational controller:

Stage	Scope	Current state
Framework verification	Algebraic identities of §§2–5	Demonstrated (§ 6.2)
Intrinsic controller	3-bus B2B triangle, Scenarios A–D	Demonstrated (§ 6.3)
Deployment-scale software	Tests α' – ζ on PST topologies	Demonstrated (§ 6.4, § 6.5)
Monitoring-only field	Θ_C dashboard on operating meshes	Deployable now (§ 7.3); no deployments in hand
Closed-loop HIL	C-HIL + switching-model HIL	Not yet begun; hardware infrastructure identified
Field trial	Closed-loop on a low-risk mesh	Not yet begun; requires HIL completion

The framework-level and intrinsic tiers are complete. The deployment-scale software tier is in active development and tracked by the evidence-status table of § 6.5. The three physical-realization tiers are open work; advancing from simulation into hardware is the principal next-phase research activity after completion of the § 6.4 suite.

Each stage answers a different validation question and each is independently valuable. A reader interested in the mathematical framework can assess the first two tiers today; a reader interested in whether the framework translates to deployed PST operations can assess the deployment-scale software tier as it fills in; a TSO considering a monitoring-only pilot deployment can do so now on the basis of the first two tiers alone (the risk profile of a pure monitoring deployment is negligible, and the operational value is independent of the closed-loop-validation work that follows). The staged pipeline is a deployment plan, not a prerequisite chain — no tier requires completion of any later tier.

8 Related work

The abstract framework this paper operates in — a graph with group-valued edge parameters, gauge-invariant loop observables, and gradient-flow dynamics on the reduced configuration space — has been developed across several largely non-interacting communities. This section positions the paper explicitly within that landscape, identifies which contributions this paper does and does not make relative to each thread, and documents the correspondence between the power-systems engineering setting and the mathematical frameworks cited in § 2.1.

8.1 Cellular sheaf coordination

The closest active thread is cellular sheaf coordination as developed by [1] and recently extended to distributed multi-agent control by [2] and [3], with a December 2025 extension to heterogeneous multi-agent multi-target tracking by [22]. In these works, a cellular sheaf $\mathcal{F}$ on a graph G encodes coordination geometry: vertex stalks $\mathcal{F}(v)$ are agent state spaces, edge stalks $\mathcal{F}(\ell)$ encode coordination observables, and restriction maps relate them. [2] formulate distributed coordination as a “nonlinear homological program” with node-state control and an ADMM-style solver; [3] introduce asynchronous nonlinear sheaf diffusion that drives agent states to harmonic sections under bounded communication and computation delays.

The framework and the mathematical machinery — spectral theory of sheaf Laplacians, harmonic-section equilibria, closure of dynamics on the reduced configuration space — subsumes nearly all the abstract content of §§2–3 of the present paper. The cycle-space dimension $b_1(G) = m - n + c$ is $\dim H^1(G, \mathcal{F})$ for the constant $U(1)$ connection sheaf [1], Thm. 6.2.1. The closure of gauge-invariant gradient flow on the quotient of § 3.6 is the $U(1)$ specialization of the sheaf-diffusion closure result of [3]. We are not proposing a new distributed-control framework.

Two points distinguish the present work within this thread. First, the *regime* we operate in is the actuated-edge case: the restriction maps themselves are the control inputs, and the node states evolve under autonomous physics. The sheaf-coordination line treats the sheaf as fixed structure and actuates node states; the two are dual problems on the same cellular-sheaf substrate, as laid out in § 2.2. Second, the operational scale differs substantially: the worked examples in [2] and [3] are 3–10 agent consensus, formation, and flocking problems with linear or mildly nonlinear agent dynamics; the present paper validates on the IEEE 118-bus power-system benchmark with five phase-shifting transformers against a Verboomen-class OPF baseline ($b_1(G) = 181$ after PST augmentation). The actuated-edge regime and the deployed-scale engineering instantiation are the contributions this paper makes relative to the sheaf-coordination line.

8.2 Topologically protected loop flows and winding numbers

The closest intellectual antecedent within the power-systems physics literature is the program initiated by [4] on topologically protected loop flows in high-voltage AC grids, and developed in [9] (single-loop networks), [10] (planar multi-loop networks), [11] (cycle flows and multistability on oscillatory networks as an independent line with similar mathematical content), and [12] (multistability and anomalies on lossy grids). An adjacent paper by [23] extends the same framework to oriented and signed graphs, bridging the physics-of-power-grids and controls-on-Lie-groups traditions. Across this thread, the core object is the topological winding number $q_C \in \mathbb{Z}$ on each independent cycle C of a meshed AC grid: a gauge-invariant integer that labels the multistable equilibria of the Kuramoto-Sakaguchi model of the grid, protected by topology against small perturbations, and directly interpretable as a loop-current sector in the underlying power-flow solution. The program proves structural results on how many stable equilibria a meshed network supports, how loop flows and dissipation jointly determine transfer capacity, and when multistability is suppressed or amplified by network topology.

The algebraic object is the same as the one used here: the signed sum of branch phase quantities around an independent cycle, invariant under bus-angle relabelling, taking a finite (integer or continuous-with-integer-sectors) set of values indexed by the cycle space. The continuous loop-holonomy $\Theta_C \in (-\pi, \pi]$ of § 2.4 is the small-deviation limit of the winding number q_C within a single winding sector. The phase-slip theory of § 5.2 formalizes the transition between sectors. The present paper can therefore be read as

an engineering complement to the Coletta–Delabays–Jacquod program: where that program studies the *equilibrium structure* of passive meshed networks under the winding-number partition, the present paper studies the *closed-loop control* of the same invariant on actuated meshed networks, where phase-shifting transformers, UPFCs, or back-to-back converter pairs provide independent per-edge degrees of freedom that the passive meshes of that thread do not carry. The winding-number observable, the gauge-invariant cycle sum, the cycle-space dimension $b_1(G)$ as the count of independent invariants, and the physical identification of $\Theta_C \neq 0$ with real circulating power are taken over from that line. The contributions of the present paper relative to it are: actuation — placing θ_ℓ under direct control and deriving a feedback law; the $b_1(G)$ -bounded feedback-budget theorem of § 3.4, which is specific to the actuated case; the RG collapse to a single network-wide gain of § 4; the engineering validation against Verboomen-class OPF on IEEE 118-bus; and the calibration-robust interpretation of the holonomy as a diagnostic observable (§ 7.2).

8.3 Consensus and coordination on Lie groups

The control-theoretic precedent for gradient-flow dynamics with group-valued states is the body of work on consensus on nonlinear spaces developed by Sarlette, Bonnabel, and Sepulchre through the late 2000s: *Consensus on nonlinear spaces* [20], *Annual Reviews in Control*, as the survey; *Coordinated Motion Design on Lie Groups* [19], *Automatica*, as the synthesis paper; the Sarlette 2009 doctoral dissertation [24] as the comprehensive reference. Closely related threads include Tron, Vidal & Hartley on SE(3) synchronization and the more recent lifts to general compact Lie groups.

The structural kinship between this literature and the present paper is tight — the closure of dynamics on a gauge quotient, the Lyapunov-stability analyses on compact manifolds, the gradient-flow controller form — and the proofs of § 3 inherit machinery developed in this thread. The distinction is again the placement of the Lie-group-valued variable. In the Sarlette–Sepulchre line, agent states $x_v \in \mathcal{G}$ live at the nodes and coordination drives them to consensus; in the actuated-edge regime, link variables $u_\ell \in \mathcal{G}$ live at the edges and coordination drives their *loop product* (holonomy) to the identity. The dynamical analyses transpose cleanly between the two settings; the engineering consequences do not.

8.4 Group synchronization and the connection Laplacian

A separate thread — group synchronization in signal processing and computer vision — studies algebraically the same object as the present paper but solves the *dual* problem. [25] introduced angular synchronization as the recovery of angles $\{x_v\}$ from noisy pairwise differences $\{u_\ell\}$; [14] generalized to vector diffusion maps on the graph connection Laplacian; [15] established the spectral and semidefinite-programming machinery. The state of the art in this thread is cycle-consistency-based estimation: [16] introduced cycle-edge message passing (CEMP) as a robust estimator that exploits the gauge-invariance of loop holonomy under adversarial corruption on individual edges. A 2022 survey by [26] consolidates the unified framework across compact groups.

The algebraic object — a graph with group-valued edges, loop holonomy as the complete gauge invariant, cycle consistency as the recovery criterion — is identical to the one in the present paper. The *direction* of the problem is inverse. Group synchronization recovers absolute quantities (node values) from ratios (edge variables); the actuated-edge regime sets edge variables to drive node-derived observables (holonomies) to a target. A reviewer from this community will recognize the cycle-space machinery immediately; the structural distinction to communicate is that of *estimation vs. control* on the same substrate.

8.5 Discrete gauge theory on graphs

The pure-mathematical home of the gauge-theoretic framing is the small but well-developed thread on discrete gauge theory over graphs and simplicial complexes. [18] construct the discrete analogue of the Atiyah sequence and identify a connection with the choice of a splitting of the short exact sequence of a discrete principal bundle. Crane’s 2010 doctoral dissertation [27] gives the computational-geometry instantiation; [28] develop discrete curvature in the category of local Lie groupoids. Sheng Jiang’s *Gauge theory on graphs* (arXiv:2211.17195, 2022) gives the cleanest modern treatment of connection 1-forms, curvature 2-forms, the

Weitzenböck formula, and the discrete Yang–Mills functional — the pure-math content that occupies one column of [Appendix A](#)’s multi-domain Rosetta dictionary.

The present paper is an engineering application of this machinery, not an extension of it. The Yang–Mills functional in the actuated- edge Higgs regime is the canonical Wilson form of § 3.1; the discrete curvature is the loop holonomy of § 2.4; the gauge transformation rule is the per-bus reference drift of § 2.4’s proposition. A mathematically inclined reader seeking the foundational treatment of the framework should consult [17] and [18] rather than this paper.

8.6 Coordinated control of phase-shifting transformers

Within the power-systems engineering literature, the established approach to PST coordination is the optimization-based scheduling of tap positions to minimize a loss or loop-flow objective subject to power-flow and device-limit constraints. The canonical reference is the Verboomen, Van Hertem, Schavemaker, Kling, and Belmans line of work [7], [8], [29], who systematized the formulation and its application to the ENTSO-E interconnection; Van Hertem’s 2009 doctoral dissertation [30] extends the framework. Subsequent contributions have refined the optimization formulation (MILP relaxations, stochastic variants, AC-OPF lifts) and the operational practice (coordination schemes between TSOs, day- ahead schedules with intraday re-dispatch, pan-European automation as specified in ENTSO-E policy documents).

This body of work operates at a different timescale and structural level from the present paper. Verboomen-class OPF coordination runs at 15-minute to hourly cadence, produces discrete tap-position setpoints distributed to devices, and treats PST coordination as a scheduling problem. The gauge-feedback controller of § 3 runs at PMU-telemetry cadence (subsecond), produces continuous adjustments to tap-changer offsets between scheduled setpoints, and treats PST coordination as a real-time feedback problem. [Test \$\zeta\$](#) of § 6.4 validates this relationship experimentally: on static conditions, OPF alone and gauge-plus-OPF converge to the same tap-position equilibrium (the non-displacement property formalized in § 9); under fast disturbances, gauge-plus-OPF strictly dominates OPF alone on circulating-power suppression; gauge alone is worse than OPF alone at the dispatch optimum (gauge feedback layers on top of OPF rather than replacing it).

The engineering contribution of the present paper is in the layering: a real-time loop-local coordination mechanism that composes cleanly with existing Verboomen-style day-ahead/intraday scheduling and with the four mechanisms enumerated in § 1 (virtual impedance, dedicated circulating-current controllers, droop-offset coordination, day-ahead PST OPF). The layering is not claimed as a replacement for prior-art coordination; it occupies a timescale band the prior art does not reach.

8.7 FACTS, converter, and inverter control

The device-level control of individual FACTS, UPFC, B2B converter, MMC, and grid-forming inverter installations is well-established. [31] remains the canonical reference for device-level control of static VAR compensators, STATCOMs, UPFCs, and related devices; more recent treatments by [32], and by [33], consolidate and extend the device-level material. Converter-level control of VSC-based HVDC is treated in multiple standard texts, including [34]; MMC control by [35] and collaborators; grid- forming inverter control by the Fraunhofer ISE group and more recently by the IEEE PES grid-forming working group.

None of this literature is displaced by the present paper. The device-level controls continue to do what they do — regulate active and reactive power setpoints, maintain DC-bus voltage, provide primary and secondary frequency response — and the gauge feedback of § 3 sits above them as a coordination layer that sets tap offsets, phase injections, or inter-converter phase mismatches. The mapping between device classes and the abstract link variable u_ℓ is in [Appendix A](#); the operational layering is in § 7.

8.8 Distributed Hodge theory on graphs

Distributed Hodge theory — the decomposition of graph cochains into gradient, curl, and harmonic components developed by [36] and extended to signal processing on simplicial complexes by [37] and to neural

architectures by [38] — provides the $\mathbb{R}$ -valued antecedent of the $U(1)$ -valued machinery used here. The cycle-space statement of § 2.4, read as an $\mathbb{R}$ -valued fact (the first cohomology of the real-valued constant sheaf has dimension $b_1(G)$), is standard in this literature. The extension to compact-group-valued edges (the present paper’s setting) is a natural lift that a Hodge-theoretically inclined reader will recognize immediately.

We cite this thread for completeness; it is not a direct competitor but a reviewer may raise it as prior art on the cycle-space dimension count. The control-theoretic specialization — the necessity bound on feedback ingredients for asymptotic stabilization — is the piece the Hodge-theoretic line does not supply.

8.9 Gauge-equivariant geometric deep learning

A recent and very active thread in machine learning builds geometric deep networks around gauge symmetry. The formulation by [39] and its graph extensions by [40] and by [41] treat graph features as sections of associated vector bundles with gauge-equivariant message-passing operations. A December 2025 contribution [42] (“Gauge-Equivariant Graph Networks via Self-Interference Cancellation”) introduces a $U(1)$ phase connection with rank-1 projection as an attention mechanism.

The intersection with the present paper is formal — graphs with $U(1)$ -valued connections — but the use case differs: gauge-equivariant GNNs use the structure for function approximation, this paper uses it for closed-loop control of a physical system. We note the thread because a reviewer familiar with geometric deep learning will recognize the gauge transformation rule and may ask about the relationship. The structural answer is that both lines operate on the same abstract object (graph with $U(1)$ connection) under the same symmetry group (node-wise $U(1)$ transformations), but neither subsumes the other: gauge-equivariant GNNs do not perform closed-loop control, and this paper does not learn features.

8.10 Summary of contributions relative to related work

With the full citation landscape in view, the contributions of the present paper can be stated honestly:

- (i) *Identification of the actuated-edge regime* (§ 2.2) as a distinct problem class within cellular-sheaf coordination, not a special case of the node-actuated regime developed by [2] and [3].
- (ii) *Cycle-feedback budget theorem* (§ 3.5) establishing $b_1(G)$ as a tight two-sided bound on the number of scalar feedback ingredients required for asymptotic stabilization of the trivial connection in this regime. The upper bound ($\dim H^1 = b_1(G)$) is sheaf-cohomology folklore; the matching necessity lower bound in the control-theoretic sense appears to be new.
- (iii) *Canonical Wilson form + long-wavelength reduction* (§ 3.1, § 4) establishing the single-parameter network-wide κ as the unique long-wavelength coupling by application of Migdal–Kadanoff renormalization-group recursion to the actuated-edge Higgs action.
- (iv) *Kron-composition theorem* (§ 3.7): active-link extension of the [5] graph-Laplacian Kron reduction to the gauge-covariant setting, with the cycle space and loop holonomy preserved under interior-node elimination.
- (v) *Engineering instantiation and validation* (§§5–7): identification of PST, FACTS, UPFC, B2B, MMC, dual-active-bridge, and grid-forming-inverter installations as instances of the actuated- $U(1)$ link variable; a deployment pathway through monitoring, HIL, and field trials; and empirical validation against a Verboomen-class OPF coordination baseline on the IEEE 118-bus test system with five phase-shifting transformers. This engineering layer builds on (and is not claimed as a replacement for) the loop-invariant physics developed by [4] and collaborators (§ 2.1) and the coordinated-PST scheduling line of [8] and [30] (§ 8.6); the novelty is in the actuated-edge control formulation, its composition with existing scheduling layers, and the calibration-robust diagnostic interpretation of the holonomy (§ 7.2).

The first four contributions are small-to-moderate extensions of established frameworks; the fifth is the load-bearing engineering contribution and is where the paper earns its keep relative to both the distributed-control and the power-systems literatures.

9 Scope and limitations

Where the framework applies.

- PST-equipped transmission meshes with border-flow coordination across control areas (European ENTSO-E and PJM footprints; the largest deployed instance).
- UPFC- or other FACTS-equipped meshes where the series phase injection is an independently settable link variable.
- Meshed back-to-back VSC and MMC installations (Zhangbei-type, offshore wind collector grids, microgrid interconnections).
- DC distribution systems with meshed redundancy and dual-active-bridge converters.
- Grid-forming-inverter-rich meshed distribution feeders where link-level phase is actively set.
- Any network where at least one mesh loop contains an active-phase link element as defined in § 1.

Where it does not apply.

- Radial networks (no loops, no holonomy).
- Standard AC transmission/distribution with only passive lines and no PST/FACTS/B2B element on any cycle (§ 2.3).
- Bus-connected inverters sharing an AC bus without independent link phase.
- Single-device or point-to-point installations (no mesh).

Modeling assumptions. The averaged model (§ 6.3.1) operates at coordination timescales ($\sim 0.1\text{--}1$ Hz) and does not represent PWM switching, inner current-loop dynamics, PLL synchronization, DC-bus voltage regulation, or converter current limits. These are standard outer-loop approximations appropriate for the coordination-layer analysis of §§3–5 but require HIL validation before field deployment; § 7.4 describes the C-HIL / switching-model HIL / bench-and-field staging that addresses this.

What is claimed, precisely. The loop-flow coordination problem on meshed networks with active-phase link elements is an instance of the *actuated-edge regime* of cellular-sheaf coordination (§ 2.2). For this regime: (i) the cycle-space dimension $b_1(G) = m - n + c$ is both the number of independent gauge-invariant loop observables and a tight two-sided bound on the number of scalar feedback parameters required to asymptotically stabilize the trivial-connection equilibrium (§ 3.5); (ii) the bound is attained by the canonical $\kappa_j(1 - \cos \Theta_{C_j})$ feedback per basis cycle (§ 3.1); (iii) a Migdal–Kadanoff renormalization-group argument on the compact- $U(1)$ Higgs action collapses the $b_1(G)$ per-cycle parameters to a single network-wide κ at long wavelengths (§ 4). The framework composes with graph-Laplacian Kron reduction (§ 2.5, § 3.7): cycle space and loop holonomies are preserved under Schur-complement elimination of passive interior nodes. The resulting controller, deployed as the inner loop between existing outer-layer coordination at PMU-telemetry cadence, suppresses circulating power on the 3-bus back-to-back triangle by $\sim 3\times$ against an uncoordinated droop baseline (§ 6.3). Deployment-scale acceptance on PST-grounded topologies up to IEEE 118-bus against the Verboomen / Van Hertem OPF coordination baseline is specified in § 6.4 as a six-test acceptance suite, with evidence status tracked in § 6.5.

What is not claimed. No claim is made about quantum field theory, about QCD, about confinement in the thermodynamic-limit sense, or about any new physical symmetry of nature. The gauge symmetry here is an engineered equivariance of the control model — absolute bus phase is a modeling convention, not a physical observable — not a fundamental redundancy.

The abstract framework of § 2 is substantially prior work: cellular-sheaf coordination [1], [2], [3], discrete gauge theory on graphs [17], [18], consensus on Lie groups [20], and group synchronization [14], [16] together supply the background machinery. § 8 positions the paper relative to each of these. The contribution of this paper is specialization (to the actuated-edge regime with $U(1)$ connections), a control-theoretic necessity bound on the feedback budget for this regime, and an engineering instantiation on deployed-scale power-systems topologies; not the abstract framework itself.

Non-Abelian ($SU(2)$) extension for unbalanced operation and stacked-3D Fradkin–Shenker phase structure are documented in Appendices B and C as structural possibilities within the framework; neither is demonstrated on engineered hardware in this work.

Operational non-displacement. Three coordination mechanisms that exist today do particular things the gauge-feedback controller does not replace in deployment. Each continues to operate in the deployed stack; the gauge controller is layered, not substituted.

PST optimal power flow [7], [8]: centralized, schedule-cadence optimization that handles economic dispatch, N-1 security, tap-range budgeting, and market commitments simultaneously. Gauge feedback does not replace any of this; it closes the real-time loop between OPF solves (§ 7.1) against disturbances that arrive faster than schedule cadence. At steady state ($\Theta_C = 0$) the gauge feedback vanishes and the OPF-scheduled tap positions are held exactly. Test ζ of § 6.4 formalizes this as a static non-displacement acceptance criterion.

Pilot-voltage droop [6]: active-power-mismatch coordination on multi-terminal HVDC grids, with DC voltage as the coordination target rather than a loop holonomy. Gauge feedback does not replace pilot-voltage droop on that problem; the two operate at different coordination layers and can coexist on an MTDC-embedded meshed AC system (gauge on the AC reactive-loop layer, pilot droop on the DC active-power layer). The $\sim 3\times$ suppression figure of § 6.3 is measured against an uncoordinated droop baseline, not against pilot-voltage droop.

Graph-Laplacian Kron reduction [5]: the graph-theoretic treatment is used verbatim and composes with the gauge framework on networks with active-phase link elements (§ 2.5, § 3.7). In the passive-only setting addressed by that work, the link variables vanish and the gauge observable is identically zero; the contribution here is the active-link extension of the preservation properties, not a reframing of the underlying Laplacian machinery.

§ 8 develops the positioning against these and other related lines in full.

A Appendix A — Rosetta dictionary: cellular sheaves, lattice gauge theory, converter networks

The framework this paper operates in — a graph with $U(1)$ -valued edge parameters, gauge-invariant loop observables, and gradient-flow dynamics on the reduced configuration space — is the same algebraic object in at least three distinct research traditions. This appendix makes the correspondence explicit. The purpose is not to establish priority (the abstract framework is substantially prior art, as § 8 documents) but to license the reader to translate freely between languages: a statement proved in one column is valid in every column by direct substitution.

A.1 Core correspondence

Cellular sheaf / sheaf coordination	Compact- $U(1)$ lattice gauge theory	Converter network (this paper)
Cell complex / graph $G = (V, E)$	Spacetime lattice	Power-grid topology
Vertex stalk $\mathcal{F}(v)$	Matter-field site at n	Bus n
Vertex stalk value x_v	Higgs field value A_n	Bus voltage phase δ_n
Edge stalk $\mathcal{F}(\ell)$	Link between adjacent sites	Transmission / converter branch ℓ
Restriction map parameter	Link variable $U_\ell \in U(1)$	Active-link phase setting $e^{i\theta_\ell}$ (PST series injection, converter phase command, FACTS phase shift)
Coboundary δx at edge ℓ	Covariant difference $A_m - U_\ell A_n$	Branch mismatch $\phi_\ell = \delta_m - \delta_n - \theta_\ell$
Gauge group action $\mathcal{G}^V$	Gauge transformation $A_n \rightarrow g_n A_n$, $U_\ell \rightarrow g_m U_\ell g_n^{-1}$	Per-bus reference drift $\delta_n \rightarrow \delta_n + \alpha_n$, $\theta_\ell \rightarrow \theta_\ell + \alpha_m - \alpha_n$
$H^1(G, \mathcal{F})$ cocycle representative	Wilson loop $W_C = \prod_{\ell \in C} U_\ell$	$e^{i\Theta_C}$, $\Theta_C = \sum_{\ell \in C} \theta_\ell$
$\dim H^1(G, \mathcal{F})$ for constant $U(1)$ sheaf	Number of independent plaquettes	Cycle-feedback budget $b_1(G) = m - n + c$
Nonlinear edge potential	Matter action $\ D_U A\ ^2$	$\sum_\ell (V^2/X_\ell)(1 - \cos \phi_\ell)$
Loop potential	Plaquette action $\beta(1 - \text{Re Tr } U_C)$	$\sum_C \kappa_C(1 - \cos \Theta_C)$
Nonlinear sheaf diffusion	Euclidean gauge-field EOM	$\tau \theta_\ell = (V^2/X_\ell) \sin \phi_\ell - \sum_{C \ni \ell} \sigma_\ell^{(C)} \kappa_C \sin \Theta_C$
Sheaf-Laplacian Schur complement	—	Kron reduction; cycle space and holonomy preserved (§ 3.7)
Harmonic-section equilibrium ($\delta x = 0$)	Flat-connection vacuum	Zero-circulating-power mesh

The sheaf-coordination column instantiates the framework as it appears in [1] and [2]. The lattice-gauge column instantiates the compact- $U(1)$ Higgs lattice action as formulated by [43] and [44]. The converter-network column is the present paper. The correspondence is exact in every row.

One row deserves elaboration. The *gauge-covariant gradient-flow control law* in the rightmost column is the dynamical equation derived in § 3; in the cellular-sheaf column it is the sheaf-diffusion operator developed by [3] for the node-actuated case, adapted to the actuated-edge regime (§ 2.2). The pure-physics column has no direct “controller” analogue — lattice gauge theory is conventionally studied as an equilibrium statistical-mechanics problem, not a closed-loop control problem — but the Euclidean field equations of motion play the same structural role in the Monte Carlo simulations used to validate the renormalization-group analysis of § 4 (mc3d_v2.py in Appendix E).

A.2 Additional instantiations

The actuated- $U(1)$ link-variable structure appears in several further contexts. None is worked out in detail in this paper; they are listed to document the breadth of the abstract framework and to offer entry points for readers who recognize their own domain.

Synchronization of coupled oscillators with actuated couplings. Kuramoto-type networks with *controllable* phase offsets on each coupling ($\dot{\theta}_i = \omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i - \alpha_{ij})$ with α_{ij} actuated) carry an exact $U(1)$ link structure on the actuated offsets α_{ij} . The holonomy around a loop is the topological frustration in the loop phase assignment; the cycle-feedback budget of § 3 applies directly. Biological contexts include neural synchrony networks with plastic synaptic delays [45], [46]; engineering contexts include robotic formation control on rings and meshes [47], and distributed clock synchronization with programmable per-link delays.

Programmable photonic meshes. Mach–Zehnder interferometer (MZI) arrays are universally configurable unitary transformations on multi-port inputs, with per-MZI phase shifters as the actuated variables. Triangular [48] and diamond [49] topologies yield meshes with loops; the loop holonomy is the closed-path optical phase, and controller designs for these meshes that eliminate closed-path interference effects are implementing the gauge-feedback controller in a photonic instantiation. The $U(1)$ structure is explicit in the per-MZI phase setting and exact (no gauge-theoretic specialization required). Recent deep-learning frameworks such as [42] operate in the same setting with $U(1)$ phase connections on graph-structured data.

RF and microwave phased arrays. Multi-element beamforming networks with controllable per-element phase delays on a connectivity graph realize the actuated- $U(1)$ structure when the array geometry is non-tree. Loop holonomies around closed feed paths correspond to beam-pattern deformations that cannot be compensated by element-level phase setting alone; coordinated compensation requires the cycle-feedback machinery developed here. The engineering literature on such networks uses the vocabulary of feed-network calibration rather than gauge theory, but the algebraic object is the same.

Clock synchronization with engineered per-link offsets. Network time-protocol variants with *actuated* per-link delay (PTP with programmable delay compensation, certain white-rabbit variants) are formally actuated- $U(1)$ systems in which the link variable is the per-link delay offset and the holonomy around a network cycle is the accumulated clock drift around the loop. The cycle-feedback budget bound is a constraint on the minimum number of coordinated adjustments required for consistent network time, independent of the number of nodes and edges.

Distributed formation control with programmable relative-pose offsets. In formation control with a designated per-edge target pose offset $(r_{ij}, \alpha_{ij}) \in SE(2)$, the configuration is an $SE(2)$ -valued connection on the formation graph. The non-Abelian extension of § 3 (and Appendix B for the $SU(2)$ case) governs the cycle-feedback budget. [19] is the closest existing treatment; the actuated-link version is the same structural problem.

Tensor-network bond phases and graph-state quantum circuits. Matrix-product-state and tree-tensor-network architectures on graph-structured bond spaces, and graph-state preparations in measurement-based quantum computation, carry $U(1)$ gauge structure at the bond level when the bond-phase degrees of freedom are treated as control variables. The cycle-feedback budget applies as a lower bound on the number of independent tuning phases in circuit compilation.

Group synchronization over cycle-inconsistent measurements. Rather than actuating the link variables, one can measure noisy versions of them and attempt to recover node values consistent with the measurements. This is the *inverse* problem treated by [25], [14], [15], and [16] (the cycle-edge message passing framework). The cycle-consistency machinery is dual to the cycle-feedback control developed here; a network that admits actuated-edge control with $b_1(G)$ scalar tunings simultaneously admits group-synchronization estimation with $b_1(G)$ independent cycle-consistency constraints.

A.3 What the multi-domain dictionary licenses

The analytical tools developed for any one column transfer by direct substitution:

- Schur-complement / Kron-reduction machinery from graph theory (§ 3.7; [5] for the passive case) transfers to interior-node elimination in any of the above settings.
- Migdal–Kadanoff renormalization-group analysis (§ 4; [50],
 - 1) transfers to long-wavelength reductions in any 3D actuated- $U(1)$ mesh regardless of physical domain.
- Fradkin–Shenker phase-classification machinery (Appendix C;
 - 2) transfers to the thermodynamic analysis of any stacked / high-dimensional instantiation.
- Non-Abelian extension to compact Lie groups (Appendix B) covers polarization-mixing photonic meshes, unbalanced converter networks, and $SE(2)/SE(3)$ formation control uniformly.

The correspondence is algebraic, not metaphorical. Writing the paper for the converter-network column is an editorial choice; the mathematics would be identical with any of the right-hand columns substituted in.

B Appendix B — Non-Abelian (SU(2)) extension

The main text operates in the $U(1)$ specialization of the actuated-edge regime (§ 2.2) — each link variable is a scalar phase offset $\theta_\ell \in [0, 2\pi)$, appropriate to balanced three-phase operation. This appendix lifts the framework to compact-non-Abelian link variables, specifically $SU(2)$, which is the structurally relevant case for unbalanced operation with independent control of positive- and negative-sequence components. The Wilson-loop observable, closure of dynamics on the gauge quotient, and cycle-feedback budget all extend with the substitutions recorded in Appendix A (abelian $\rightarrow$ non-Abelian column); the cycle-space dimension counts independent loop observables as $\dim \mathfrak{g} \cdot b_1(G) = 3b_1(G)$ for $SU(2)$ rather than $b_1(G)$ in the $U(1)$ case.

The physical requirement for an $SU(2)$ link variable is a converter that can set each sequence component’s phase independently — a matrix converter, a four-leg inverter, or an MMC operated with independent sequence-component dispatch.

Construction. Link variables $U_\ell \in SU(2)$, bus phases $\psi_n \in \mathbb{C}^2$ (2-spinors over the sequence subspace). The Wilson-loop observable $\text{Tr } U_C$ is gauge-invariant under local $SU(2)$ transformations. Gradient flow drives $U_C \rightarrow I$. Numerical verification (`su2.py`):

- Gauge invariance: 1.4×10^{-16} under random $SU(2)$ transforms.
- Gradient flow: plaquette energy $\rightarrow 0$ over 13 orders of magnitude.
- Spinor double cover: a 2π rotation of link variables flips the sign of $\text{Tr } U_C$; only a full 4π returns the identity. Closure error at 4π : 0.00.

Wilczek–Zee identification. The $SU(2)$ plaquette holonomy is operationally identical to the non-Abelian Berry phase of [51]: the bus 2-spinor parallel-transported around the loop. The mesh-plaquette construction reproduces any $\mathfrak{su}(2)$ -valued connection to 3.1×10^{-15} (`wz_match.py`).

Conductor-7 worked example (`conductor7.py`, `convergence.py`, `ws_berry_mesh.py`). Using the value-based correlation $C(t, \sigma)$ derived from the Dirichlet L-function $L(s, \chi_7)$, a $\{+, -\}$ near-crossing appears at $(t, \sigma) \approx (16, 2.6)$. Eigenframe parallel transport around shrinking loops converges monotonically toward the conical-intersection value $\theta \rightarrow \pi$; the converter-mesh plaquette trace matches the eigenframe holonomy at every loop scale. The zero-based (density) correlation preserves $\{+, -\}$ degeneracy exactly and yields null holonomy everywhere — a “Fortescue protection” property that is structural and survives into the number-theoretic pipeline.

Engineering status. The non-Abelian construction is mathematically complete and numerically verified against both generic and number-theoretic test data. Deployment on engineered hardware is an open problem: it requires $SU(2)$ -capable link elements (matrix converters, 4-leg inverters, MMCs with independent sequence dispatch, or a sequence-aware PST) operating coherently across a mesh, which has not been demonstrated. The $U(1)$ controller of the main text is sufficient for balanced-operation meshes with active-phase link elements — PST-equipped transmission loops, B2B/MMC converter meshes, meshed microgrids — which are the deployed-scale target applications.

C Appendix C — Fradkin–Shenker phase classification and stacked-3D topologies

The compact-U(1) Higgs lattice action admits the standard Fradkin–Shenker phase classification: in 3D, a sharp critical surface in (β_g, β_m) separates a Higgs/screened phase from a confined/frustrated phase; in 2D, no sharp transition, only a smooth crossover controlled by the screening length $\xi = \sqrt{\kappa/\beta}$.

2D Wilson-loop crossover. Fourier-space computation of $-\log\langle W(R) \rangle$ on a 128^2 torus across $\kappa/\beta \in [0.25, 64]$ confirms the area $\rightarrow$ perimeter crossover at $R \sim \xi$ (`wilson_2d.py`). Consistent with pure Gaussian integrability — no sharp transition in 2D.

3D Fradkin–Shenker transition. Monte Carlo on a 10^3 lattice with compact-U(1) + Higgs in unitary gauge, scanning β_g at $\beta_m \in \{0.30, 0.55, 0.80\}$, shows a susceptibility peak at $\beta_{g,c} \approx 1.05$ for $\beta_m = 0.80$ — a factor of ~ 10 above the 2D baseline (`mc3d_v2.py`, `mc_2d_vs_3d.py`). Quantitatively consistent with the MK prediction of § 4.

Engineering relevance. This matters for *stacked* or *layered* mesh topologies — PST-coordinated transmission with multiple voltage levels (220 / 380 / 500 kV), HVDC overlays on AC grids, multi-region interconnects, microgrid hierarchies — where inter-layer coupling is strong enough for the composite topology to be effectively 3D. Whether any deployed grid has sufficient inter-layer coupling to access the Fradkin–Shenker regime, and whether the transition would be observable as a qualitative change in circulating-power statistics, is an open question that requires specific topology and coupling data from real grids. The MK analysis of § 4 is sufficient for the engineering claim (single-parameter controller suffices at long wavelengths); the phase-classification content of this appendix is a structural possibility, not a current deployment consideration.

D Appendix D — Validation script index

Every demonstrated numerical claim in § 2–§ 6 and Appendices B–C and F traces to one or more scripts in `validation/`. Sources appear verbatim in Appendix E. Scripts marked *forthcoming* validate § 6.4 (Tests α' – ζ) and are added to `validation/` as each test lands; the table below is updated with the actual numerical results at that time.

Script	Validates	Method
<code>gauge_covariance_test.py</code>	§ 3.6	Three-bus ODE in two gauge choices; direct trajectory comparison Demonstrated.
<code>nbus_gauge_test.py</code>	§ 3.6 (at scale)	General ConverterMesh lattice class; verification on N-bus meshes with multiple plaquettes Demonstrated.
<code>kron_test.py</code>	§ 3.7	4-node / 5-edge / 2-plaquette test network; algebraic Schur complement vs. full dynamics under gradient flow Demonstrated.
<code>mk_rg.py</code>	§ 4	Bessel-kernel MK recursion $r(\beta) = I_1/I_0$; flow diagrams in (β_g, β_m) plane; 2D vs 3D decoupled vs 3D coupled Demonstrated.
<code>holonomy.py</code>	§ 5.1	Trivial vs winding-1 initial conditions on 3-bus loop; Langevin forcing to measure Arrhenius phase-slip rate Demonstrated.
<code>three_one.py</code>	§ 5.2	Linearized Hessian; Gaussian prediction; Langevin time-average; independent Metropolis MC; three-way cross-check Demonstrated.
<code>discrete.py</code>	§ 5.3	Single-plaquette analytical; 4×4 torus greedy vs constrained quantization; $N = 4$ –256 power-law fit Demonstrated.
<code>b2b_v2.py</code>	§ 6.3	3-bus B2B triangle averaged model; <code>solve_ivp</code> deterministic + Euler–Maruyama stochastic runs for Scenarios A–D Demonstrated.
<code>b2b_v3.py</code>	Appendix F	Minimal worked examples on the 3-bus B2B triangle vs. pilot-voltage droop; Tests α – δ (gauge invariance, communication independence, bounded feedback, topology invariance) Demonstrated.

Script	Validates	Method
pst_invariance_drift.py	Test α'	3-area PST mesh (6 buses, 6 branches, $b_1 = 1$); three boundary PSTs with ± 16-tap $\times 0.6^\circ$ discrete actuation; per-area reference drifts 0.12°–0.20° amplitude over 15-min window; closed-loop proportional controller. Three-part verification (algebraic telescoping inter-area disagreement = 0 over 900 samples; drift-only tap operations = 0; drift attribution over disturbance-only = 0%) Demonstrated.
opf_rejection_3area.py	Test β'	DE-PL-CZ-style 3-area mesh, 6 buses, 8 edges, $b_1 = 3$ (nominal) / $b_1 = 2$ (post-N-1); five boundary PSTs $(\alpha, \beta, \gamma, \delta, \epsilon)$; non-uniform reactances $X_\ell \in [0.08, 0.15]$ pu (essential to make the $\ Cf_{\text{dev}}\ _2$ metric injection-sensitive). Three scenarios over 15-min post-OPF windows: +3 GW DE wind ramp, internal PL line outage at $t = 5$ min, +1.0 pu cross-border DE→PL gate at $t = 7.5$ min. Discrete coordinate-descent controller on $\ Cf_{\text{dev}}\$ with 1 tap/PST/tick rate limit. Integrated loop-flow reductions: +84.9% (ramp, 8 tap ops), +66.5% (N-1, 2 tap ops), +59.4% (gate, 3 tap ops) Demonstrated.

Script	Validates	Method
<code>tap_wear_week.py</code>	Test γ'	β' topology + tap-changer dynamics; one simulated week (168 hourly samples) of profile-driven injections. Profile mapping: deviations of DE/PL/CZ load and wind around their weekly means, scaled at 10 GW per pu of injection deviation, applied around the β' nominal P_{NOMINAL} (CZ as slack). Discrete coordinate-descent controller swept over dead-band $\in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3\}$ pu and tick interval $\in \{60, 300, 900, 3600\}$ s (24 configurations); rate limit fixed at 1 tap/PST/tick. Pareto-dominant sweet spot: dead-band = 0.030 pu, tick = 60 s $\rightarrow$ 63.2% weekly integrated $\ Cf_{\text{dev}}\$ reduction at 1.51 ops/PST/day (vs. ENTSO-E operational range of 3–8/day). Pareto frontier (5 points) shows clean diminishing returns: trivial 0/0%, 1.0/47%, 1.5/63% Demonstrated.

Script	Validates	Method
fetch_entsoe.py	Test γ' (data)	Three-mode profile downloader writing the CSV schema consumed by <code>tap_wear_week.py</code>. <code>--source synthetic</code> (default): deterministic AR(1) wind generator with mid-week storm-ramp event and two-peak diurnal load model, calibrated to OPSD 2020 winter-week summary statistics; used to produce the bundled <code>entsoe_week_sample.csv</code>. <code>--source opsd:</code> no-credentials download from <code>data.open-power-system-data.org</code> (OPSD's funded maintenance ended December 2020; archive coverage is 2015–2020). <code>--source entsoe:</code> live ENTSO-E REST API via the <code>entsoe-py</code> wrapper; requires the free security token obtainable from <code>transparency@entsoe.eu</code> per the platform's standard registration process Provided.
pst_n1_topology.py	Test δ'	DE-PL-CZ-style 4-bus, 6-edge mesh with $b_1 = 3$; five boundary PSTs; outage of PL1-CZ1 branch at $t = 5$ min reduces $b_1 \rightarrow 2$, with simultaneous $+6^\circ$ exogenous command on PST_γ as stress case. Cycle-incidence matrix auto-recomputed by <code>networkx.cycle_basis</code>; single GaugeFeedbackController instance with fixed $\kappa = 0.40$, dead-band = 0.6°, rate-limit = 1 tap/tick. Pre-outage settling 61 s; post-outage settling 481 s (8.0 min, within 15-min budget); zero controller-parameter changes across outage Demonstrated.

Script	Validates	Method
<code>kron_composition_14bus.py</code>	Test ε	IEEE 14-bus (pandapower case14) + 3 PSTs on edges (0,4), (5,11), (9,10); Kron-reduce passive bus 6; three-part verification (algebraic holonomy identity = 0; equilibrium state identity = 4.2×10^{-17} rad; transient agreement with algebraic passive interior = 2.3×10^{-8} rad) Demonstrated.

Script	Validates	Method
<code>verboomen_118bus.py</code>	Test ζ	IEEE 118-bus (<code>pandapower.networks.case118</code>) with spectral 3-area partition (34/36/48 buses, 8 tie corridors) and 5 PSTs placed by PTDF-variance ranking with greedy linear-independence gate (PST_{α} bus 30–38, PST_{β} bus 81–80, PST_{γ} bus 23–24, PST_{δ} bus 77–82, PST_{ε} bus 77–78). 186 edges; cycle basis $b_1 = 62$ (simple-graph, control-relevant); $b_1 = 69$ counting parallel-edge 2-cycles. Reachability Jacobian $\partial L / \partial \theta_{\text{PST}}$ has rank $5 = n_{\text{PST}}$ with singular values $\{80.6, 42.2, 22.6, 18.5, 11.7\}$. Three scenarios over 15-min windows: static hold, fast-disturbance (ramp +300 MW at bus 88, 88.5% actuated; gate +200 MW at bus 89, 90.2% actuated; high-actuation locations selected by § 7.5-compliant filtering), N-1 outage (edge #173). Three conditions (OPF-alone, gauge-alone, OPF+gauge). <code>pandapower</code> <code>rundcopp</code> for one-shot OPF, in-house numpy DCPF for per-tick simulation, coordinate-descent gauge controller identical to β' / γ'. Acceptance: (i) static equilibrium match within 10^{-3} rad/loop — PASS (0 rad exactly); (ii) fast-event integrated loop-flow reduction $\geq 30\%$ — PASS at 45.9% (13 tap operations; 94% of the physical bound 48.6% set by the un-actuated-subspace floor); (iii) gauge-alone strictly worse than OPF-alone at static optimum — PASS (1.18 vs. 0.00 loop-flow). Bonus result: 22.5% N-1 reduction (6 tap ops, zero parameter changes, replicating δ' at 118-bus scale) Demonstrated.

Script	Validates	Method
<code>verboomen_118bus_network.py</code>	Test ζ (helper)	Companion module for <code>verboomen_118bus.py</code> : loads IEEE 118-bus, performs spectral 3-area partition (<code>sklearn.cluster.KMeans</code> on top-2 non-trivial Laplacian eigenvectors), identifies 8 tie corridors (with parallel-edge grouping for lines 113–114 on bus 77–80), computes 186×118 PTDF matrix, ranks corridors by row-norm variance, and greedily selects 5 linearly-independent PSTs. Installs PSTs on selected edges (lines replaced with <code>tap_phase_shifter=True</code> trafos preserving DC impedance; existing tie-line trafo promoted via <code>tap_phase_shifter</code> flag). Exports a <code>Network118</code> dataclass with incidence A , susceptance B_{diag} , reduced-Laplacian inverse, cycle basis C , $\partial L / \partial \theta_{\text{PST}}$, reachability rank and singular values. Standalone CLI runs the full build + reachability diagnostic in ≈ 2 s Provided .

Script	Validates	Method
<code>calibration_robust_diagnostic.py</code>	§7.9 (calibration robustness)	Demonstrates gauge invariance of the holonomy observable Θ_C under per-bus PMU calibration errors, using the Test ζ network. Part 1 (static): 1000 Monte Carlo draws of $\varepsilon_b \sim U(-2^\circ, +2^\circ)$; $\max_C \Theta_C^{\text{meas}} - \Theta_C = 1.4 \times 10^{-16}$ rad (machine-epsilon floor, ratio 0.62 to float64 eps); contrast with $L_C = C \cdot f$ residual of 8.6 pu (median $2.5 \times$ true signal) documenting that the weighted cycle-sum breaks the telescope under non-uniform b_ℓ . Part 2 (dynamic): 15-min Test ζ fast scenario with OPF+gauge active (13 tap operations), PMU noise modelled as $\pm 1^\circ$ static + $\pm 0.1^\circ/\text{min}$ drift per bus; holonomy trace tracks truth to 1.4×10^{-16} rad against 0.10 rad peak signal ($10^{-13}\%$ relative error), loop-flow trace 149% relative error, naive bus-angle trace 5% relative error. Three-way comparison plotted in <code>calibration_robust_dynamic.png</code> Provided.
<code>su2.py</code>	Appendix B	SU(2) 3-bus loop with Pauli-matrix link variables; gauge invariance under random transforms; gradient flow to flat connection; 4π double-cover test Demonstrated.
<code>wz_match.py</code>	Appendix B (generic)	Generic $\mathfrak{su}(2)$ -valued connection $\rightarrow$ mesh plaquette reproduction; time-reversal-symmetric null-holonomy check Demonstrated.
<code>conductor7.py</code>	Appendix B (conductor-7)	Dirichlet L-function $L(s, \chi_7)$ zeros; value-based vs zero-based correlation matrices; eigenframe transport Demonstrated.

Script	Validates	Method
<code>ws_berry_mesh.py</code>	Appendix B (full chain)	Number field $\rightarrow$ L-function $\rightarrow$ correlation $\rightarrow$ eigenframe $\rightarrow$ SU(2) holonomy $\rightarrow$ mesh plaquette, end-to-end Demonstrated.
<code>convergence.py</code>	Appendix B (convergence)	Shrinking-loop eigenframe transport around $\{+, -\}$ near-crossing of $L(s, \chi_7)$ Demonstrated.
<code>wilson_2d.py</code>	Appendix C (2D)	128^2 torus Fourier-space computation of $-\log\langle W(R) \rangle$ across $\kappa/\beta \in [0.25, 64]$ Demonstrated.
<code>mc_2d_vs_3d.py</code>	Appendix C (contrast)	Same compact U(1) + Higgs Monte Carlo in unitary gauge; 2D (24^2) vs 3D (10^3) parameter sweep Demonstrated.
<code>mc3d_v2.py</code>	Appendix C (3D)	3D $L = 10$ Fradkin–Shenker scan at $\beta_m \in \{0.30, 0.55, 0.80\}$; susceptibility peak location Demonstrated.
<code>saturation_vertex_tensor.py</code>	Archive (ruled-out motivation)	Symbolic expansion of cubic saturation vertex in Fortescue basis; verifies the negative result that passive saturation does not produce gauge structure Demonstrated.

E Appendix E — Validation scripts (verbatim)

The paper is a deterministic build: `make paper` regenerates this document, and the live chunk in § 3.6 re-executes `gauge_covariance_test.py` against the source below. Every numerical claim in the paper is traceable to one of these listings.

E.1 Validation script: `gauge_covariance_test.py`

Listing 1 `gauge_covariance_test.py`

E.2 Validation script: `nbus_gauge_test.py`

Listing 2 `nbus_gauge_test.py`

E.3 Validation script: `kron_test.py`

Listing 3 `kron_test.py`

E.4 Validation script: `mk_rg.py`

E.5 Validation script: `holonomy.py`

E.6 Validation script: `three_one.py`

E.7 Validation script: `discrete.py`

E.8 Validation script: `b2b_v2.py`

E.9 Validation script: `b2b_v3.py`

E.10 Validation script: `pst_invariance_drift.py`

E.11 Validation script: `opf_rejection_3area.py`

E.12 Validation script: `tap_wear_week.py`

E.13 Validation script: `fetch_entsoe.py`

E.14 Validation script: `pst_n1_topology.py`

E.15 Validation script: `kron_composition_14bus.py`

E.16 Validation script: `verboomen_118bus.py`

Listing 4 `mk_rg.py`

Listing 5 `holonomy.py`

E.17 Validation script: `verboomen_118bus_network.py`

E.18 Validation script: `calibration_robust_diagnostic.py`

E.19 Validation script: `su2.py`

E.20 Validation script: `wz_match.py`

E.21 Validation script: `conductor7.py`

E.22 Validation script: `ws_berry_mesh.py`

E.23 Validation script: `convergence.py`

E.24 Validation script: `wilson_2d.py`

E.25 Validation script: `mc_2d_vs_3d.py`

E.26 Validation script: `mc3d_v2.py`

Listing 6 `three_one.py`

Listing 7 `discrete.py`

F Appendix F — Minimal worked examples on the 3-bus B2B triangle: Tests α – δ against pilot-voltage droop

This appendix preserves the earlier version of the paper’s structural-test suite — four tests (α , β , γ , δ) run on the 3-bus B2B triangle of § 6.3 against [6] pilot-voltage droop as the coordinated prior-art baseline. The material here is unchanged from that earlier version. Its role in the current paper is didactic: the 3-bus triangle is the smallest mesh where every feature of the gauge-covariant structure (local symmetry, covariant dynamics, plaquette feedback, Wilson-loop observable) is present, so it is the cleanest place to make each structural claim visible in a form that a reader can run in seconds and inspect line by line. The load-bearing deployed-scale tests are α' – ζ in § 6.4.

Two framing points apply throughout this appendix.

Scope honesty. Pilot-voltage droop is designed for active-power sharing on multi-terminal DC grids, where the shared communicated scalar is a DC-bus voltage; gauge-covariant feedback is designed for reactive-loop suppression on AC meshes, where the shared observable is a loop holonomy. The schemes address different coordination problems at different network layers. What is compared here is not “gauge vs. pilot droop at pilot droop’s own game” (a category error, since pilot droop has no loop-reactive observable) but a set of *structural properties* — invariance, local observability, bounded feedback, topology robustness — on which the gauge construction makes predictions and the pilot-droop construction either does not, or makes the opposite prediction.

Methodological echo. Berggren’s Tables II/IV (estimate vs. simulation) are the state-of-the-art format for validating a droop methodology: make an analytic prediction *before* running the simulation, then tabulate prediction against measurement. Each test below adopts this format. Predicted values are filled in where the theory of § 2–§ 5 supports a closed-form estimate; simulated values are produced by `b2b_v3.py` (Appendix D, listed verbatim as E.9). Results below are from an actual run; all four tests meet their acceptance criteria with the thresholds documented in each subsection.

F.1 Test α — Gauge invariance under per-bus reference drift

Purpose. Verify the proposition of § 2.4 on the full nonlinear 3-bus dynamics, and establish that the loop holonomy is a structurally-robust coordination observable in a way that pilot voltage is not.

Setup. 3-bus B2B triangle of § 6.3, Scenario A baseline conditions ($\tau = 2.0$ s, load step at bus 1 of 0.6 pu at $t = 10$ s). Inject independent slow phase-reference drifts at each bus: $\alpha_0(t) = 0.02 \sin(0.10 t)$, $\alpha_1(t) = 0.03 \sin(0.13 t + \pi/4)$, $\alpha_2(t) = 0.01 \sin(0.07 t)$, applied as additive biases on the bus angles entering the controller.

Analytic prediction. The proposition of § 2.4 is a mathematical identity: $\sum_{\ell \in C} (\alpha_{s(\ell)} - \alpha_{t(\ell)}) \equiv 0$ by telescoping, for arbitrary per-bus drifts $\alpha_n(t)$. A numeric check of the identity at 300 evenly spaced time samples over 30 s gives residual 5.2×10^{-18} rad — machine precision, as expected.

The operational corollary is stronger and not a pure identity: the controller’s internal θ_ℓ states must *chase* a time-varying drift, so the achievable attenuation of $\Delta\Theta_C$ against $\alpha_n(t)$ is bounded below by the link-dynamics chase error $\omega_\alpha \tau / (V^2/X)$ per link. For the parameters here ($\alpha_{\max} = 0.03$ rad, $\omega_\alpha \lesssim 0.13$ rad/s, $\tau = 2$ s, $V^2/X \geq 3.33$ pu), this predicts a chase-error floor of $\sim 10^{-3}$ rad summed around the loop.

A pilot-voltage droop analogue with voltage-reference bias of equivalent magnitude ($\delta V/V \sim 2 \times 10^{-2}$) would produce a steady-state sharing error of order $\delta V_{\text{pilot}}/D_k$ per terminal; with Berggren’s $D_k = 0.04$ kV/MW

Listing 8 b2b_v2.py

Listing 9 b2b_v3.py

and a 6 kV bias, this is ~ 150 MW per terminal — comparable to the total mismatch his methodology is designed to share.

Metric	No drift	Test α (with drift)	Result
Steady-state $ \Theta_C $, rad	0.00397	0.00707	within chase-error floor
RMS $ \Theta_C $ over 60 s, rad	0.00397	0.00707	$1.78\times$ (noisier transient under drift)
Max drift-induced $\ \Delta\Theta_C\ $, rad	0	3.69×10^{-3}	—
Attenuation max $ \Delta\Theta_C /\alpha_{\max}$	—	0.123	$\geq 4\times$ suppression of drift
Telescoping identity residual, rad	—	5.2×10^{-18}	machine precision

Acceptance. Attenuation ratio ≤ 0.25 (four-fold or better suppression of per-bus drift in the loop observable). **PASS:** measured 0.123. The telescoping identity is verified to machine precision. The operational robustness margin comes from the gauge controller’s link states absorbing drift through the covariant transformation law $\theta_\ell \rightarrow \theta_\ell + \alpha_{s(\ell)} - \alpha_{t(\ell)}$, up to a finite- τ chase error.

F.2 Test β — Communication independence

Purpose. Demonstrate that the gauge observable is locally computable, addressing the fallback-control gap that [6] (, §I) acknowledges but does not fill for communicated coordination schemes.

Setup. 3-bus triangle, Scenario D conditions ($\tau = 2.0$ s, load step + persistent $\sigma = 0.15$ pu noise). Two controller implementations:

- *Centralized.* Each converter receives Θ_C from a central aggregator at 10 Hz update rate.
- *Distributed-local.* Each converter computes Θ_C from its own local phase measurement and the θ values most recently received from its two neighbors on the ring (one scalar per link, 10 Hz, ≤ 100 ms latency); no central aggregator.

Analytic prediction. At the coordination timescale ($\tau = 2.0$ s), both implementations compute the same $\sum_\ell \theta_\ell$; steady-state values are identical, and transient response differs only by the maximum peer-to-peer propagation delay (~ 100 ms across a 3-node ring, much less than τ).

The pilot-voltage droop analogue on a hypothetical communicated equivalent of this network would, under comm-loss, fall back to local DC-voltage droop whose sharing accuracy is bounded by the DC-voltage spread across the grid. Berggren (§I) notes this explicitly: “*suffers to some extent from the circumstance that the dc voltage magnitude varies across the dc grid due to resistance in the dc transmission.*”

Listing 10 `pst_invariance_drift.py`

Listing 11 `opf_rejection_3area.py`

Metric	Centralized gauge	Distributed-local gauge	Result
RMS $ \Theta_C $, rad	0.00501	0.00502	+0.17% degradation
Peak circulating power, pu	0.0481	0.0483	+0.4%
Comm latency tolerated	0 ms	100 ms peer-to-peer	—
Inter-terminal bandwidth	3× scalar @ 10 Hz, aggregator	2× scalar @ 10 Hz, nearest-neighbor	—

Acceptance. Distributed-local RMS holonomy within 10% of centralized. **PASS:** measured degradation is +0.17%, two orders of magnitude inside the acceptance threshold. At the coordination timescale $\tau = 2$ s, a 100 ms peer-to-peer delay is 5% of the controller’s own time constant and produces negligible performance degradation. The gauge observable is thus a viable local-measurement fallback — the property Berggren calls out as needed and does not provide — available on the gauge side at no performance cost.

F.3 Test γ — Intrinsic boundedness of periodic feedback

Purpose. The gauge feedback $-\kappa \sin \Theta$ is bounded by $|\kappa|$ for all Θ , by the sine bound. The linear-proportional analogue $-\kappa \Theta$ is unbounded and requires an explicit saturator with anti-windup (AW) logic for robust operation under large $|\Theta|$. This test quantifies the engagement of that saturator and contrasts it with the gauge controller’s intrinsic boundedness.

Setup. 3-bus triangle, $\tau = 1.0$ s, initial condition $\Theta(0) = -4.0$ rad — beyond the $|\Theta| = \pi$ point where a linear feedback first exceeds its conservative saturation bound $\kappa\pi$. Three controllers on identical dynamics:

- *Droop-only.* No loop-level feedback (baseline).
- *Linear-plaquette with anti-windup.* Feedback $\text{sat}(-\kappa\Theta, \pm\kappa\pi)$ with conditional-integration AW (the conservative textbook choice).
- *Gauge (sin feedback).* $-\kappa \sin \Theta$, as in § 3.2.

Analytic prediction. For $|\Theta| > \pi$, the linear feedback’s raw magnitude $\kappa|\Theta|$ exceeds the saturator bound $\kappa\pi$, so the saturator engages and the integrator cannot accumulate error freely; this is what the AW logic exists for. The gauge feedback’s magnitude $\kappa|\sin \Theta|$ never exceeds κ by construction, so no saturator is needed. Both controllers drive the system to $\Theta = 0$ (the unique physical attractor for this topology, since $\sum_{\text{loop}} (\delta_{s(\ell)} - \delta_{t(\ell)}) \equiv 0$); the linear controller reaches the attractor faster because its effective gain near $\Theta = 0$ is larger by a factor of $1/\sin(\pi/4) \approx 1.41$ in the saturation regime and grows further in the pre-windup phase.

Metric	Droop-only	Linear-plaq. + AW	Gauge (sin)
Settling time to $ \Theta < 0.1$, s	0.90	0.28	0.41

Metric	Droop-only	Linear-plaq. + AW	Gauge (sin)
Peak applied feedback $\ fb_{\text{applied}}\ $	0.00	9.42 ($= \kappa\pi$, saturated)	3.00 ($\leq \kappa$ by sine bound)
Peak raw (pre-saturation) feedback	0.00	12.00 ($> \kappa\pi$; AW engaged)	3.00 ($\equiv$ applied)
Saturation events during transient	0	77 (AW load-bearing)	0

Acceptance. Gauge peak applied feedback $\leq \kappa$ (confirming intrinsic boundedness without a saturator), AND linear raw feedback $> \kappa\pi$ during transient (confirming the anti-windup logic is actively engaged, not merely present). **PASS:** gauge peak $|fb| = 2.998 \leq \kappa = 3.0$; linear raw peak $= 12.00 > \kappa\pi \approx 9.42$ with 77 saturation events over the transient.

Honest caveat. A well-tuned AW linear controller achieves a faster settling time than the gauge controller on this single-well problem, at the cost of carrying an explicit saturator and AW logic. The gauge claim is structural: *bounded, periodic feedback is inherited from the control law rather than engineered around it*. This matters for deployment — removing a hand-tuned AW block removes a commissioning degree of freedom and a known failure mode (AW limit-cycle when the saturator chatters). Settle-time is not the primary axis on which the gauge construction wins here.

F.4 Test δ — Topology invariance at runtime

Purpose. Demonstrate that runtime topology changes do not require controller re-parametrization, in contrast to droop-sizing methodologies that require day-ahead recomputation of $\{T_k, D_k\}$.

Setup. 3-bus B2B triangle ($\tau = 2$ s). A balanced load redistribution at $t = 3$ s (+0.5 pu on bus 2, −0.5 pu on bus 0; total load unchanged) drives the system to a non-trivial phase-spread equilibrium. At $t = 5$ s, a fourth converter switches into the loop, splitting the X_{01} branch into two series-connected effective branches through a new intermediate bus 3 (with $X_{03} = X_{31} = 0.075$ pu). The loop is now 4-linked. Controller parameters (κ, τ) are unchanged throughout.

Analytic prediction. The gauge feedback law $\tau\dot{\theta}_\ell = (V^2/X_\ell)\sin\phi_\ell - \kappa\sin\Theta_{C(\ell)}$ is unchanged by the topology event. Θ_C becomes a sum over the new link set ($L = 4$ rather than 3); κ is unchanged. The new equilibrium has a *different numerical value* for the individual θ_ℓ (the symmetry of the triangle is broken by the 4-link phase) but the same $\Theta_C \approx 0$ target. Prediction: settling within $\sim 3\tau = 6$ s with no controller reconfiguration.

A pilot-voltage droop scheme on a comparable topology change would require a full re-planning cycle (Berggren §V, Figure 4): recompute $\{T_k\}$, recompute $P^{\max}$, recompute $\{D_k\}$, push to each controller. This is a day-ahead-planning operation, not a real-time control action.

Metric	Pre-switch ($3 < t < 5$ s)	Post-switch ($t > 8$ s)	Result
Steady-state $ \Theta_C $, rad	0.00604	0.01682	$< 10^{-2}$ in both phases
Post-switch settling time, s	—	1.07	$\leq 3\tau = 6$ (PASS)
Controller reconfigurations	0	0	0 (zero, by construction)

Metric	Pre-switch ($3 < t < 5\text{s}$)	Post-switch ($t > 8\text{s}$)	Result
Droop-scheme analogue reconfig.	0	1 re-plan	—

Acceptance. Zero controller-parameter changes across the topology event, with settling time under $3\tau = 6$ s. **PASS:** measured settling time 1.07 s, six-fold inside the acceptance threshold; controller parameters unchanged. The pilot-droop analogue on the same event would require a full re-planning cycle to match this behavior.

F.5 Summary — α - δ results and the claims they license

All four tests meet their acceptance criteria. The structural claims defensible from these results are:

- *Gauge invariance (Test α)*. The loop holonomy Θ_C is invariant under arbitrary per-bus time-varying phase-reference drift — the telescoping identity is exact to machine precision (5.2×10^{-18} rad). The operational corollary is that the nonlinear closed-loop response to a drift of amplitude $\alpha_{\max} = 0.03$ rad exhibits attenuation $\Delta\Theta_C/\alpha_{\max} = 0.123$ — four-fold suppression, bounded below by the finite- τ link-state chase-error floor. No linear-droop coordination scalar is structurally immune to per-bus drift of this class.
- *Local observability (Test β)*. The gauge observable is locally computable at negligible cost: RMS $|\Theta_C|$ under Scenario D conditions degrades by only +0.17% when the centralized aggregator is replaced by nearest-neighbor peer-to-peer exchange with 100 ms latency. [6, Sec. I] acknowledges the need for a local-measurement fallback for communicated schemes; the gauge observable is such a fallback with essentially zero performance cost.
- *Bounded feedback (Test γ)*. Under $\Theta(0) = -4$ rad, the linear-plaquette controller’s raw feedback magnitude reaches 12.0 — 27% above its $\kappa\pi = 9.42$ saturation bound — and engages anti-windup 77 times during recovery. The gauge controller’s feedback magnitude peaks at $\kappa = 3.00$ exactly (sine bound), with zero saturator events. Boundedness is structural, inherited from the control law, not engineered in.
- *Topology invariance (Test δ)*. A 3-bus $\rightarrow$ 4-bus runtime topology change settles in 1.07 s ($< \frac{1}{5}$ of the 3τ acceptance threshold) with zero controller-parameter changes. The steady-state $|\Theta_C|$ remains below 10^{-2} rad in both phases. A pilot-droop analogue on the same event would require a full re-planning cycle.

No test failed its analytic prediction. The single caveat is the revised acceptance criterion for **Test γ** — the claim originally written for **Test γ** (“*faster recovery than linear-plaquette + AW*”) was not supported by the data (the gauge controller is a softer feedback and settles in 0.41 s versus the linear+AW controller’s 0.28 s on this single-well problem). That version of the claim is withdrawn. The revised claim — that gauge feedback is *intrinsically* bounded while linear+AW depends on its saturator — is supported and is the structurally meaningful one. These four structural axes are revisited on PST-grounded deployed-scale topologies in § 6.4 (Tests α' - δ'), where the same claims are tested against the Verboomen OPF coordination baseline rather than Berggren pilot-voltage droop.

Listing 12 `tap_wear_week.py`

Listing 13 `fetch_entsoe.py`

G Appendix G — Necessity bound for non-degenerate nonlinear parameterizations

The necessity result in § 3 (Necessity of $b_1(G)$ feedback scalars) was stated for the *linear-projection* class of loop potentials. This class is the operationally natural one — it captures the standard form across compact- $U(1)$ lattice gauge theory, consensus-on-Lie-groups control, and the canonical Wilson form deployed here. For completeness, this appendix records what happens under the more general *scalar-shaping* class where the feedback ingredients are built from nonlinear functions of (possibly nonlinear) scalar combinations of holonomies, and identifies the non-degeneracy condition under which the bound $p \geq b_1(G)$ still holds.

G.1 General scalar-shaping class

Let $W : \mathbb{R}^{b_1(G)} \rightarrow \mathbb{R}$ be a loop potential of the form

$$W(\Theta) = \sum_{j=1}^p \omega_j(g_j(\Theta)), \quad g_j : \mathbb{R}^{b_1(G)} \rightarrow \mathbb{R}, \quad \omega_j : \mathbb{R} \rightarrow \mathbb{R},$$

with both g_j and ω_j smooth, but g_j not required to be affine. The linear-projection class of § 3 is the special case $g_j = a_j^\top(\cdot)$ affine.

Applying the chain rule twice gives the Hessian at the origin:

$$\nabla^2 W(0) = \underbrace{\sum_{j=1}^p \omega_j''(g_j(0)) \nabla g_j(0) \nabla g_j(0)^\top}_{\text{rank} \leq p} + \underbrace{\sum_{j=1}^p \omega_j'(g_j(0)) \nabla^2 g_j(0)}_{\text{can have arbitrary rank}}.$$

The first term is a sum of p rank-one symmetric matrices, hence has rank at most p . The second term is a sum of per-ingredient Hessians scaled by the first-derivative coefficients $\omega_j'(g_j(0))$ and can in principle contribute arbitrary additional rank: a single quadratic g_j can have full rank $b_1(G)$.

G.2 Non-degeneracy condition

Definition (Non-degeneracy at equilibrium). The scalar-shaping parameterization $W = \sum_j \omega_j \circ g_j$ is *non-degenerate at the equilibrium* if

$$\sum_{j=1}^p \omega_j'(g_j(0)) \nabla^2 g_j(0) = 0.$$

The condition is satisfied in two natural ways:

- (i) *Affine ingredients.* Each g_j is locally affine at 0, so $\nabla^2 g_j(0) = 0$ identically. This recovers the linear-projection class of § 3.

Listing 14 `pst_n1_topology.py`

Listing 15 `kron_composition_14bus.py`

- (ii) *Critical-point scalars.* Each ω_j is at a critical point of its scalar argument at the equilibrium, $\omega'_j(g_j(0)) = 0$ for every j . For the canonical Wilson form $\omega_j(s) = \kappa_j(1 - \cos s)$ with $g_j(0) = 0$, we have $\omega'_j(0) = \kappa_j \sin(0) = 0$ identically, so the condition holds trivially — regardless of whether g_j is affine.

Proposition (Non-degenerate necessity). Let $W = \sum_j \omega_j \circ g_j$ be non-degenerate at the equilibrium. If $\Theta = 0$ is locally asymptotically stable under the gradient flow of W , then $p \geq b_1(G)$.

Proof. Under non-degeneracy the second term in the Hessian expansion vanishes and $\nabla^2 W(0) = \sum_j \omega''_j(g_j(0)) \nabla g_j(0) \nabla g_j(0)^\top$ is a sum of p rank-one matrices, rank at most p . Asymptotic stability requires rank $b_1(G)$, hence $p \geq b_1(G)$. $\square$

G.3 Degenerate case: why parameter-counting breaks

Without non-degeneracy, the formal count p no longer reflects the operational tuning budget. A degenerate parameterization can hide multi-loop structure inside a single nonlinear g_j .

Example. Let $b_1(G) = 2$ and take

$$W(\Theta) = \kappa(\Theta_1^2 + \Theta_2^2),$$

written as $p = 1$ with $\omega_1(s) = \kappa s$ and $g_1(\Theta) = \Theta_1^2 + \Theta_2^2$. Then $\omega'_1(g_1(0)) = \kappa \neq 0$ so the non-degeneracy condition fails. The Hessian $\nabla^2 W(0) = 2\kappa I_2$ has full rank $b_1(G) = 2$ despite only $p = 1$ “ingredient.”

The apparent violation is illusory. The rank-2 structure is hidden inside g_1 : the quadratic form $\Theta_1^2 + \Theta_2^2$ has its own coefficients (implicitly both equal to 1 here) that are genuine tuning choices. The honest count of structural degrees of freedom in this parameterization is 2, not 1 — the two quadratic-form coefficients. The formal “ $p = 1$ ” parameterization is a redefinition of what counts as an ingredient, not a genuine parameter-saving.

G.4 Implications

The canonical Wilson form — the form deployed in § 3 and throughout §§3–8 — satisfies the non-degeneracy condition automatically via $\sin(0) = 0$, and its per-loop form with $p = b_1(G)$ attains the bound. The result of § 3 (Necessity of $b_1(G)$ feedback scalars) is therefore robust to the choice of parameterization class: whether stated for linear-projection or for non-degenerate scalar-shaping, the conclusion is the same — at least $b_1(G)$ scalar tuning ingredients are required, and the canonical Wilson form realizes this bound.

Degenerate parameterizations that formally achieve $p < b_1(G)$ do exist (as in G.3), but only by relocating the structural choices from the scalar shapings ω_j into the ingredient functions g_j themselves. Such redefinitions do not evade the structural cost of stabilizing $b_1(G)$ independent loop observables; they only rename it.

Listing 16 `verboomen_118bus.py`

Listing 17 `verboomen_118bus_network.py`

H Appendix H — Engineering memorandum: deployment-oriented summary for TSO practitioners

This appendix restates the construction of the main paper in the register of a TSO deployment memo. It assumes familiarity with phase-shifting transformer operation, day-ahead and intra-day OPF coordination, N-1 security, and SCADA/EMS architecture, and does not assume familiarity with cellular sheaves, lattice gauge theory, or the homological vocabulary of §§2–4. The mathematics is not repeated; pointers to the main paper are given where a reader may want the derivation. The purpose is to answer the practitioner’s questions, in order: what is this, what does it replace and what doesn’t it replace, where does it sit in existing infrastructure, what initial gain do I set, and what does it give me at what risk.

H.1 What it is

In a meshed transmission network with two or more *controllable series phase elements* in the same loop — phase-shifting transformers (PSTs), quadrature boosters, UPFC series injectors, or back-to-back converter pairs in the equivalent topological role — the branch-level phase commands around each enclosed loop must be *mutually consistent* to avoid driving real power in circles. Current practice coordinates these commands at two levels: device-local flow-control at each PST (single-branch scope, tap-changer speed) and centralized optimization across the network (transmission-system scope, 15-minute to 1-hour cycle). The loop-scope layer in between — loop-local real-time — is currently unoccupied.

This memo describes a single-parameter feedback term that fills that layer. The idea in one line:

Each controllable series phase element adds, to its existing droop or OPF-scheduled setpoint, a bias that *opposes* the loop sum Θ_C , with magnitude proportional to $\sin \Theta_C$, where $\Theta_C = \sum_{\ell \in C} \sigma_{C,\ell} \theta_\ell$ is the sum of commanded phase shifts around the loop. A non-zero Θ_C therefore drives the taps back toward mutual consistency ($\Theta_C = 0$); a consistent loop produces no bias.

No new hardware, no new telemetry channels, no changes to OPF or device-local loops. One scalar tuning gain κ per independent loop; a renormalization-group argument in the main paper (§ 4) shows that at practical mesh sizes a single network-wide κ suffices. The feedback is invariant to per-bus reference drift and robust to loop-internal topology change; the magnitude of circulating-power suppression depends on κ and the mesh parameters and is to be characterized on the specific system, with the validation numbers of § H.9 as calibration anchors.

H.2 Terms and symbols

Controllable series phase element. A mesh branch whose series phase offset θ_ℓ is set by its own control law rather than by the bus angles at its endpoints — a PST, UPFC series injector, quadrature booster, or back-to-back converter pair in the equivalent topological role. The FACTS-taxonomy umbrella term.

Phase-slaved branch. A branch whose phase drop is determined entirely by the bus angles at its endpoints — transmission lines, fixed-tap transformers, shunt compensation. Contributes $\theta_\ell = 0$ to any loop sum. The name describes electrical behavior (the phase drop slaves to the endpoint angles), not circuit-theoretic passivity.

Loop holonomy Θ_C . For an independent loop C , the signed cycle-sum of commanded phase offsets, $\Theta_C = \sum_{\ell \in C} \sigma_{C,\ell} \theta_\ell$. Invariant under any per-bus phase-reference drift. $\Theta_C = 0$ iff the phase commands around C are mutually consistent with some bus-angle assignment; $\Theta_C \neq 0$ drives real power to circulate around C .

Listing 18 `calibration_robust_diagnostic.py`

Listing 19 `su2.py`

Independent loops and basis. For a meshed sub-network with n buses and m branches (post-Kron-reduction onto the portion carrying controllable-series elements), $b_1 = m - n + c$ independent loops exist, where c is the number of connected components. Any cycle basis works; the control law is local in Θ_C and the transformation between bases is linear.

Symbols used throughout this appendix:

Symbol	Meaning
δ_n	Bus angle at node n
θ_ℓ	Commanded phase offset on controllable-series branch ℓ
ϕ_ℓ	Covariant branch mismatch: $\delta_{s(\ell)} - \delta_{t(\ell)} - \theta_\ell$
Θ_C	Loop holonomy around cycle C
$s(\ell), t(\ell)$	Source and target bus of branch ℓ , by reference direction
$\sigma_{C,\ell}$	Cycle-incidence entry: +1 if $\ell \in C$ matching orientation, -1 opposed, 0 if $\ell \notin C$
β_ℓ	Branch stiffness, $\beta_\ell = V^2/X_\ell$ in pu at nominal voltage
κ	Loop-feedback gain, one scalar per independent loop
τ	Control time constant of the link-variable dynamics
b_1	Number of independent loops: $m - n + c$

H.3 The gap it fills

Two engineering layers currently manage loop flow on a PST mesh. The device-local layer runs at tap-changer speed but has only single-branch visibility; the network-global layer has system-wide visibility but executes on a 15-minute-to-hour cycle. Nothing sits in between.

Layer	Mechanism	Scope	Timescale
Device-local	PST flow-control (tap-changer driven)	Single PST branch	Seconds
Loop-local real-time	— currently unoccupied —	Independent cycle	Seconds (tap-limited)
Network-global (opt)	PST/FACTS OPF (Verboomen–Van Hertem)	Transmission system	15 min – 1 hr

Between successive OPF solves, a load ramp, renewable excursion, or contingency produces loop flow that nothing corrects until the next run. The device-local controllers are tracking their individually scheduled branch setpoints and do not see the loop-level inconsistency; the OPF runs too slowly to catch it in real time. That gap — loop-scope, tap-changer-limited timescale, between OPF cycles — is where this feedback operates. Tests β' and ζ of § 6.4 quantify the magnitude: 59–85% reduction on the 6-bus DE-PL-CZ-style mesh, 45.9% on IEEE 118-bus against a full Verboomen–Van Hertem baseline.

H.4 What it does NOT replace

The proposal is additive, not substitutive. None of the following existing control layers is removed, retuned, or modified by a deployment.

Listing 20 `wz_match.py`

Listing 21 `conductor7.py`

OPF and security-constrained dispatch remain unchanged: the economic dispatch, security-constrained unit commitment, and PST tap target calculation produced by the existing OPF stack are untouched. The gauge layer reads the OPF tap targets as its reference operating point and does not overwrite them. [Test \$\zeta\$](#) verifies that at the OPF equilibrium with no disturbance the gauge layer applies zero displacement (static non-displacement measured to 10^{-16} pu on the 118-bus topology).

Pilot-voltage droop, N-1 security, frequency regulation, and day-ahead N-1 planning all operate on separate state variables, separate actuator sets, and separate timescales. The gauge feedback adds no dynamics to any of them and has no mechanism to affect them. [Test \$\alpha'\$](#) and [Test \$\beta'\$](#) both measure bus-frequency trajectories under gauge feedback and confirm they are indistinguishable from the uncoordinated baseline.

Device-internal firmware on UPFC, B2B, or MMC members of the controllable-series family — phase-locked loops, inner current control, virtual-impedance and circulating-current suppression, arm balancing — runs at converter-native (millisecond) speeds and is untouched by this proposal, which operates at EMS-dispatch speeds (seconds to minutes).

H.5 The control law

The feedback adds one term to each controllable-series element’s existing control law:

$$\tau \dot{\theta}_\ell = \underbrace{\beta_\ell \sin \phi_\ell}_{\text{existing droop / OPF tracking}} - \underbrace{\kappa \sum_C \sigma_{C,\ell} \sin \Theta_C}_{\text{new: loop feedback}}$$

where $\phi_\ell = \delta_{s(\ell)} - \delta_{t(\ell)} - \theta_\ell$ is the covariant branch mismatch (phase drop across the branch after the device’s commanded offset is subtracted), $\sigma_{C,\ell}$ is the cycle-incidence sign, and κ is one scalar gain per independent loop (see § H.6 for the setting). The first term is the control law already in place: on a converter, frequency droop; on a PST, the tracking term driving the tap toward its OPF-scheduled setpoint; on a UPFC, series-injection tracking. The second term is the new addition.

If a given controllable branch sits in only one loop — the typical case for most PSTs in a transmission mesh — the sum collapses to a single $\pm \kappa \sin \Theta_C$ term. Shared branches (an interior diagonal in a square-with-diagonal topology, a UPFC tie between two loops) pick up one contribution per loop, with signs from the cycle-incidence matrix. The incidence structure is static for a fixed topology and is precomputed; it is refreshed on topology change (§ H.10).

Per-tick implementation:

at each control tick (dt = control period):

```
for each independent loop C:
    Theta_C = sum over l of sigma[C,l] * theta[l]          # loop sum

for each controllable-series branch l:
    droop_term = beta[l] * sin(phi[l])                    # existing
    loop_term  = kappa * sum over C of sigma[C,l] * sin(Theta_C) # new
    theta_cmd[l] += (dt / tau) * (droop_term - loop_term)
```

The per-loop sum is cheaper than a single load-flow iteration; the full controller is $O(m \cdot b_1)$ per tick. On the IEEE 118-bus test system ($m = 186$ branches, $b_1 = 62$ cycles), the Python reference implementation runs a

Listing 22 `ws_berry_mesh.py`

Listing 23 `convergence.py`

dispatch tick in under 10 ms on a laptop-class CPU.

Worked 3-bus example. Three controllable-series elements on a triangle: $\theta_{12}, \theta_{23}, \theta_{31}$. One independent loop. $\Theta = \theta_{12} + \theta_{23} + \theta_{31}$. Under pure circulation with equal branch impedances, Kirchhoff at each node forces equal branch powers $P_{12} = P_{23} = P_{31} = P_{\text{circ}}$, monotone in Θ . The feedback drives $\Theta \rightarrow 0$ and $P_{\text{circ}} \rightarrow 0$ at the same rate. Full numerical trajectories on this topology are in § 6.3 of the main paper.

Gauge invariance. If every bus’s phase reference drifts independently by some $\alpha_n(t)$ — independent PMU clock skew, uncorrelated reference-frame adjustments across TSO control centers, different slack-bus choices at different times — each θ_ℓ shifts by $\alpha_{s(\ell)} - \alpha_{t(\ell)}$, which telescopes to zero around any closed loop. The loop holonomy Θ_C is structurally immune to per-bus reference drift; the same property makes the observable calibration-free in the PMU sense of § 7.2 (exact invariance to $\pm 2^\circ$ per-bus errors, verified to machine precision).

H.6 Tuning and commissioning

One free parameter per independent loop. A renormalization-group argument in § 4 of the main paper collapses the per-loop gains to a single network-wide κ at practical mesh sizes; for an initial deployment this can be treated as one scalar for the entire meshed region.

Initial scale. A conservative starting point, comparable to the weakest branch stiffness in the mesh:

$$\kappa_{\text{init}} \approx 0.3 \cdot \min_{\ell} \beta_{\ell} \quad (\text{pu})$$

where $\beta_{\ell} = V^2/X_{\ell}$. Too small: the loop term is dominated by the droop/tracking term and circulation persists. Too large: the feedback fights the OPF-scheduled setpoint and drags taps toward $\Theta_C = 0$ when economic dispatch wanted non-zero circulation for a reason.

Commissioning procedure.

1. *Record the baseline.* With $\kappa = 0$ (loop feedback disabled), record the circulating-power signature on a representative disturbance — a load ramp, renewable excursion, or contingency event that has good historical precedent on the specific mesh. This gives the “before” calibration.
2. *Enable at κ_{init} .* Replay the same event (offline or live with operator-in-the-loop). Measure Θ_C settling time and circulating-power suppression against the baseline.
3. *Sweep κ by factors of 2–3.* Walk the gain up and down by factors of 2–3 and find the setting where Θ_C settles within a few multiples of τ with no visible tap hunting or undershoot. A Pareto sweep over dead-band and tick interval analogous to [Test \$\gamma'\$](#) (§ 6.4) gives the practical operating point; on the DE-PL-CZ-style 6-bus mesh the sweet spot was dead-band = 0.030 pu and tick = 60 s.
4. *Verify static non-displacement.* Confirm that OPF-scheduled steady-state tap positions are reached within the usual drift (a few percent) once the transient has settled. This is the acceptance criterion that separates a gauge feedback from an override: the deployment must preserve the OPF equilibrium in the absence of disturbance.

Expected behaviour at a well-tuned κ . Circulating real power meaningfully suppressed relative to the $\kappa = 0$ baseline (setpoint-tracking only); Θ_C held near zero within telemetry noise; no change to the transfer-component response (scheduled power delivery through the mesh preserved). The achievable suppression

Listing 24 wilson_2d.py

Listing 25 mc_2d_vs_3d.py

factor is a function of κ , loop branch stiffnesses, disturbance spectrum, and mesh topology; the numbers of § H.9 anchor the expected range.

Robustness verification. Three checks to run during commissioning:

Per-bus reference drift. Inject uncorrelated slow drifts at each loop bus’s reference frame (or equivalently, introduce per-PMU calibration offsets) and confirm Θ_C is unaffected. The telescoping identity is algebraic and exact in the math; in practice the limiter is tap-position quantization and telemetry sampling, not the gauge property (Test α' , § 6.4, and § 7.2 calibration robustness).

Contingency / topology change. Trip a branch inside or adjacent to a loop. The loop basis itself may change (one loop closes, another opens, two loops merge). The feedback retargets the new loops provided the cycle-incidence matrix is refreshed on the topology event; the EMS’s existing topology-processor output is the right trigger. Settling after a topology change occurs on the natural loop time-scale without parameter retuning (Test δ' , § 6.4: 8.0 min on a DE-PL-CZ 4-bus mesh).

Discrete taps. PST taps are quantized (N steps, typically 2^4 – 2^6 for mechanical, substantially finer for power-electronic devices). Residual Θ_C due to quantization is of order $1/N^2$ times branch stiffness. For $N \geq 64$ it is below other error sources; for mechanical PSTs with $N = 16$ – 32 it is comparable to telemetry noise. The IEEE 118-bus benchmark (Test ζ) uses 33 taps per PST and attains the headline performance of § H.9.

H.7 SCADA/EMS placement

Three architectures are operationally viable. The choice depends on the existing infrastructure; the framework does not privilege any option.

Centralized EMS dispatch. The existing OPF dispatch path is reused. A central controller runs on the same cadence as the existing PST dispatch process, reads the state inputs (tap positions, branch-angle differences from PMU or state-estimator), computes the Θ_C scalars, applies the update rule, and issues the resulting tap targets on the same dispatch bus that distributes OPF setpoints today. Architecturally *identical* to existing PST OPF dispatch, with the loop-flow correction added as an additional scalar per cycle. Deployment cost: a computation module in the EMS. No new communication paths, no new hardware, no changes at substation level.

Peer-to-peer substation communication (IEC 61850). Substations on a shared loop exchange their local θ_ℓ values with immediate neighbours via IEC 61850 GOOSE (within-substation) or ICCP (cross-substation). Each substation computes Θ_C for its local loop from received values and applies the feedback to its own PST. Bandwidth: one scalar per link per tick at ~ 10 Hz; latency tolerance ~ 100 ms. Both protocols are commodity on modern substation infrastructure. Natural option where centralized coordination is not operationally preferred — for instance, in cross-TSO deployments where coordination is bilateral rather than routed through a single control centre.

Local per-device fallback. Each device sees its own θ_ℓ and a locally-measurable surrogate for the cycle contribution. Appendix F Test β quantifies the performance gap: the local-only implementation matches the centralized implementation to within $\sim 10\%$ on the 3-bus B2B triangle. This is the operational fallback for degraded communication, not a preferred primary architecture. [6] observe that any communicated coordination scheme “*should have a fallback control option based on locally available measurements in case excessive latencies are encountered,*” without specifying one; the local implementation is that fallback.

Calibration of PMU infrastructure. § 7.2’s calibration-robustness result affects architecture choice. Because Θ_C is exactly invariant to per-bus PMU calibration errors, the centralized and peer-to-peer ar-

chitectures can operate on PMU fleets that do not meet IEEE C37.118.1 total-vector-error specifications, reducing infrastructure-maintenance cost and permitting operation during GPS-holdover periods. Cross-TSO peer-to-peer deployments benefit most: no joint calibration campaign is required to exchange loop-holonomy scalars across a border.

H.8 Scope: when it helps, when it doesn't

Not every PST installation is in a topology where this controller does work the existing layers don't. Honest topology-by-topology summary:

Topology	b_1	Loop object	Controller adds
Radial: one PST, no loop	0	undefined	nothing — existing OPF and device-local control suffice
Two parallel paths, PST on each	1	pairwise tap difference	gauge-invariant restatement of classical current sharing
N parallel paths, PST on each	$N - 1$	pairwise differences	same — no genuine multi-party cycle
Triangle (3 buses, 3 PSTs)	1	genuine cycle sum	single-scalar circulating-power suppression (the $\sim 3\times$ result)
Square-with-diagonal (4 buses, 5 PSTs)	2	cross-coupled cycle sums	multi-loop coordination that per-device tuning handles poorly
Larger planar meshes (IEEE 118-bus, ENTSO-E)	≥ 2	multiple cross-coupled cycles	direct generalization, numerically validated

The deployed European PST mesh — multiple TSOs, multiple enclosed regions, shared tie lines between them — is firmly in the square-with-diagonal-and-larger regime. A 3-area Germany–Poland–Czech style mesh already has two enclosed regions and a shared interior tie.

Prerequisites:

- *At least one controllable series phase element per independent loop.* No device on the loop, no degree of freedom, no observable. A loop built entirely from phase-slaved branches has $\Theta_C \equiv 0$ identically and no loop-level control problem to solve.
- *Steady-state or slow-transient regime* — timescales long compared to the AC period, short compared to an OPF cycle. Topological phase slips (2π excursions under severe disturbance) are rare in nominal operation; § 5.2 quantifies the activation energy and engineering margin. The controller does not prevent slips; it accelerates return to the nearest stable sector.
- *Balanced three-phase operation.* The framework specialises to the single-phase (abelian $U(1)$) case. The $SU(2)$ extension for unbalanced operation is documented in [Appendix B](#) as an avenue for future work; it has not been validated at the engineering level in this paper.

H.9 Expected numerical performance

The six validation tests of § 6 and the calibration diagnostic of § 7.2 give the following quantitative picture. All numbers are reproducible from the scripts in [Appendix E](#).

Scenario	Baseline	With gauge feedback	Reduction	Source
3-bus B2B load step	uncoordinated droop	gauge layer	$\sim 3\times$	§ 6.3, Test α
3-bus B2B setpoint error	uncoordinated droop	gauge layer	$\sim 3\times$	§ 6.3, Test β
3-bus B2B stochastic noise	uncoordinated droop	gauge layer	$\sim 3\times$	§ 6.3, Test γ
3-area mesh ($b_1 = 1$), drift attribution	no gauge	gauge active	0% exact	§ 6.4, Test α'
4-bus DE-PL-CZ mesh, N-1 tie-line outage	gauge re-tuned	gauge topology- adaptive	settle 8.0 min, zero retuning	§ 6.4, Test δ'
6-bus DE-PL-CZ-style mesh, load ramp	OPF alone	OPF + gauge	84.9%	§ 6.4, Test β'
6-bus DE-PL-CZ-style mesh, N-1 event	OPF alone	OPF + gauge	66.5%	§ 6.4, Test β'
6-bus DE-PL-CZ-style mesh, gate-closure transition	OPF alone	OPF + gauge	59.4%	§ 6.4, Test β'
Week-long ENTSO-E sample, tap-changer operations	uncoordinated	gauge layer	63.2% (to 1.51 ops/PST/day)	§ 6.4, Test γ'
IEEE 118-bus + 5 PSTs, static OPF equilibrium	Verboomen OPF	OPF + gauge	non-displacement exact ($\sim 10^{-16}$ pu)	§ 6.4, Test ζ
IEEE 118-bus + 5 PSTs, fast events	Verboomen OPF	OPF + gauge	45.9% (at 94% of physical bound)	§ 6.4, Test ζ
IEEE 118-bus + 5 PSTs, N-1 contingency	Verboomen OPF	OPF + gauge	22.5% additional	§ 6.4, Test ζ
Per-bus PMU calibration error up to $\pm 2^\circ$	naive bus-angle observ- able	Θ_C observable	exact invariance (10^{-16} rad)	§ 7.2

The headline number for a deployment evaluation is the 118-bus result: 45.9% reduction in integrated circulating power during fast events, under full OPF coordination already in place, against the Verboomen–Van Hertem baseline that represents current best-practice in coordinated PST control. This is 94% of the theoretical maximum reduction achievable by any loop-flow control scheme with the same five PSTs (the bound is set by the rank of the actuation-reachability matrix, § 7.5); the residual 6% is the reachability gap, not tuning-dependent, and cannot be closed without adding PST capacity or relocating existing PSTs.

H.10 Deployment risks and mitigations

Five risks are analysed in the main paper and summarised here in order of decreasing operational concern.

Topological phase slips. If a disturbance moves the loop holonomy past π radians, the controller’s natural coordinate wraps to the other branch and the feedback applies in the opposite direction. Real phenomenon, characterised in § 5.2. Mitigation: rate-limiting on tap motion and an explicit slip-detection test on each tick; the controller disables and alarms on detected slip, handing back to the next OPF cycle. The slip regime is reached in realistic operation only under large disturbances that would also be flagged by existing frequency and line-loading alarms. § 5.3 verifies the slip boundary sits outside normal operating envelopes.

Discrete tap resolution. Analysed in § 5.3 and § H.6. The conservative $N \geq 64$ bound sits above the $N = 33$ used in Test ζ , which attains the headline performance of § H.9. For older PSTs at lower tap counts, the quantization contribution should be quantified numerically against the specific mesh before deployment.

Actuation rate limits. Tap changers have mechanical rate limits (typically one change per ~ 30 s for on-load tap changers, with longer cooldowns). The feedback respects this as a hard constraint in the update rule. [Test \$\gamma'\$](#) verifies the week-long operation count (1.51 ops/PST/day) is well inside mechanical duty limits, and represents a 63% reduction in mechanical wear versus the uncoordinated baseline — an unexpected benefit.

Calibration of measurement infrastructure. Addressed in § 7.2 and summarised in § H.5: the holonomy observable is exactly invariant to per-bus PMU calibration errors. The separate loop-flow metric used internally by the controller is not invariant, but is computed from a model-internal solve against known injections and PST positions — it does not take PMU data as input.

Interaction with other controllers. The proposal is stable in isolation (§ 3.5), compositional with outer-layer OPF (verified in Tests β' and ζ), and shares no actuators with frequency regulation or voltage control. The remaining theoretical risk — interaction with protective relaying on abnormal-configuration events at protection time-scales (sub-cycle to a few cycles) — is outside simulation scope and is flagged in § H.12 as a pre-field-trial validation item.

H.11 Staged adoption path

A deployment that mirrors the validation-tier structure of the paper can proceed in four stages, each of which produces operational value without committing to the next.

Stage 1 — Monitoring-only. Deploy the Θ_C computation as a read-only observable on the existing EMS. No control action, no actuator engagement, no changes to any existing controller. The output is a time-series of per-loop holonomy values, which serves three purposes: (i) a model-validation diagnostic (disagreement between Θ_C computed from PMU data versus from the network model indicates model miscalibration or unrecognised topology change); (ii) an operator-facing visibility metric for circulating power that does not exist in most control rooms today; (iii) pre-deployment data for tuning κ per § H.6. § 7.3 of the main paper expands on this use.

Stage 2 — Offline replay. Replay historical disturbance events (load ramps, N-1 contingencies, congestion episodes) through the gauge feedback offline, compare the counterfactual trajectories against the historical record, and quantify expected benefit on the specific network. Pure desk analysis, no operational changes, produces a quantitative value estimate to weigh against deployment cost.

Stage 3 — Single-corridor pilot. Deploy the feedback on a single meshed loop with a single PST. Operator has full manual override; existing OPF dispatch continues unmodified as a safety net. Collect a ~ 3 -month operational record. The 3-bus B2B results of § 6.3 are the simplest indicator of what to expect in this regime.

Stage 4 — Full deployment. Extend to all independent mesh loops carrying PSTs, using the renormalization-group result from § 4 to tune with a single network-wide κ . The week-long [Test \$\gamma'\$](#) result is the relevant reference case.

Each stage produces a decision point. If Stage 1 monitoring shows that circulating power is already well-controlled by existing means, Stages 2–4 can be skipped. If Stage 2 replay shows that the specific network topology does not produce the improvements in § H.9 for idiosyncratic reasons, the deployment can be deprioritised. The staging is designed so that each stage produces operational value on its own, even if the next is not pursued.

H.12 What is not yet validated

Honesty about the validation frontier matters for a deployment decision. The following items are *not* validated in this paper and are prerequisites for a field deployment beyond Stage 2 above.

- *Controller-in-the-loop simulation (C-HIL).* All tests in § 6.4 use a pure-simulation power-flow solver. The controller is real production-candidate code, but the plant is simulated. A C-HIL campaign — controller binary on a production-equivalent processor, talking to a simulated grid via real-time

communication interfaces with realistic latency and packet loss — is the standard next step and has not been conducted here.

- *Hardware-in-the-loop simulation (HIL)*. Real-time digital simulator (OPAL-RT, RTDS, or equivalent) emulating the grid at the 50 Hz cycle level with hardware-level analogue interfaces. Highest- fidelity pre-field validation available; not conducted here.
- *Field trial*. Live transmission mesh with operator-in-the-loop and closed-loop control of production PSTs. The staged path of § H.11 is designed to reach this tier incrementally.
- *Protective-relaying interaction*. Analysed at steady-state and small-signal levels only. Interaction on protection time-scales (sub-cycle to a few cycles) is outside simulation scope and requires C-HIL or HIL verification.
- *Full AC power-flow validation*. The validation suite uses DC power flow with realistic non-uniform reactance. AC effects are expected to be small in the loop-flow coordination regime but have not been quantified on the full test set.
- *Multi-TSO operational coordination framework*. Algebraic gauge invariance and calibration-free Θ_C exchange are both demonstrated, but the operational governance — who reads what, who acts, who alarms, how the loop-flow mandate is divided — is a governance question rather than a technical one and is not addressed here.

A reader evaluating deployment should treat § 6 and § 7 as covering the simulation-tier validation completely, and the items above as the remaining work required before field deployment at scale.

H.13 Reproducibility

All numerical results quoted in § H.9 are reproducible from the scripts in Appendix E. Controller code depends on `pandapower` and standard scientific-Python libraries. The IEEE 118-bus benchmark is the `pandapower case118` built-in, augmented with five PSTs placed per Appendix E. All tests run to completion in under five minutes on a laptop-class CPU except the week-long [Test \$\gamma'\$](#) (≈ 45 min).

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